

AI 모델 성능 개선 결과

(관심 수위 이상)

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(1 , 3 시간 뒤 시점 수위 오차 및 정확도 & 10분 단위 분석)

1. Parallel LSTM 성능 개선 결과

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
 - smoothing : 궁내교 (1) 대곡교 (2)
 - 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)
-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+) 누적 강우 컬럼 추가

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교 _Ti	궁내교 _Ti	대곡교 누적강우	궁내교 누적강우
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0	0.0	0.0

관심 수위 전체 데이터

- 학습 , 검증 : 총 67개 中 61개 데이터 파일 사용
 - 테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)
- + 학습에서 검증에 사용된 2개 이벤트 사용
- 학습 : 2014 ~ 20230629 (43개)
 - 검증 : 20230713 ~ 250813 (12개)
 - 테스트 : 2550916 ~ 251006 (6개)

관심 수위 20년 이후 데이터

- 학습 , 검증 : 총 40개 中 35개 데이터 파일 사용
 - 테스트 : 학습 데이터에서 제외한 5개 데이터 파일 (4개 이벤트)
- + 학습에서 검증에 사용된 2개 이벤트 사용
- 학습 : 2020 ~ 20240920 (25개)
 - 검증 : 20240921 ~ 250919 85번 (7개)
 - 테스트 : 250919 86번 ~ 251006 (3개)

누적강우 110mm 학습데이터

: 누적강우110mm 이상 1 ~ 6은, 동일한 ' 집중호우 (12시간 110mm 이상) ' 조건에서
학습· 검증· 테스트 데이터 구성방식을달리한6가지실험시나리오中4, 5 번째시나리오

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
 - smoothing : 궁내교 (1) 대곡교 (2)
 - 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)
-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우110mm 이상추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

관심 수위 전체 데이터

- 학습 , 검증 : 총 11개 中 9개 데이터 파일 사용
- 테스트 : 학습 데이터에서 제외한 2개 데이터 파일 (2개 이벤트)
+ 학습에서 검증에 사용된 4개 이벤트 사용

조합 4

- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022 (39번 이벤트) (1개)
- 테스트 : 2022 (66.67번 이벤트) (1개)

조합 5

- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022 (35번 이벤트) (1개)
- 테스트 : 2022 (66.67번 이벤트) (1개)

궁내교

- » 단일 이벤트 기준으로 잘 맞는 것도 많지만 이벤트 간 분산이 큼
- » 22.07.12 : 거의 모든 조합에서 R² 음수 → 데이터 문제가 아니라 사상 특성 문제

- R2 평가

12시간 예측

R2	궁내교	
Test data	6시간	3시간
20.08.02	0.8759	0.7961
22.07.12	-1.9263	-1.3191
23.07.11	0.7968	0.7596
24.07.23	0.6043	0.4878
25.07.16	0.4520	0.4308
25.08.14	0.8114	0.2319

관심수위

궁내교				
관심수위	+ 누적강우	3시간 예측	X 0.98	X 1.02
0.9656	0.8893	0.8214	0.8492	0.7922
0.4201	-0.6973	-0.3291	0.2864	0.0156
0.8622	0.9031	0.9196	0.8814	0.8994
0.7308	0.8385	0.7568	0.8244	0.8400
0.7947	0.7659	0.9456	0.7627	0.8186
0.5087	0.6406	-0.9700	0.4144	0.4433

유량

궁내교
유량
0.9470
-0.3192
0.9153
0.7854
0.7869
0.7481

누적강우 110mm

궁내교	
4번 조합	5번 조합
0.7237	0.8774
0.3192	0.0586
0.8533	0.8028
0.7623	0.7051
0.8108	0.7764
0.8475	0.8673

대곡교 제외

궁내교
유량
0.8253
-1.2836
0.9267
0.7720
0.4997
0.7909

20년 이후

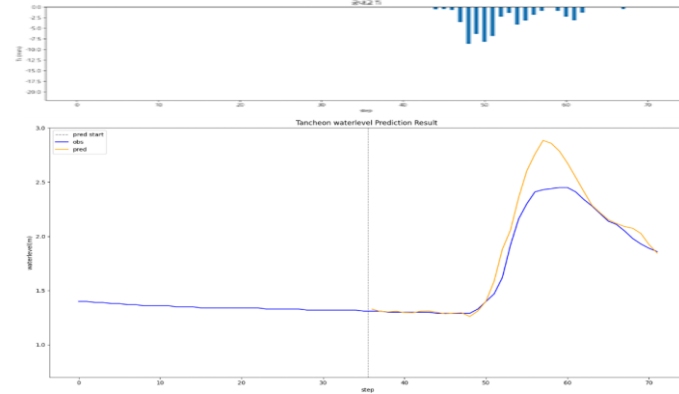
궁내교	
20년 이후	+ 누적강우
0.9513	0.9507
-0.8548	0.2329
0.8751	0.8663
0.8136	0.8524
0.8213	0.7764
0.7835	0.5660

궁내교 - 12시간 예측 (6시간)

* model_sunnam_(gn/dg)72

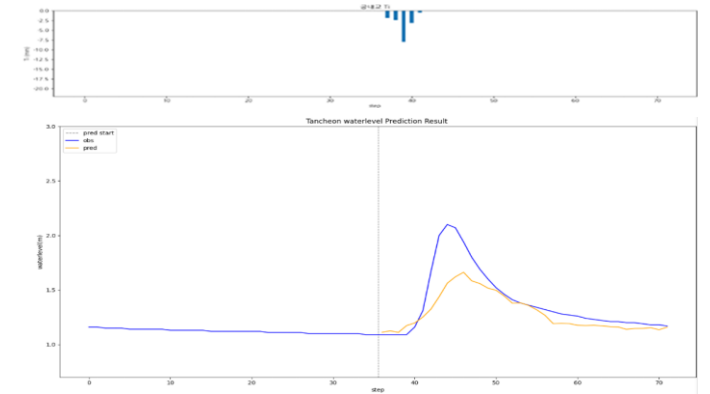
testdata_200802

R2 : 0.8759



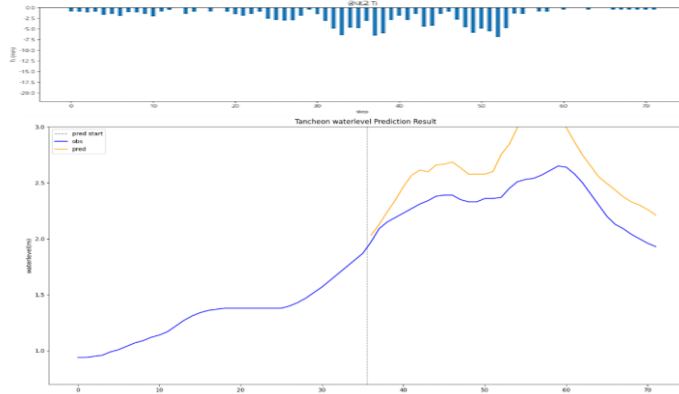
testdata_240723

R2 : 0.6043



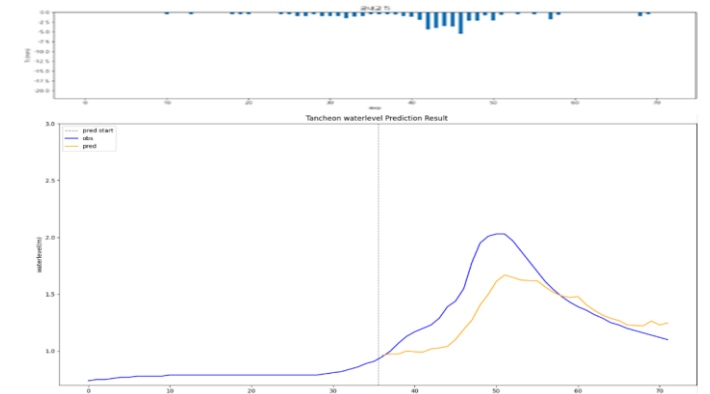
testdata_220712

R2 : -1.9263



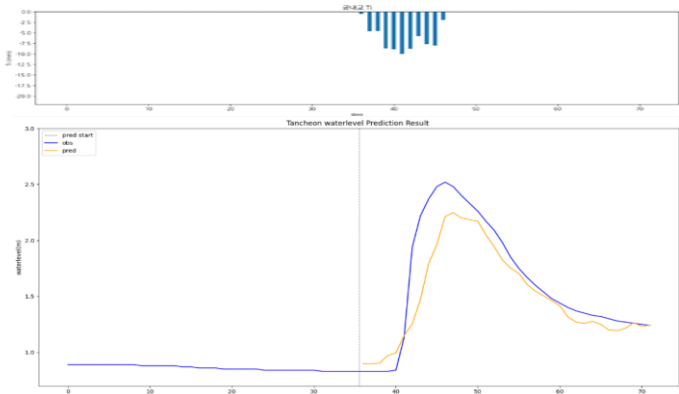
testdata_250716

R2 : 0.4520



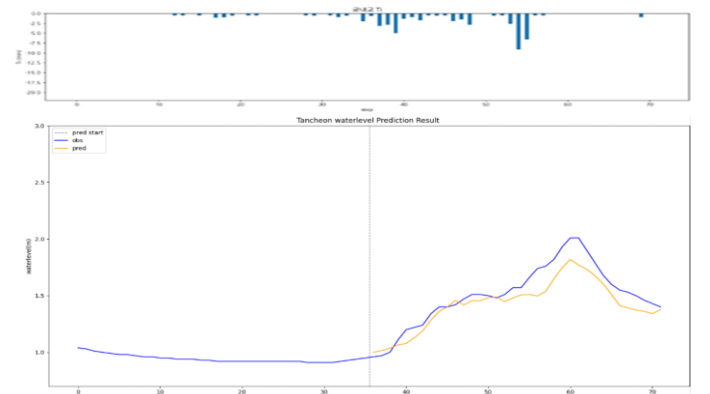
testdata_230711

R2 : 0.7968



testdata_250814

R2 : 0.8114

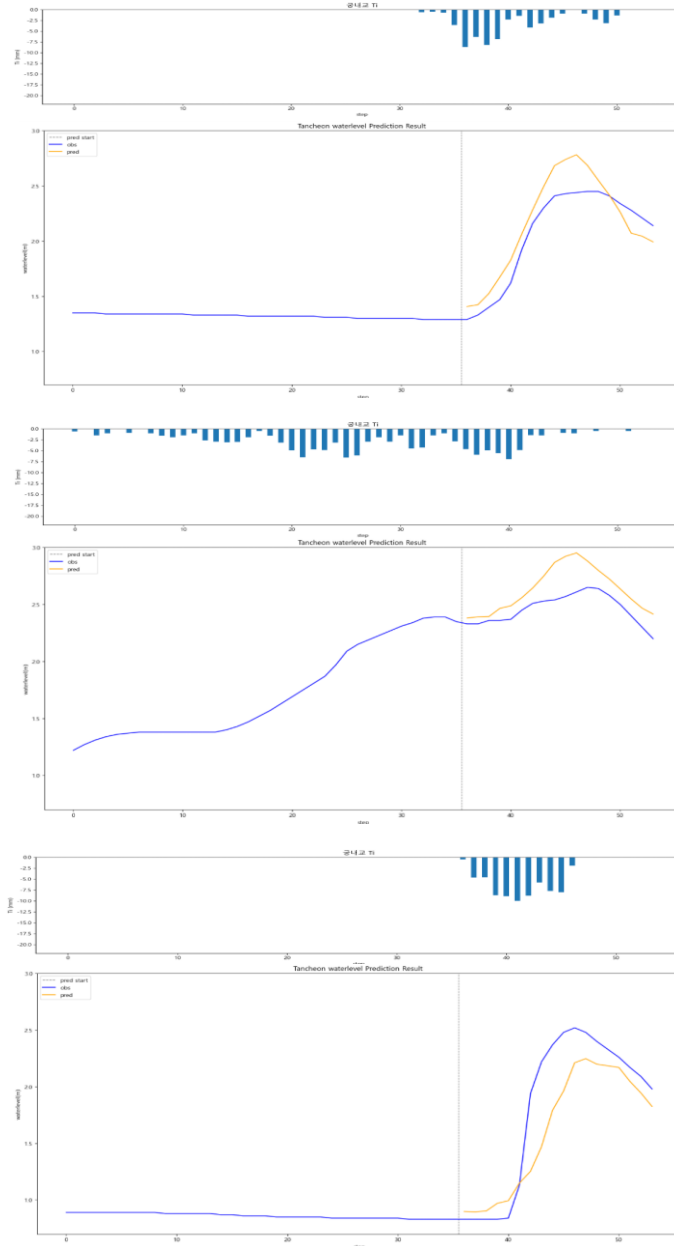


궁내교 - 12시간 예측 (3시간)

* model_sungnam_(gn/dg)72

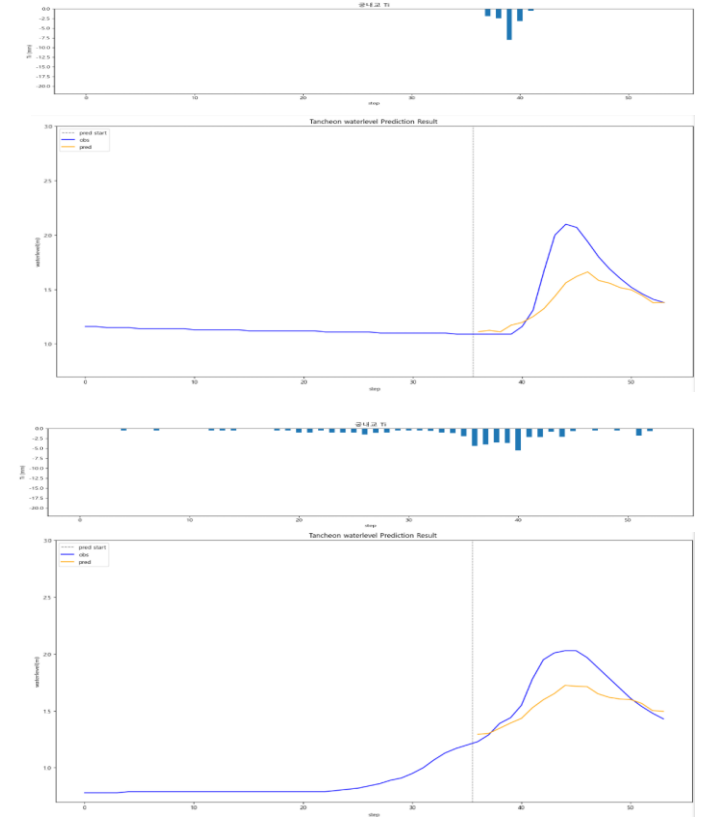
testdata_200802

R2 : 0.7961



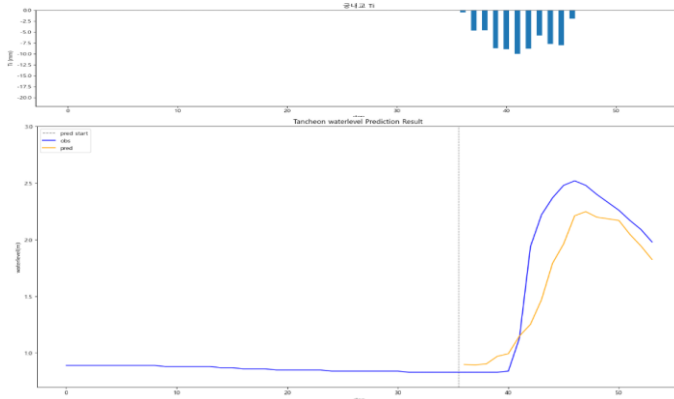
testdata_240723

R2 : 0.4878



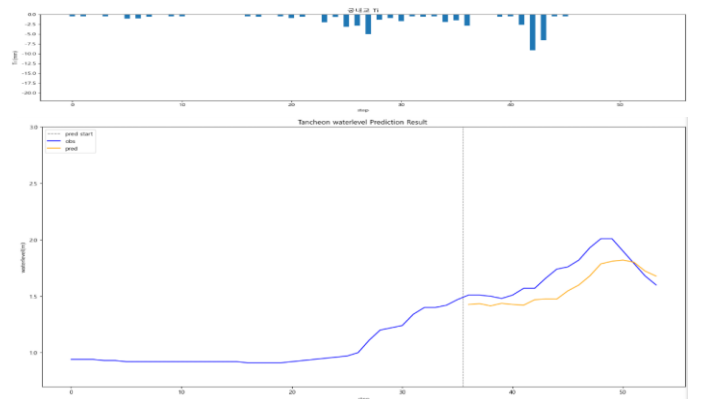
testdata_220712

R2 : -1.3191



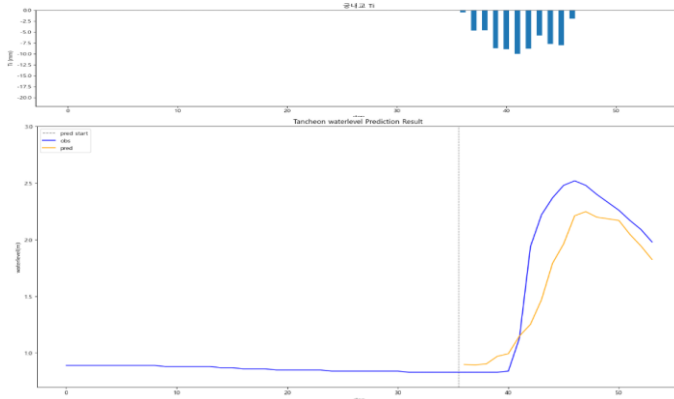
testdata_250716

R2 : -2.8628



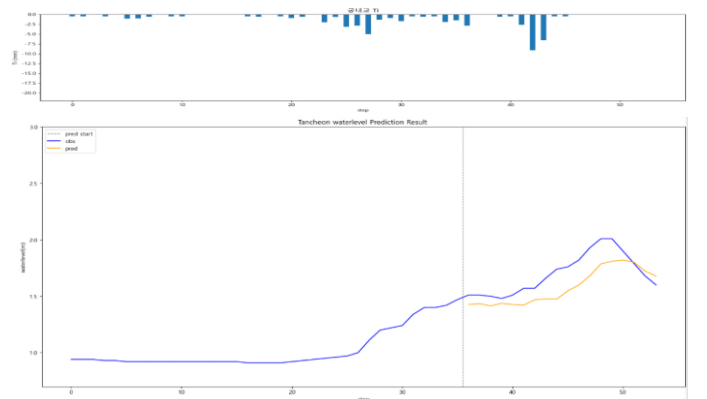
testdata_230711

R2 : 0.7596



testdata_250814

R2 : 0.2319

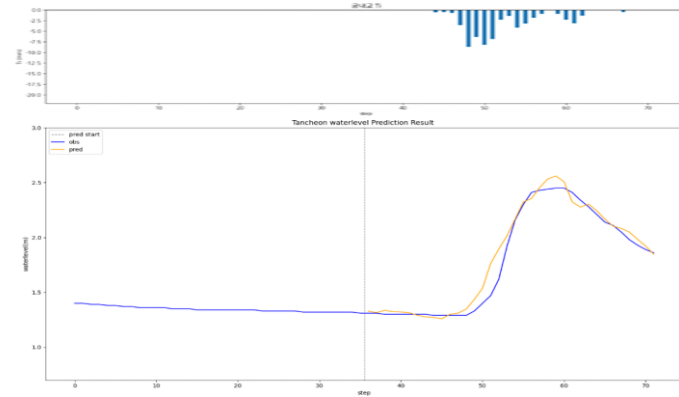


궁내교 - 관심수위

* model_(gn/dg)_wl18_3_2

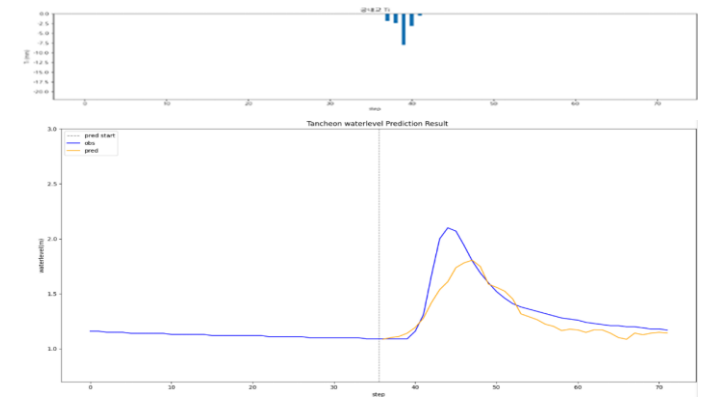
testdata_200802

R2 : 0.9656



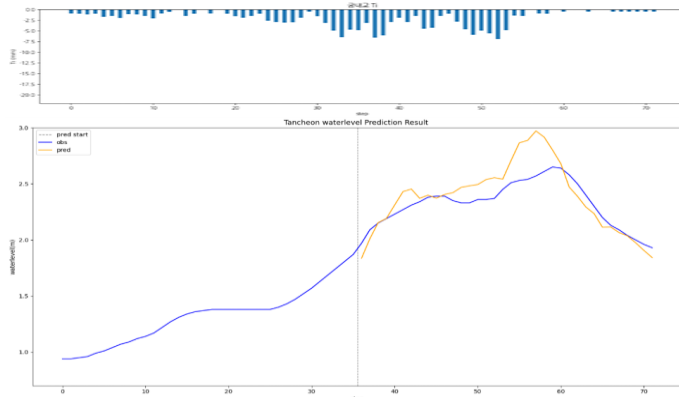
testdata_240723

R2 : 0.7308



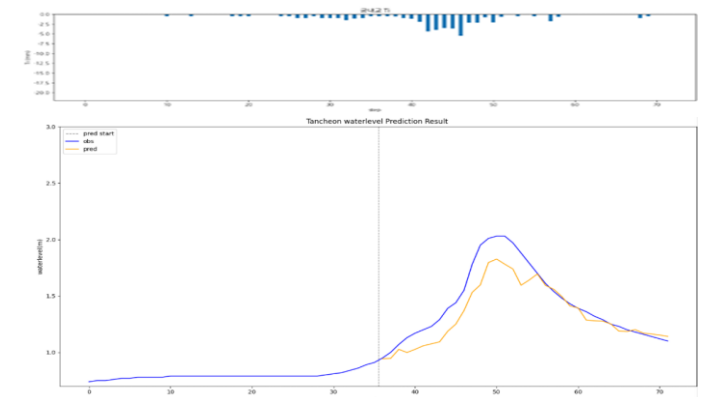
testdata_220712

R2 : 0.4201



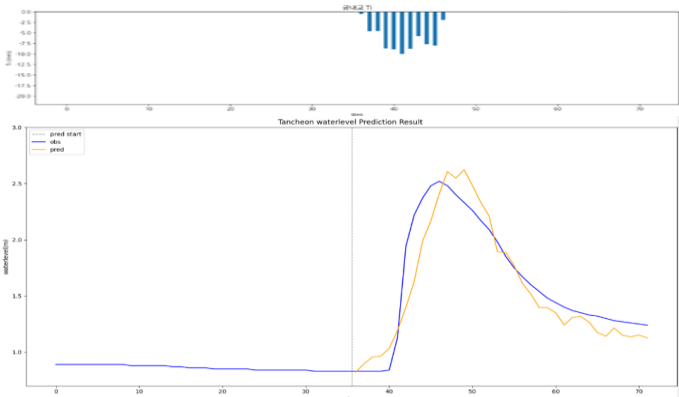
testdata_250716

R2 : 0.7947



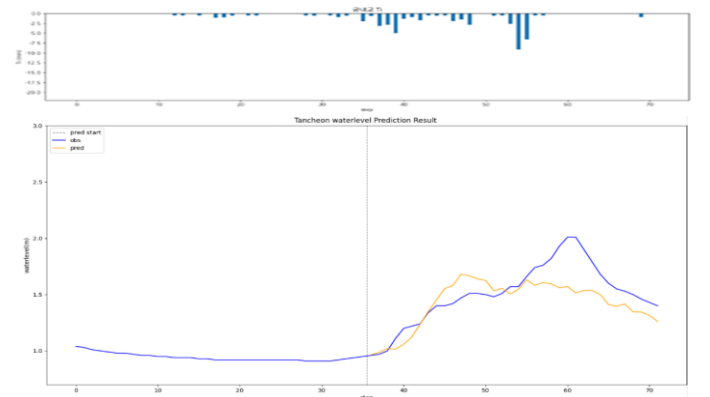
testdata_230711

R2 : 0.8622



testdata_250814

R2 : 0.5087

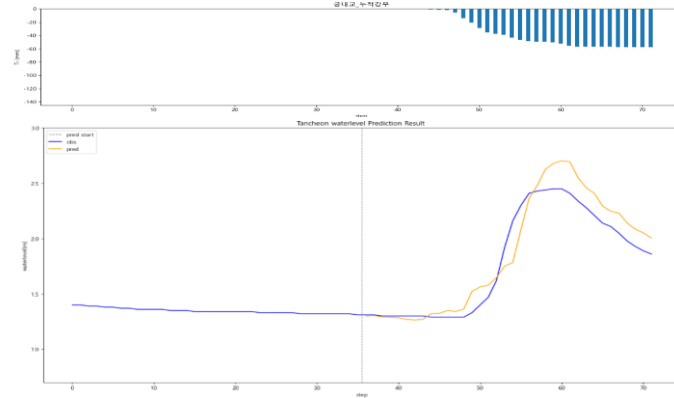


궁내교 - 관심수위 (+ 누적강우)

* model_(gn/dg)_rf

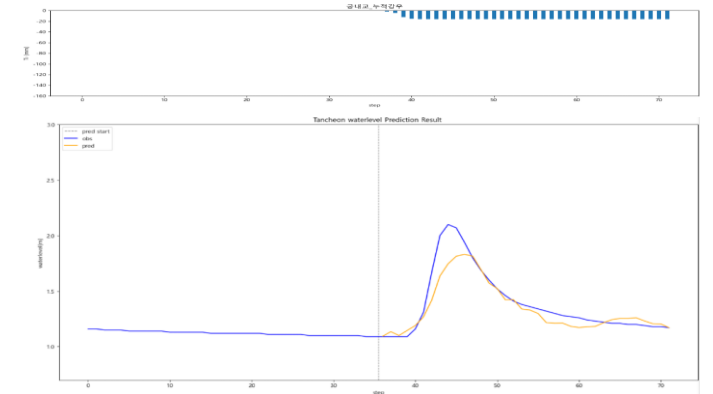
testdata_200802

R2 : 0.8893



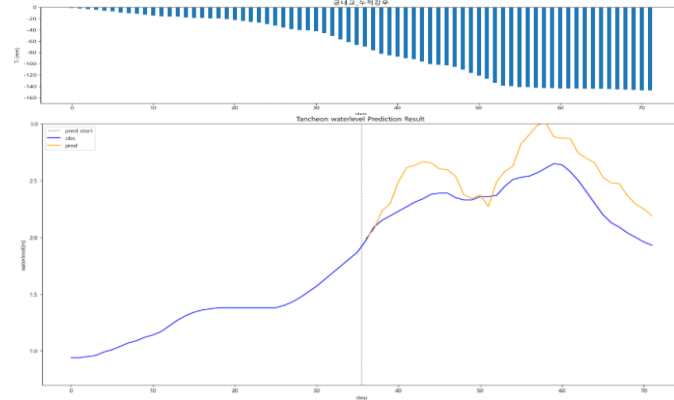
testdata_240723

R2 : 0.8385



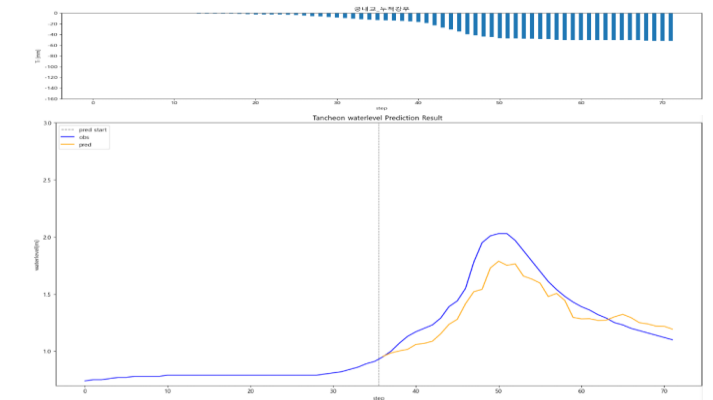
testdata_220712

R2 : -0.6973



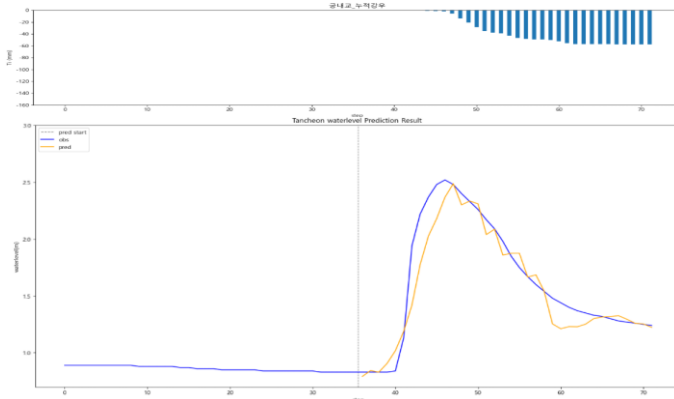
testdata_250716

R2 : 0.7659



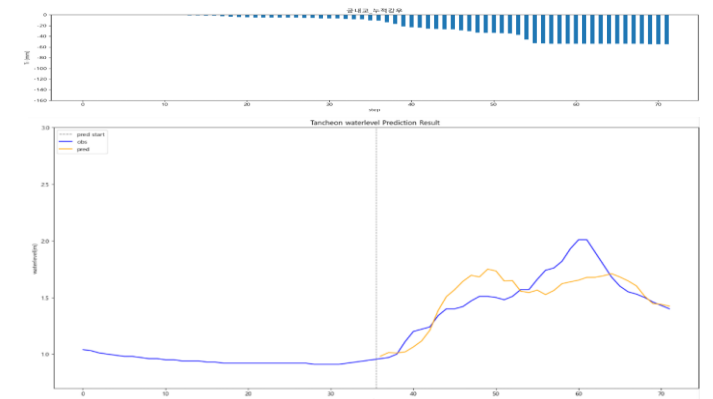
testdata_230711

R2 : 0.9031



testdata_250814

R2 : 0.6406

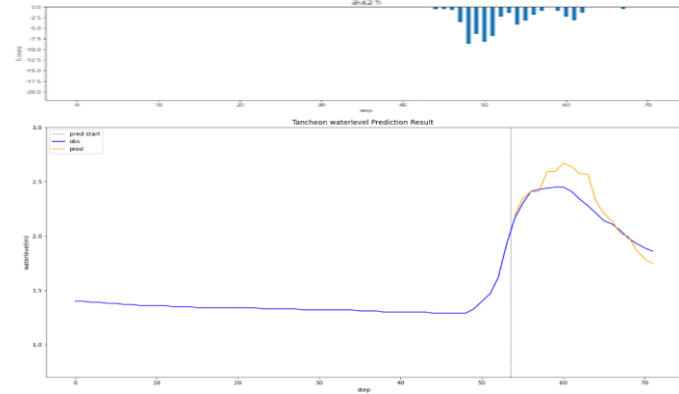


궁내교 - 관심수위 (3시간 예측)

* model_(gn/dg)_3

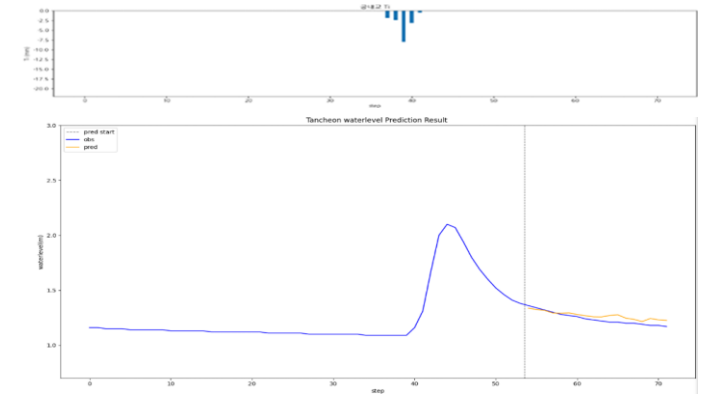
testdata_200802

R2 : 0.5417



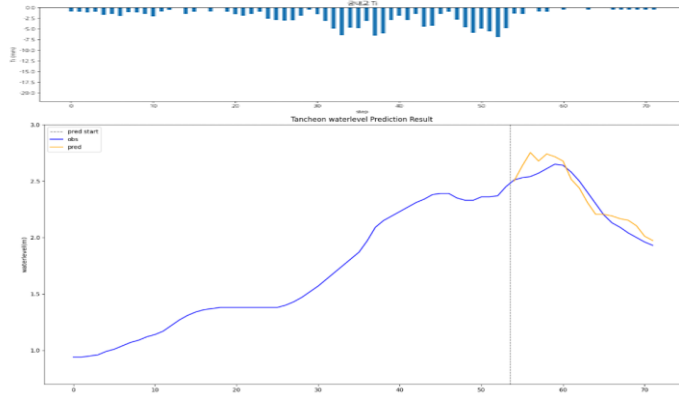
testdata_240723

R2 : 0.5351



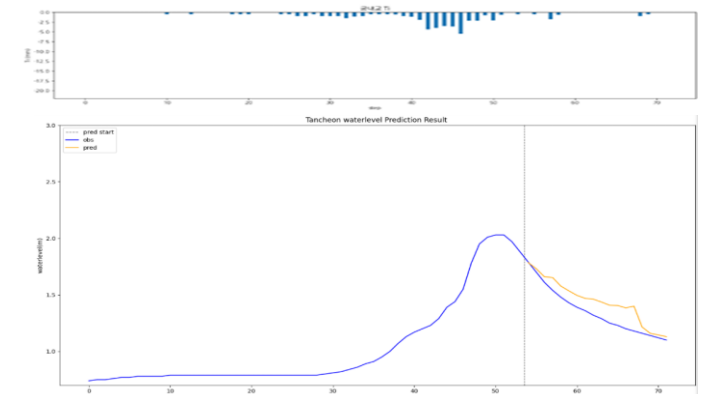
testdata_220712

R2 : 0.8669



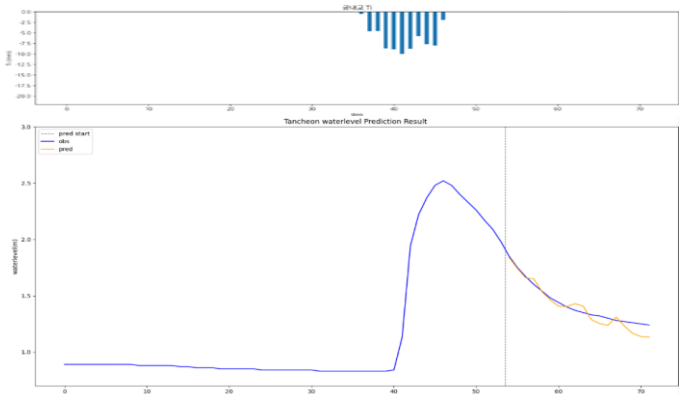
testdata_250716

R2 : 0.6605



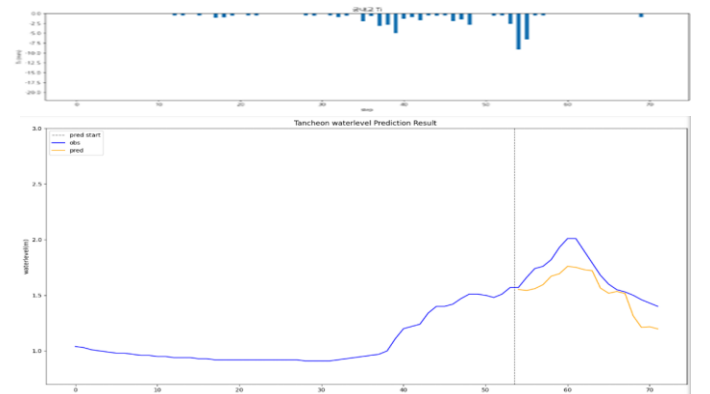
testdata_230711

R2 : 0.9006



testdata_250814

R2 : 0.2037

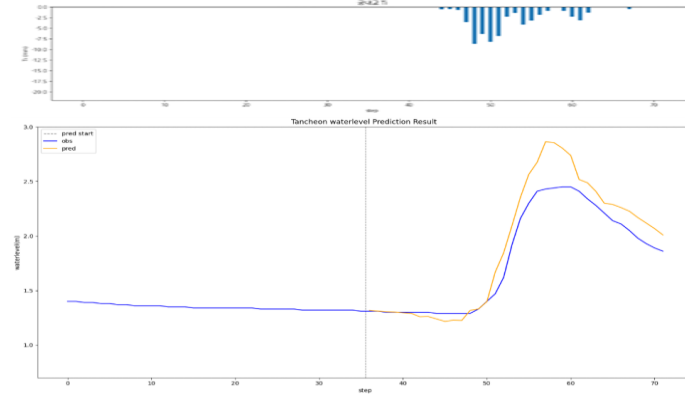


궁내교 - 관심수위 (수위 X 0.98)

* model_(gn/dg)_0.98

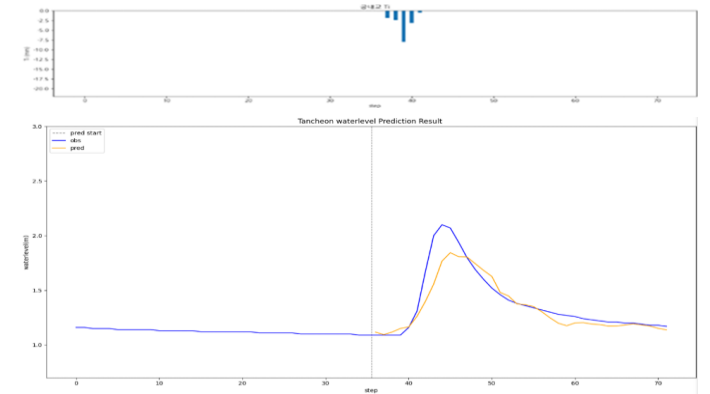
testdata_200802

R2 : 0.8492



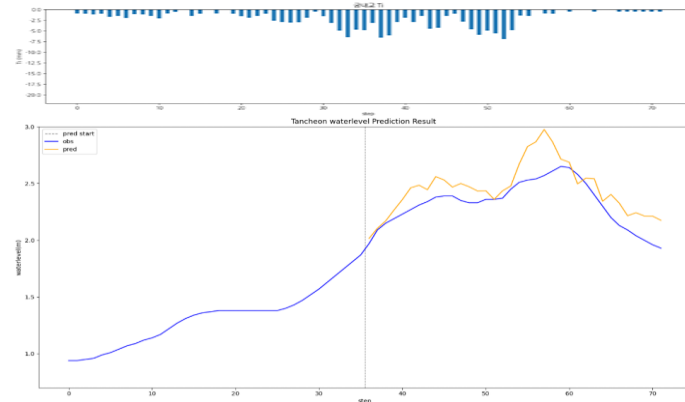
testdata_240723

R2 : 0.8244



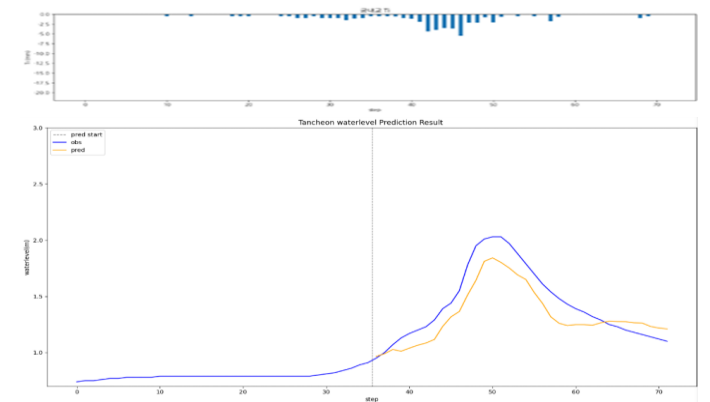
testdata_220712

R2 : 0.2864



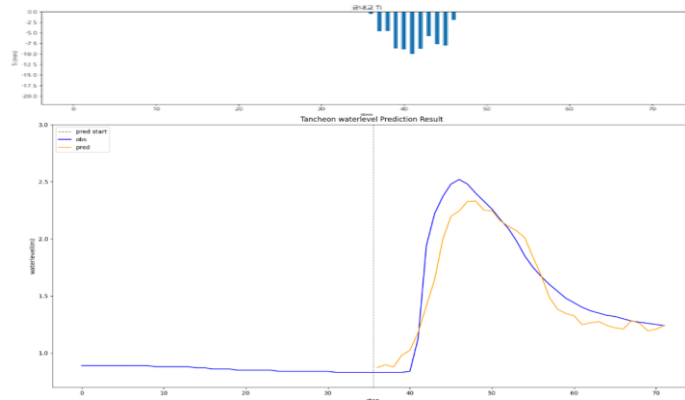
testdata_250716

R2 : 0.7627



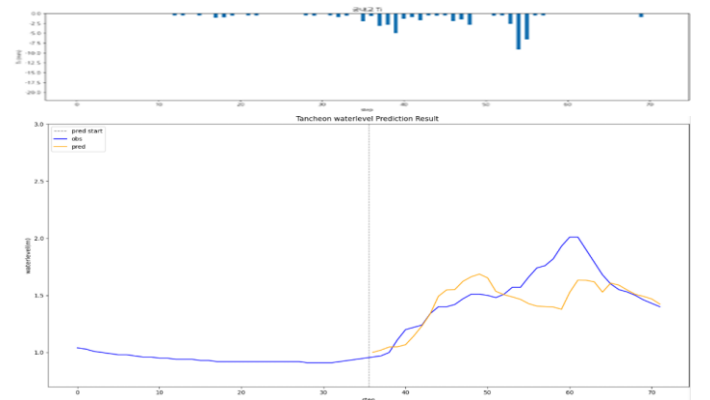
testdata_230711

R2 : 0.8814



testdata_250814

R2 : 0.4144

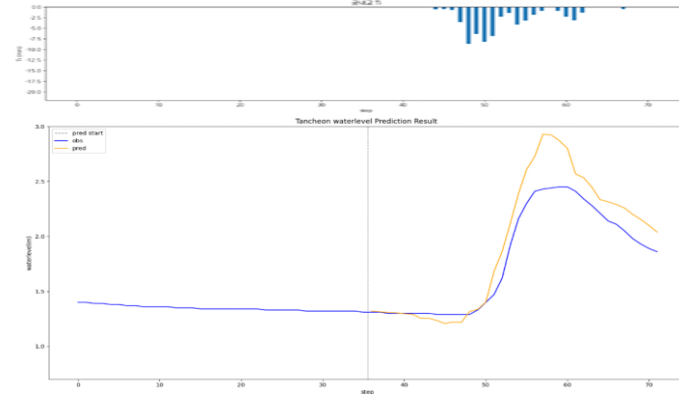


궁내교 - 관심수위 (수위 X 1.02)

* model_(gn/dg)_1.02

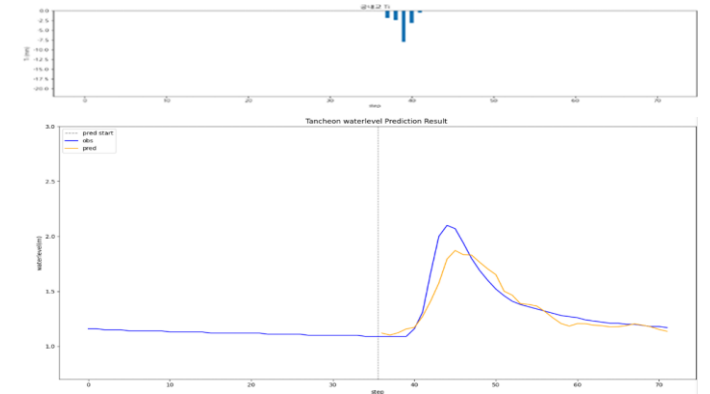
testdata_200802

R2 : 0.7922



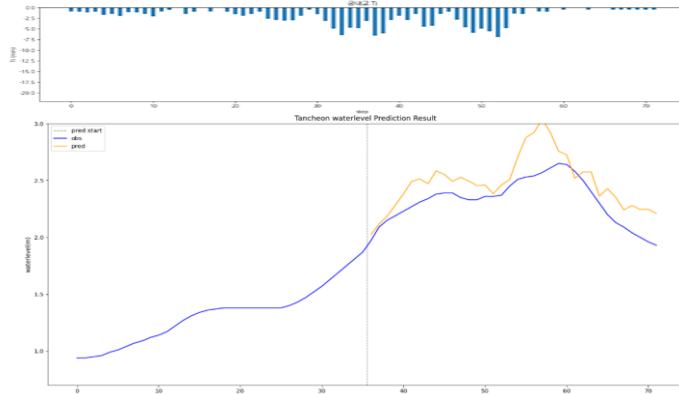
testdata_240723

R2 : 0.8400



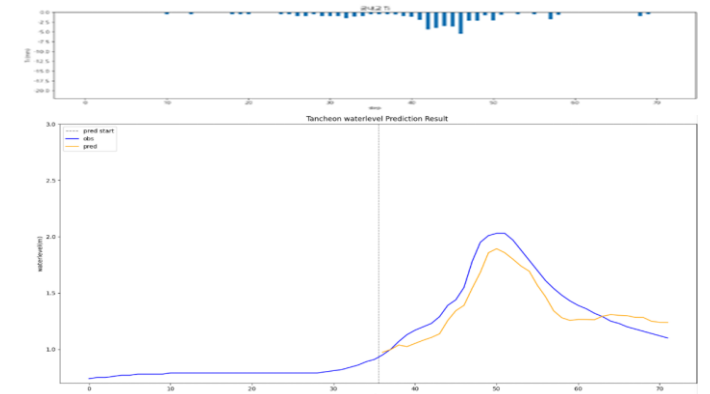
testdata_220712

R2 : 0.0156



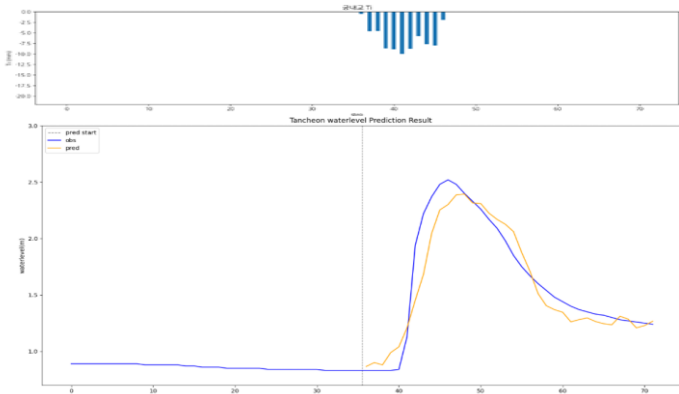
testdata_250716

R2 : 0.8186



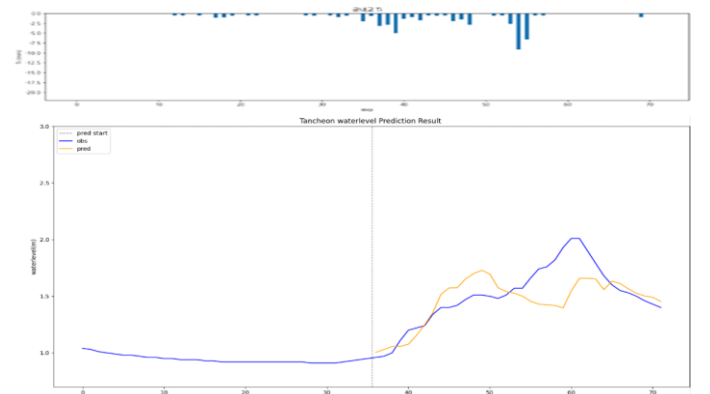
testdata_230711

R2 : 0.8994



testdata_250814

R2 : 0.4433

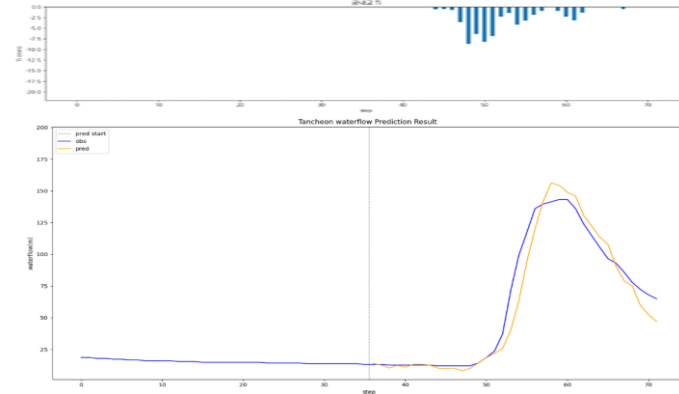


궁내교 - 관심수위 (유량 예측)

* model_(gn/dg)_flow

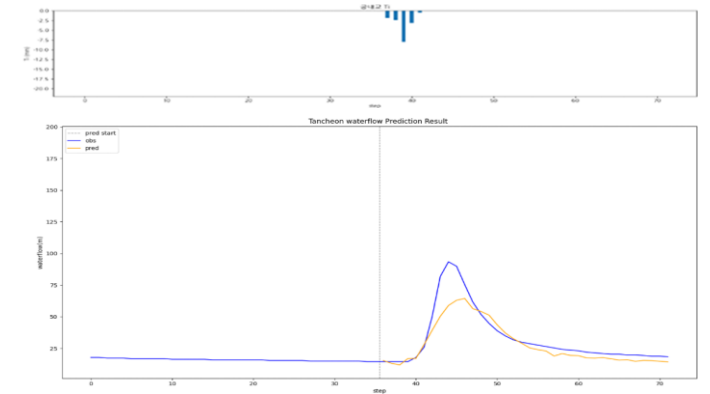
testdata_200802

R2 : 0.9470



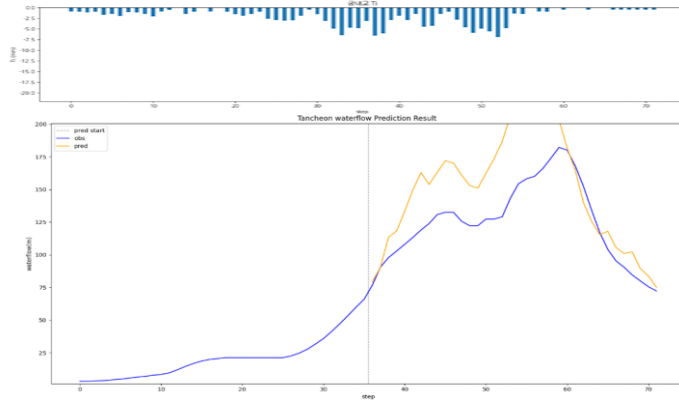
testdata_240723

R2 : 0.7854



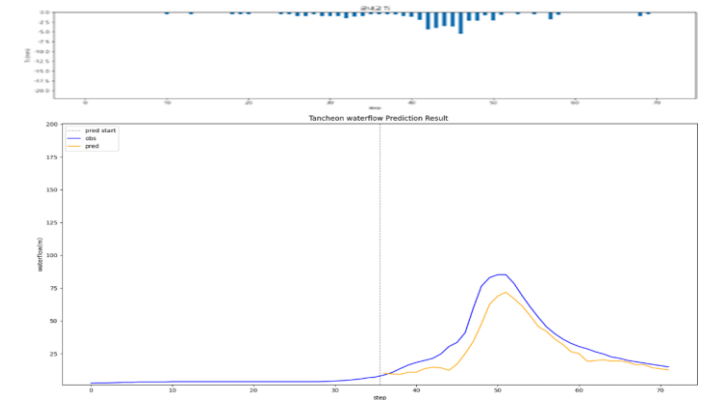
testdata_220712

R2 : -0.3192



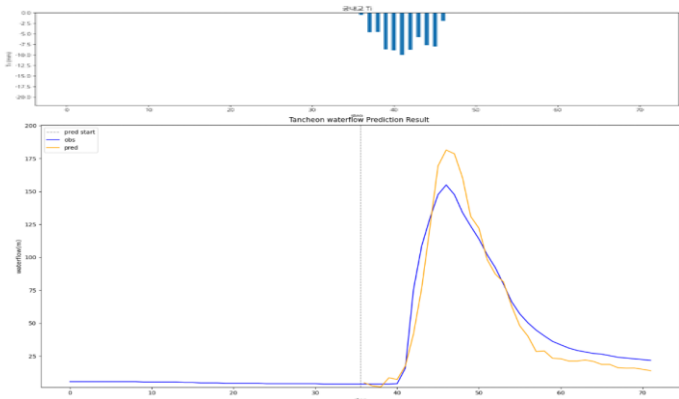
testdata_250716

R2 : 0.7869



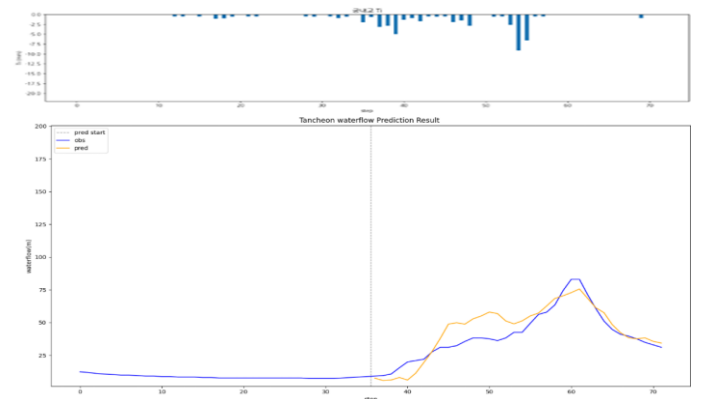
testdata_230711

R2 : 0.9153



testdata_250814

R2 : 0.7481

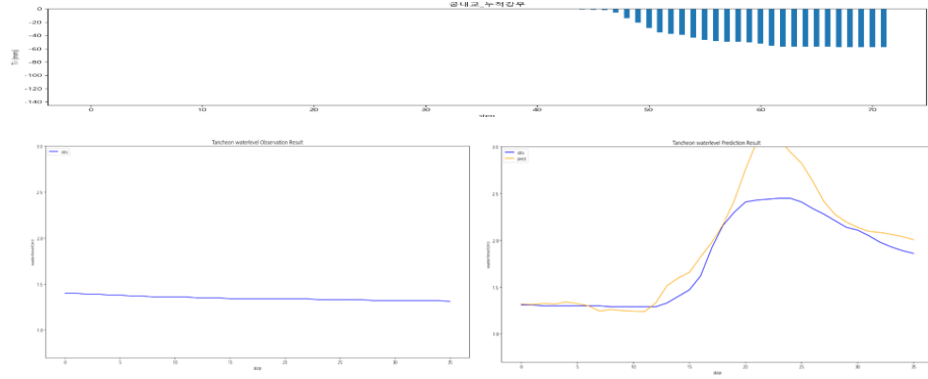


궁내교 - 관심수위 (누적 강우 110mm 이상 4)

* model_(gn/dg)_110_4_2

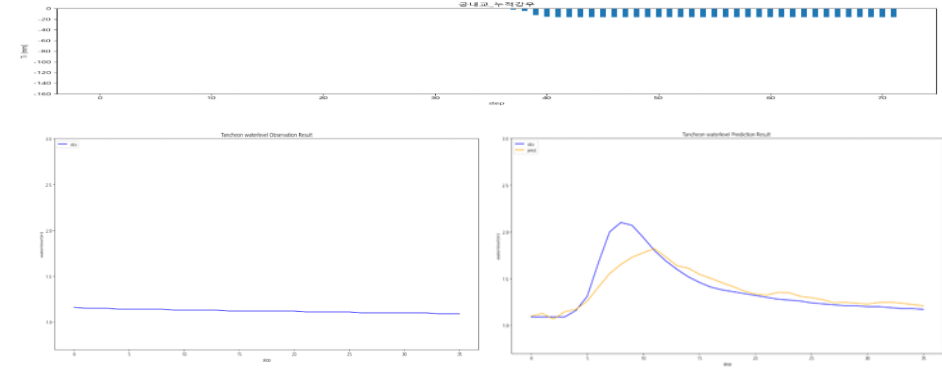
testdata_200802

R2 : 0.7237



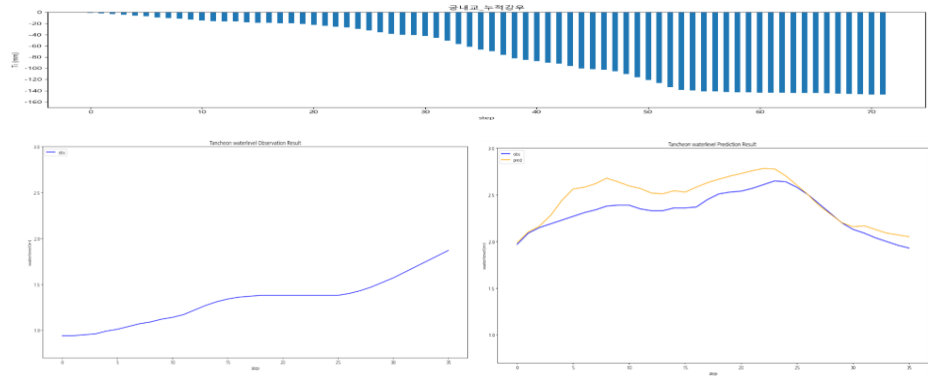
testdata_240723

R2 : 0.7623



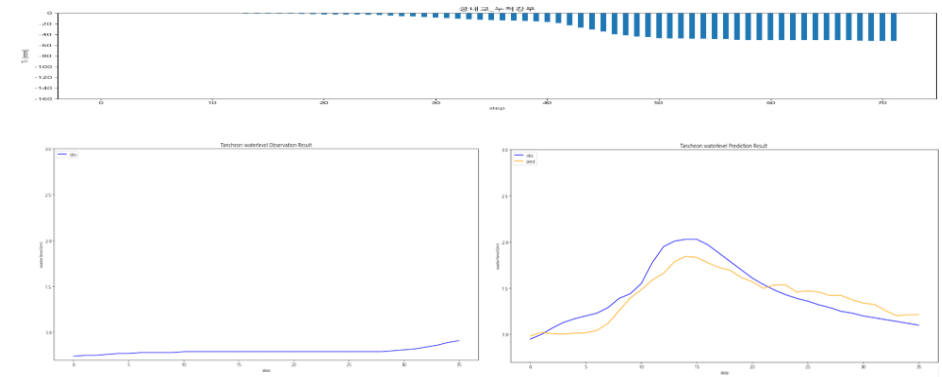
testdata_220712

R2 : 0.3192



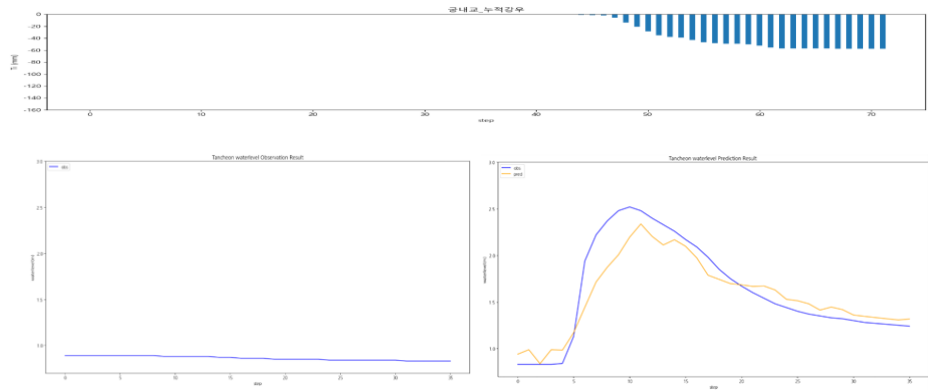
testdata_250716

R2 : 0.8108



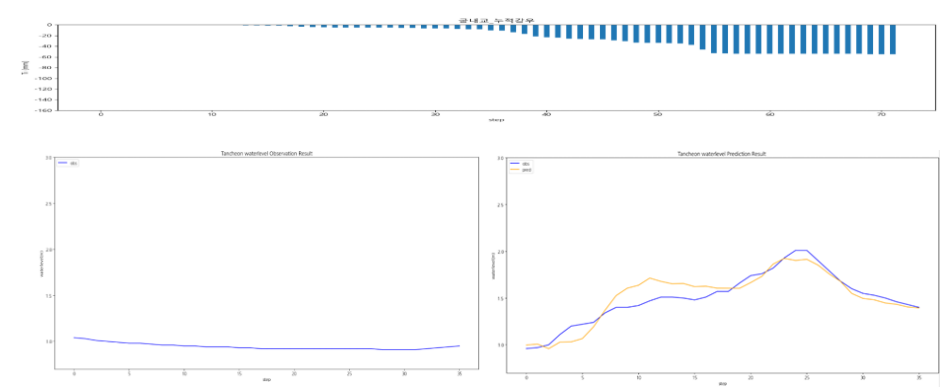
testdata_230711

R2 : 0.8533



testdata_250814

R2 : 0.8475

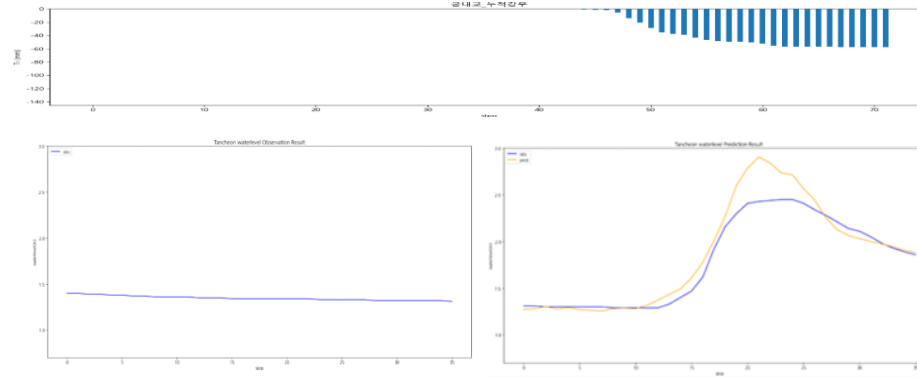


궁내교 - 관심수위 (누적 강우 110mm 이상 5)

* model_(gn/dg)_110_5_2

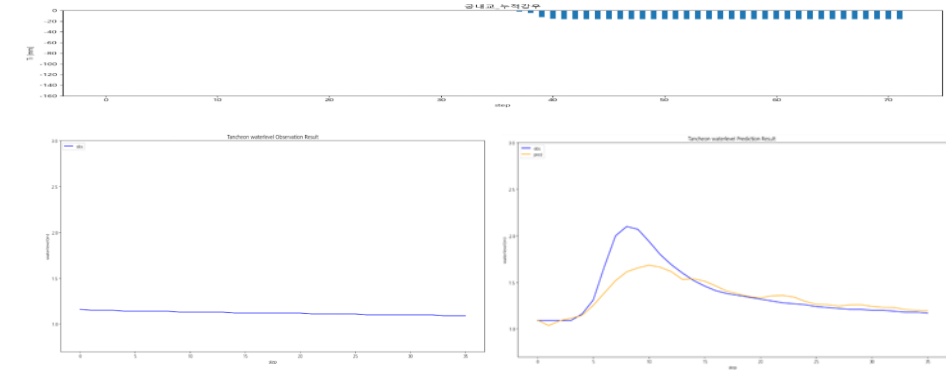
testdata_200802

R2 : 0.8774



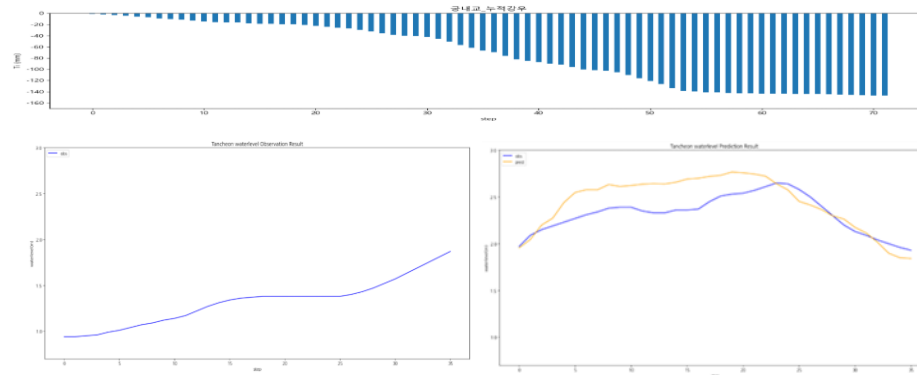
testdata_240723

R2 : 0.7051



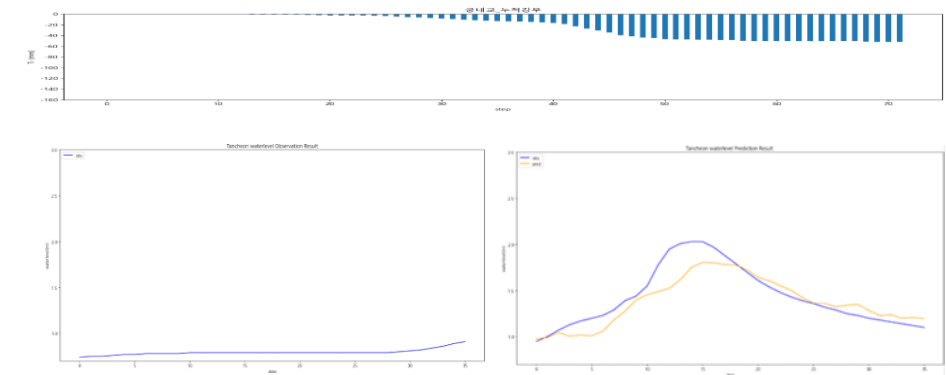
testdata_220712

R2 : 0.0586



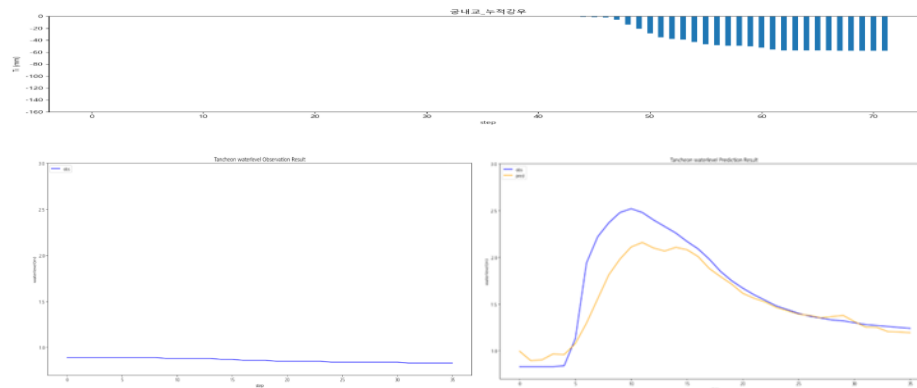
testdata_250716

R2 : 0.7764



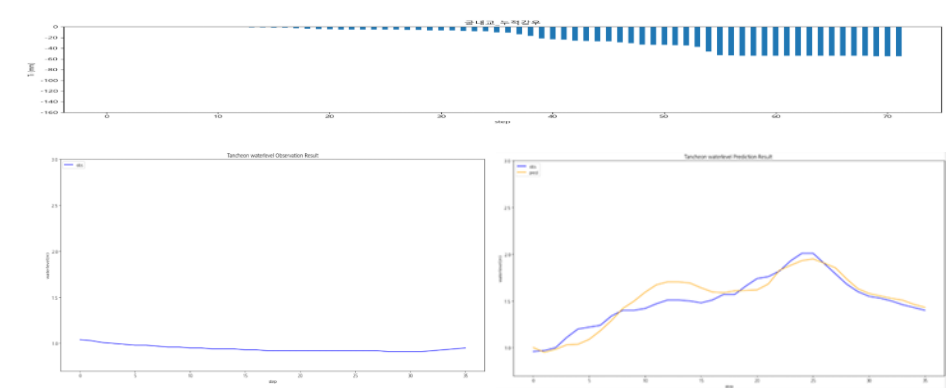
testdata_230711

R2 : 0.8028



testdata_250814

R2 : 0.8673

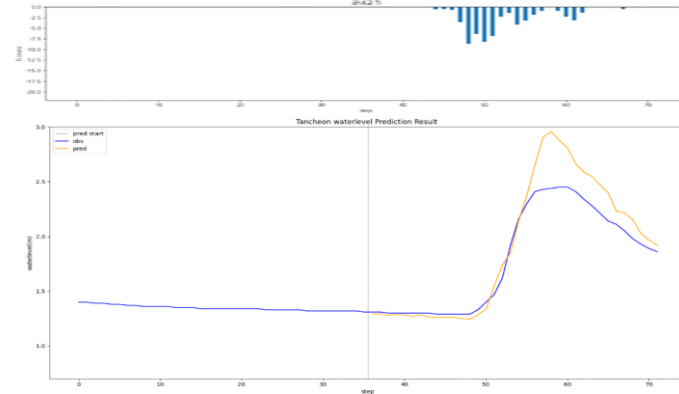


궁내교 - 관심수위 (대곡교 컬럼 제외)

* model_(gn/dg)_none

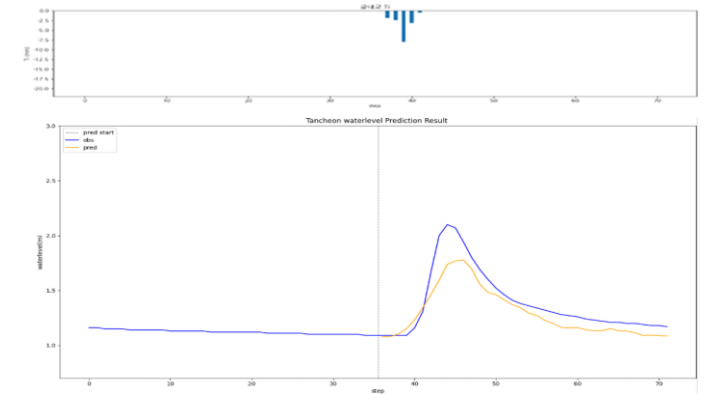
testdata_200802

R2 : 0.8253



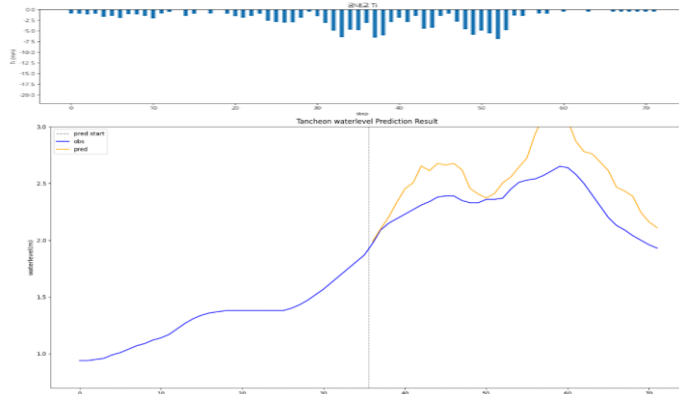
testdata_240723

R2 : 0.7720



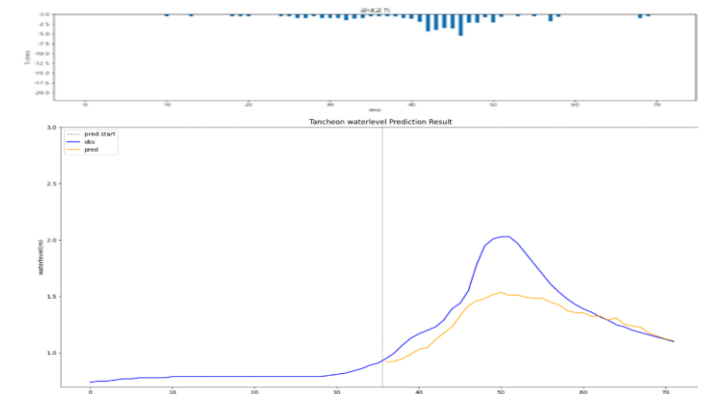
testdata_220712

R2 : -1.2836



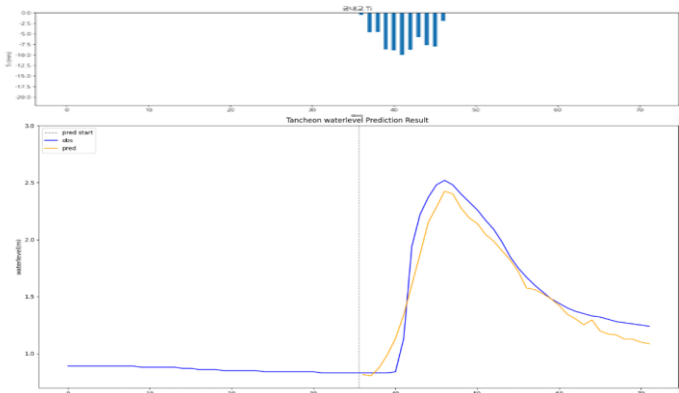
testdata_250716

R2 : 0.4997



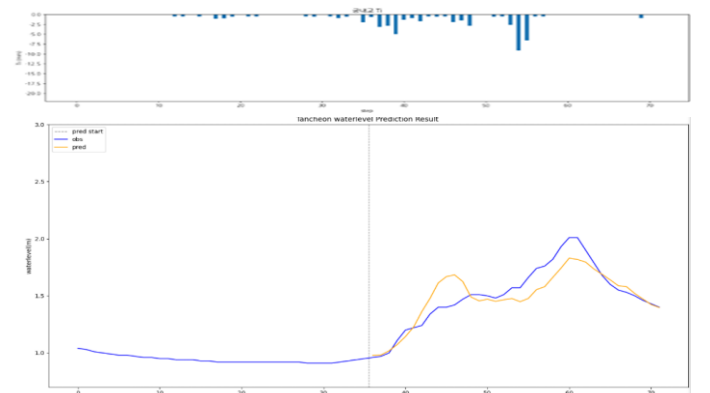
testdata_230711

R2 : 0.9267



testdata_250814

R2 : 0.7909

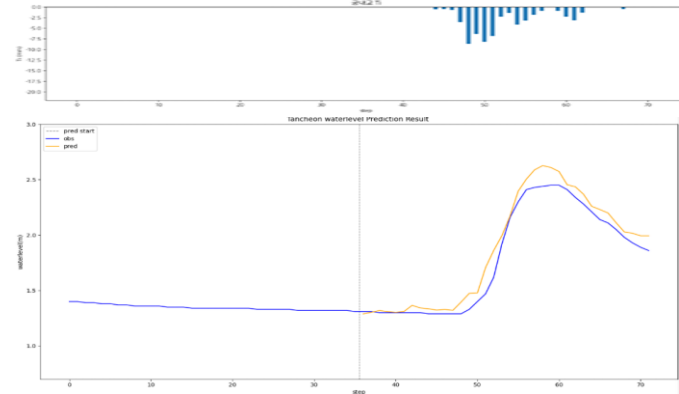


궁내교 - 관심수위 (20년 이후)

* model_(gn/dg)_20

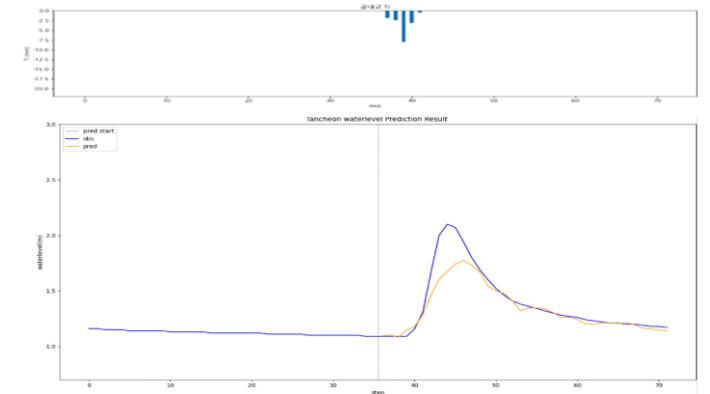
testdata_200802

R2 : 0.9513



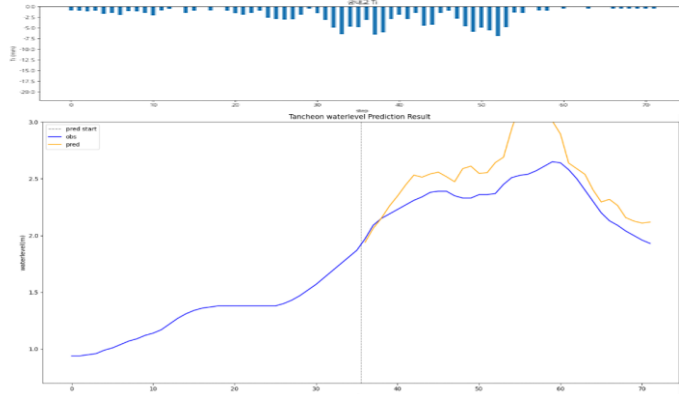
testdata_240723

R2 : 0.8136



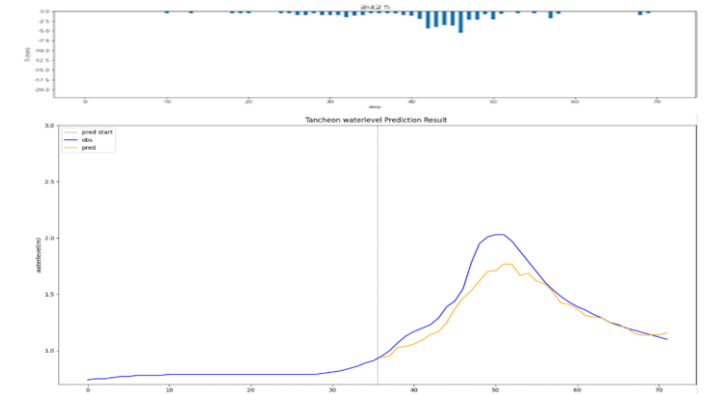
testdata_220712

R2 : -0.8548



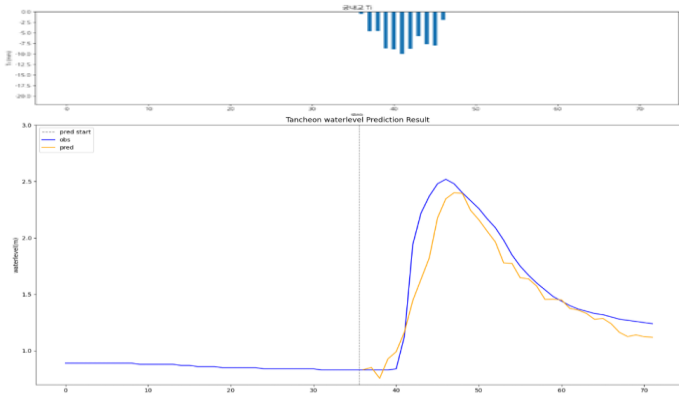
testdata_250716

R2 : 0.8213



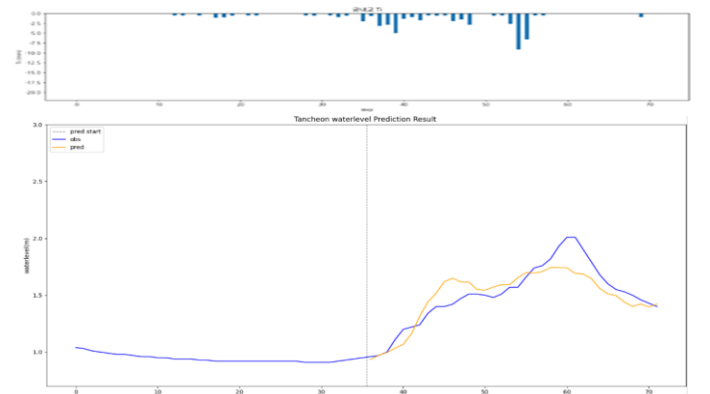
testdata_230711

R2 : 0.8751



testdata_250814

R2 : 0.7835

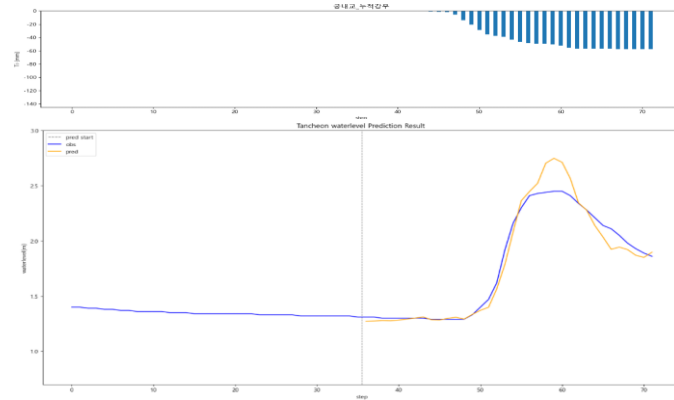


궁내교 - 관심수위 (20년 이후 + 누적 강우)

* model_(gn/dg)_rf20

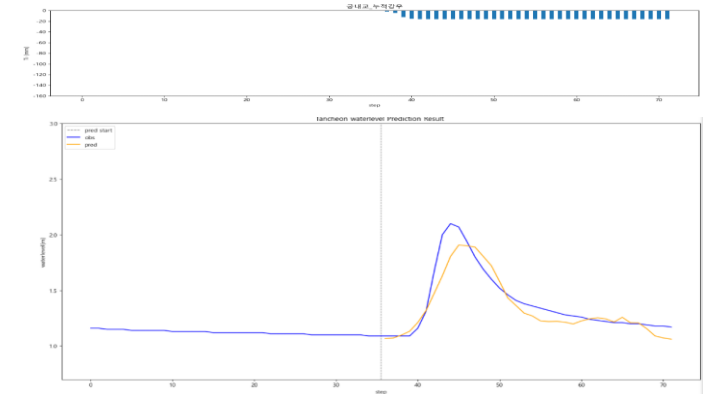
testdata_200802

R2 : 0.9507



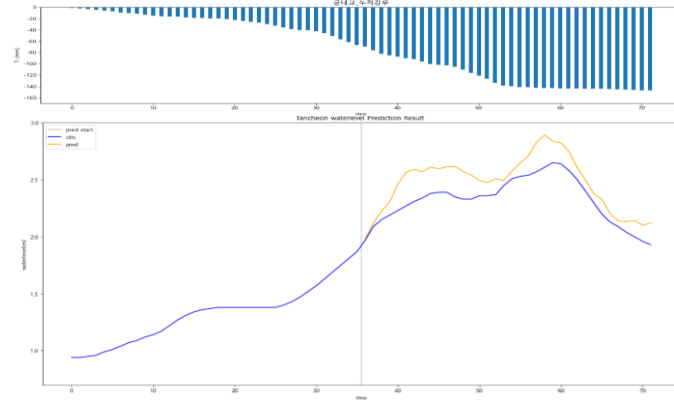
testdata_240723

R2 : 0.8524



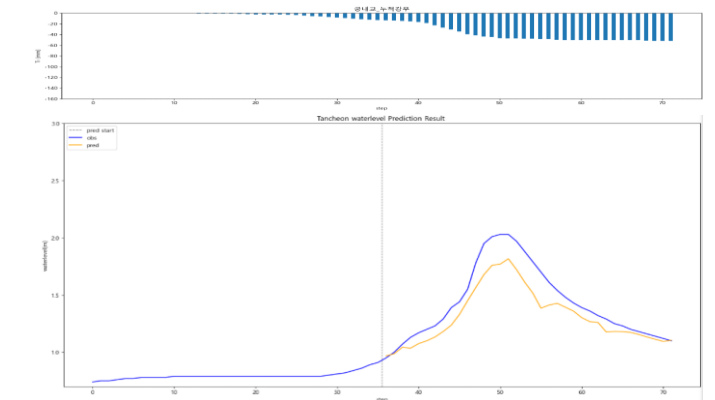
testdata_220712

R2 : 0.2329



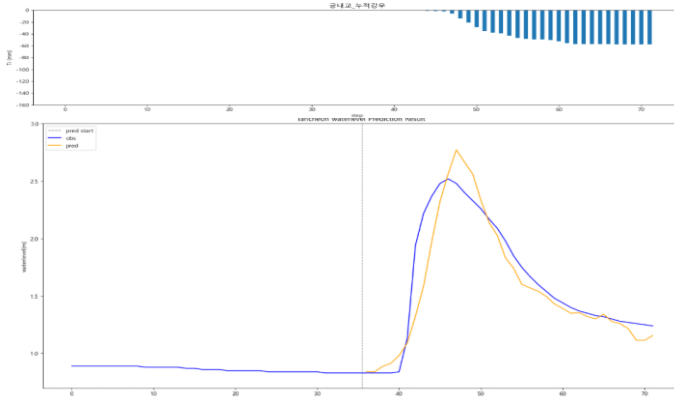
testdata_250716

R2 : 0.7764



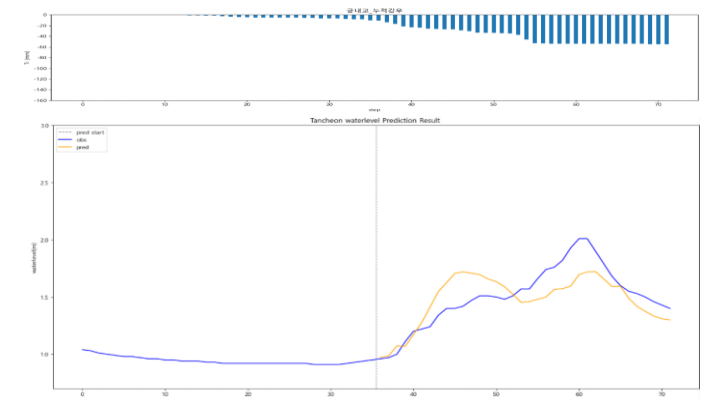
testdata_230711

R2 : 0.8663



testdata_250814

R2 : 0.5660



대곡교

- » 궁내교 대비 높은 R²
- » 특히 누적강우 및 2020년 이후 데이터 학습 구성에서 성능이 안정적 (22.07.12 이벤트에서는 예외)

- R2 평가

12시간 예측

R2	대곡교	
Test data	6시간	3시간
20.08.02	0.9934	0.4887
22.07.12	-0.4222	-1.7001
23.07.11	0.7761	0.8885
24.07.23	0.2256	0.2750
25.07.16	0.8822	0.8926
25.08.14	0.7106	0.8466

관심수위

대곡교				
관심수위	+ 누적강우	3시간 예측	X 0.98	X 1.02
0.8748	0.7441	0.9113	0.8579	0.8688
0.5503	0.6399	-0.1308	0.7718	0.5974
0.8659	0.8559	0.7288	0.8246	0.8086
0.6749	0.5656	0.5679	0.5221	0.4905
0.9470	0.9182	0.5949	0.9736	0.9726
0.8781	0.8301	0.8942	0.7718	0.7639

유량

대곡교
유량
0.8104
0.9411
0.7145
0.7932
0.8734
0.9158

누적강우 110mm

대곡교	
4번 조합	5번 조합
0.8350	0.9246
-0.0424	-0.1000
0.7456	0.8276
0.0871	0.5130
0.8305	0.9577
0.5011	0.6115

20년 이후

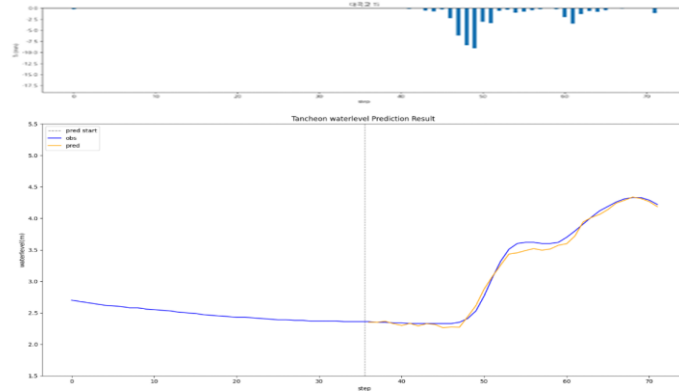
대곡교	
20년 이후	+ 누적강우
0.8845	0.8523
-0.1561	-0.5266
0.7304	0.7527
0.6077	0.6739
0.8647	0.9528
0.8840	0.8631

대곡교 - 12시간 예측 (6시간)

* model_sunnam_(gn/dg)72

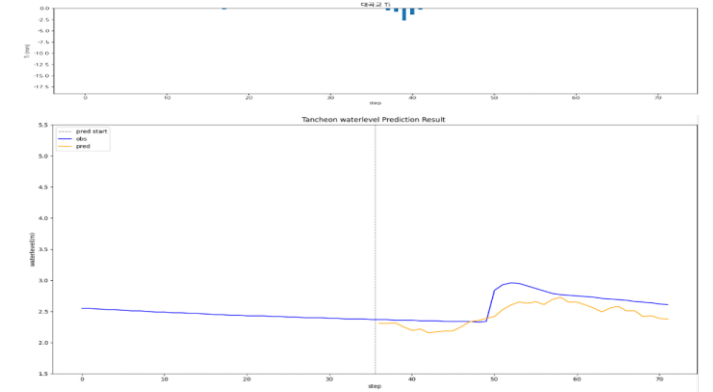
testdata_200802

R2 : 0.9934



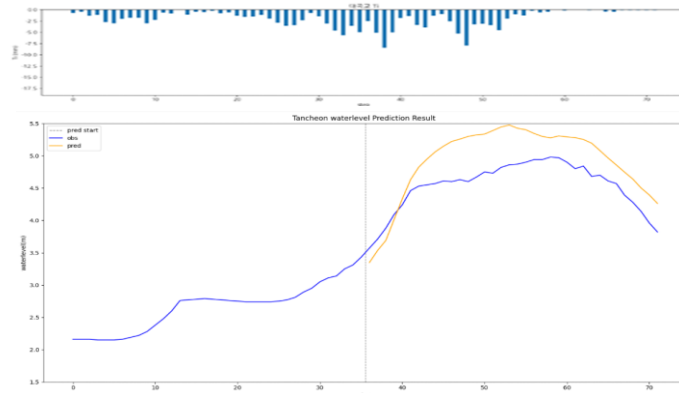
testdata_240723

R2 : 0.2256



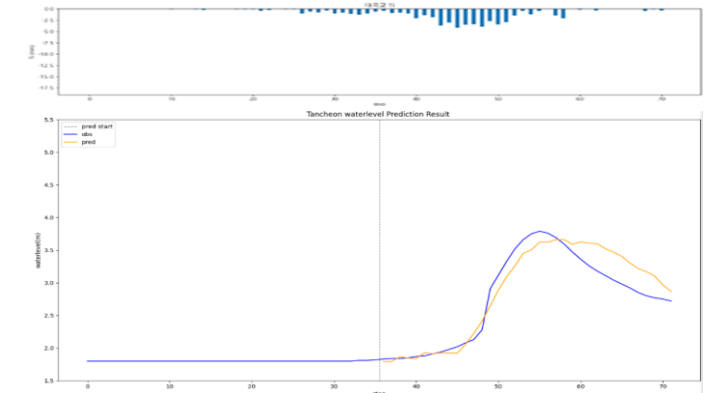
testdata_220712

R2 : -0.4222



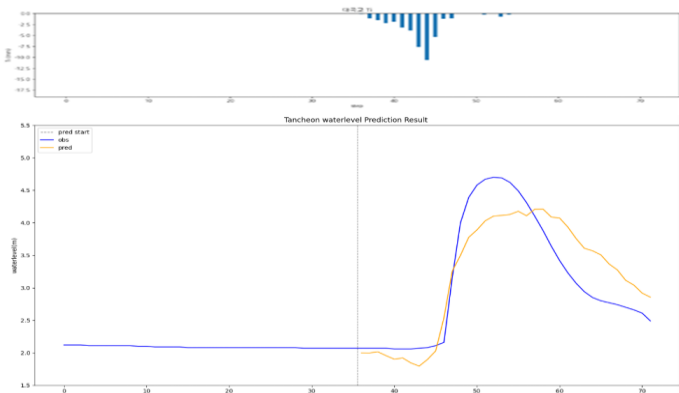
testdata_250716

R2 : 0.8822



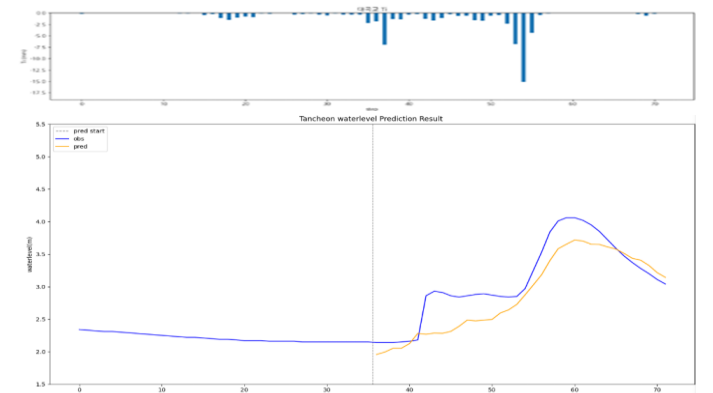
testdata_230711

R2 : 0.7761



testdata_250814

R2 : 0.7106

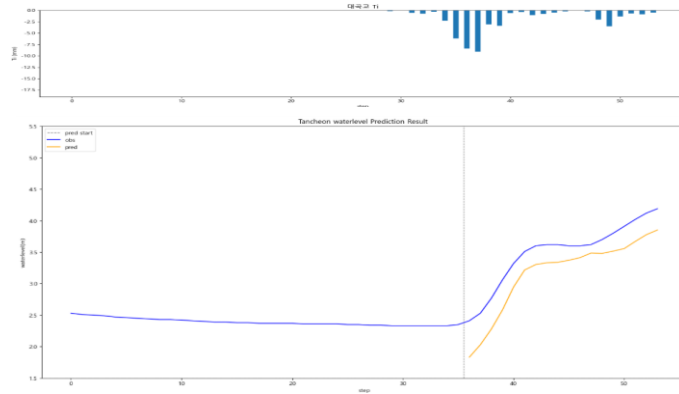


대곡교 - 12시간 예측 (3시간)

* model_sunnam_(gn/dg)72

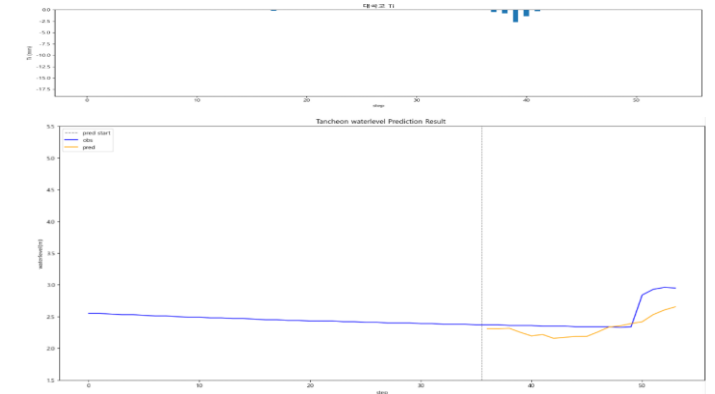
testdata_200802

R2 : 0.4887



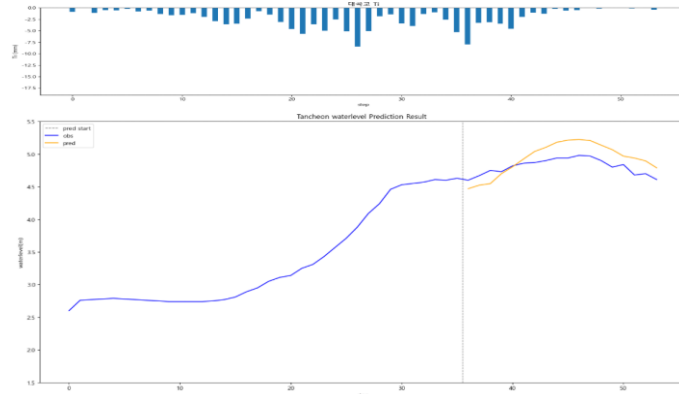
testdata_240723

R2 : 0.2750



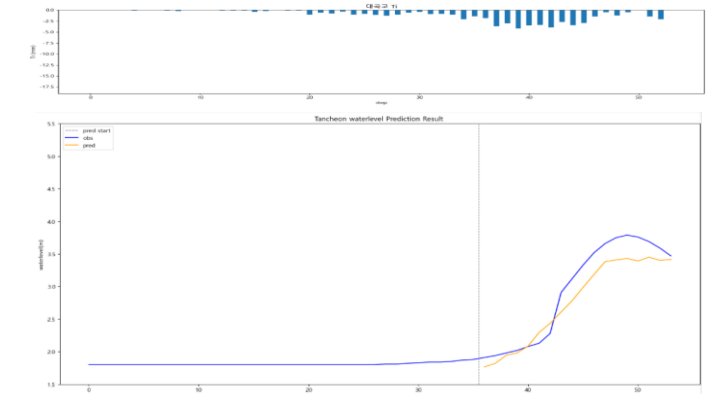
testdata_220712

R2 : -1.7001



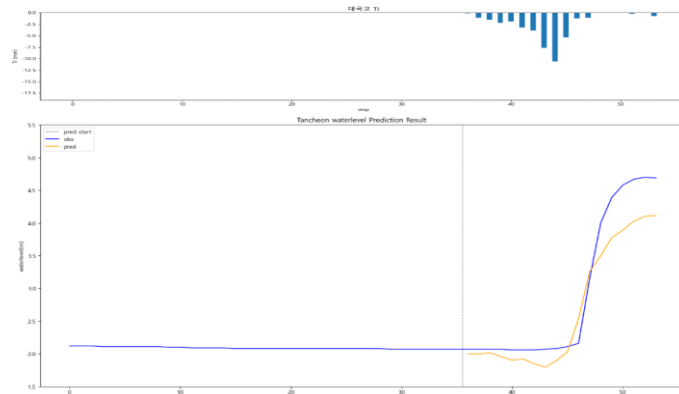
testdata_250716

R2 : 0.8926



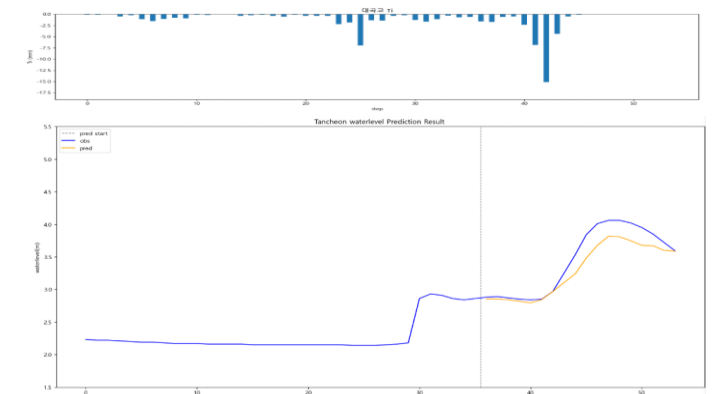
testdata_230711

R2 : 0.8885



testdata_250814

R2 : 0.8466

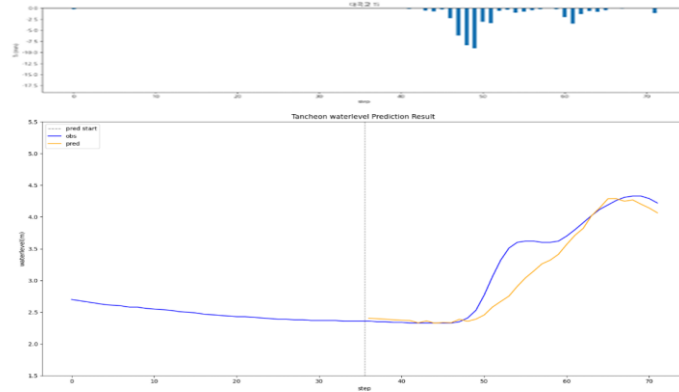


대곡교 - 관심수위

* model_(gn/dg)_wl18_3_2

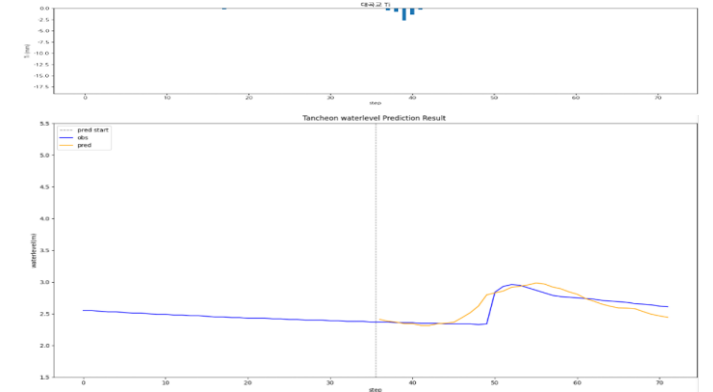
testdata_200802

R2 : 0.8748



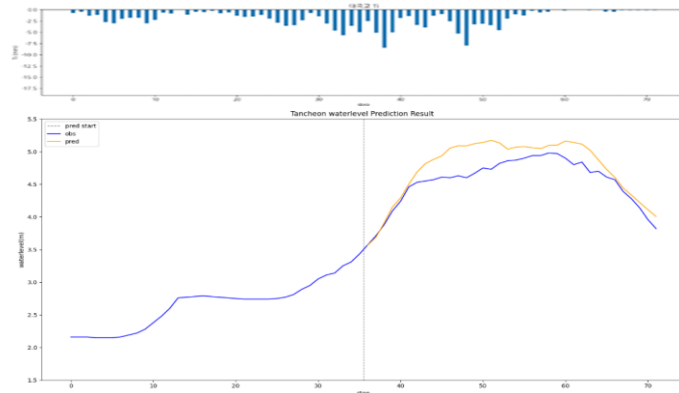
testdata_240723

R2 : 0.6749



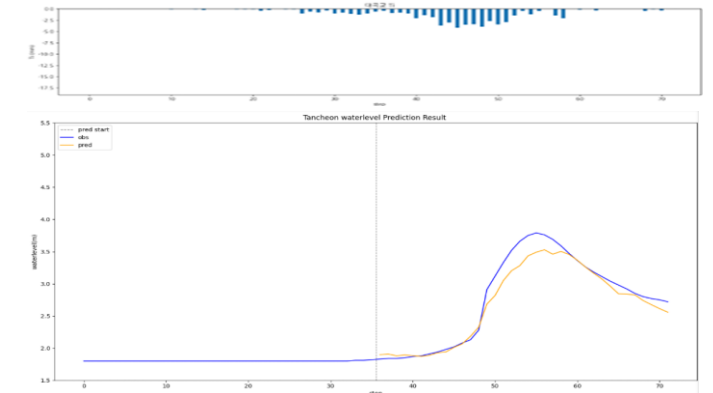
testdata_220712

R2 : 0.5503



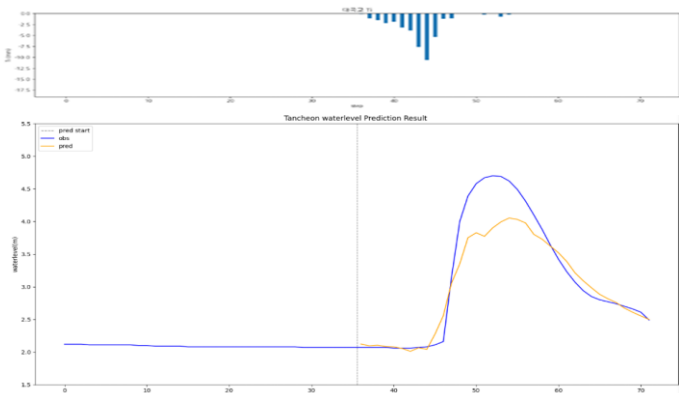
testdata_250716

R2 : 0.9470



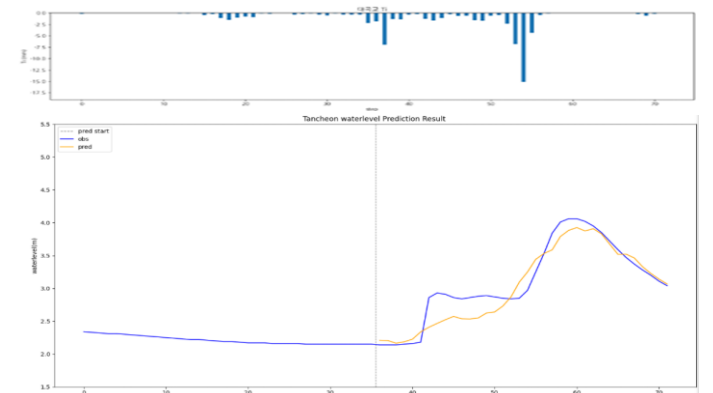
testdata_230711

R2 : 0.8659



testdata_250814

R2 : 0.8781

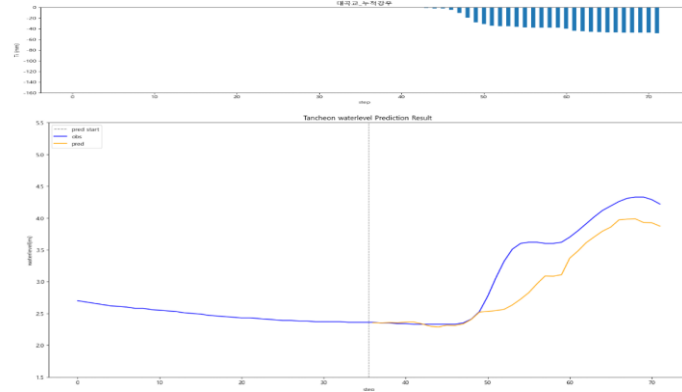


대곡교 - 관심수위 (+ 누적강우)

* model_(gn/dg)_rf

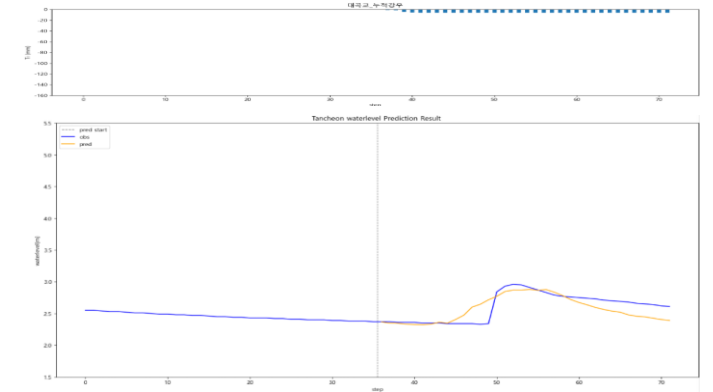
testdata_200802

R2 : 0.7441



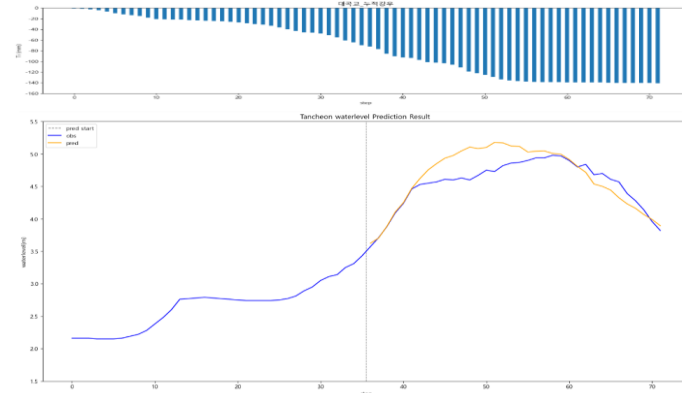
testdata_240723

R2 : 0.5656



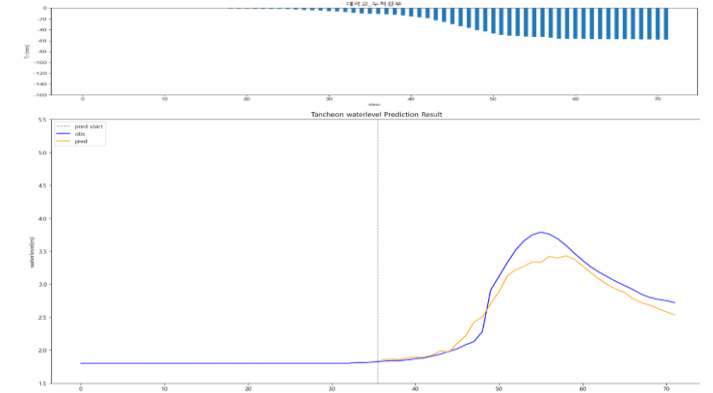
testdata_220712

R2 : 0.6399



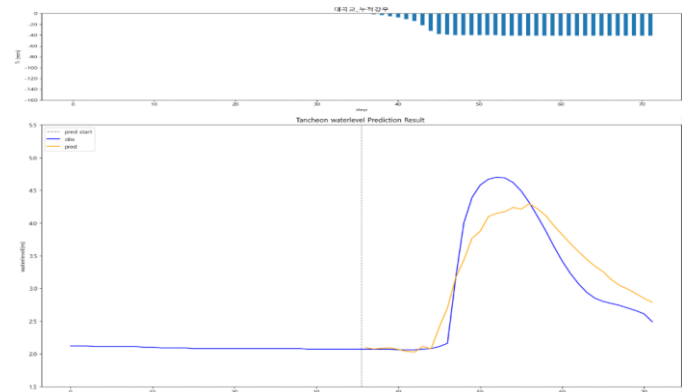
testdata_250716

R2 : 0.9182



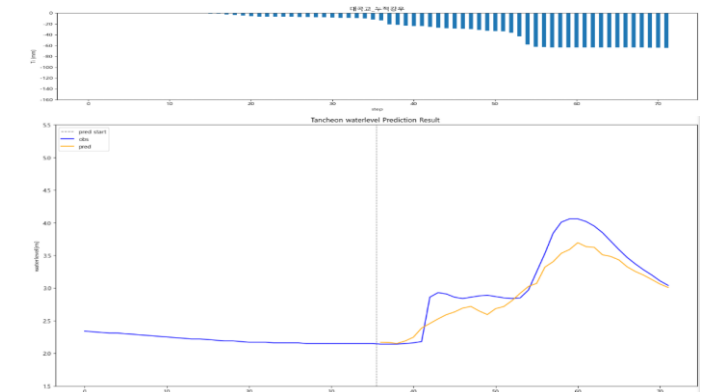
testdata_230711

R2 : 0.8559



testdata_250814

R2 : 0.8301

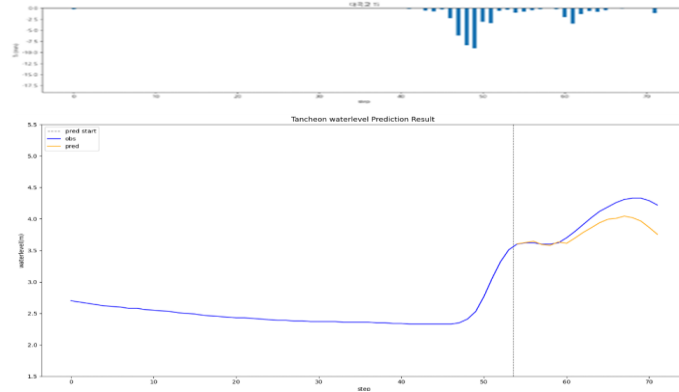


대곡교 - 관심수위 (3시간 예측)

* model_(gn/dg)_3

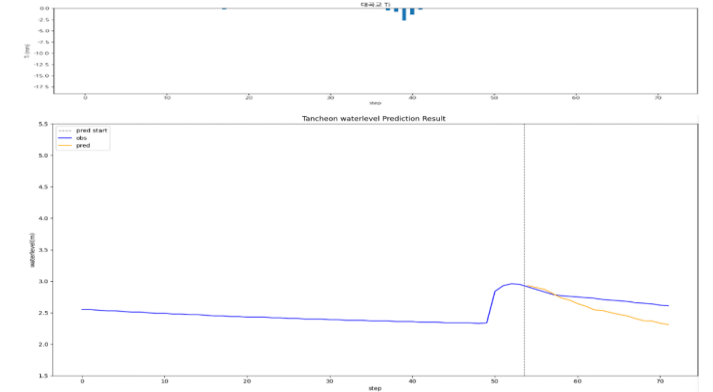
testdata_200802

R2 : 0.4341



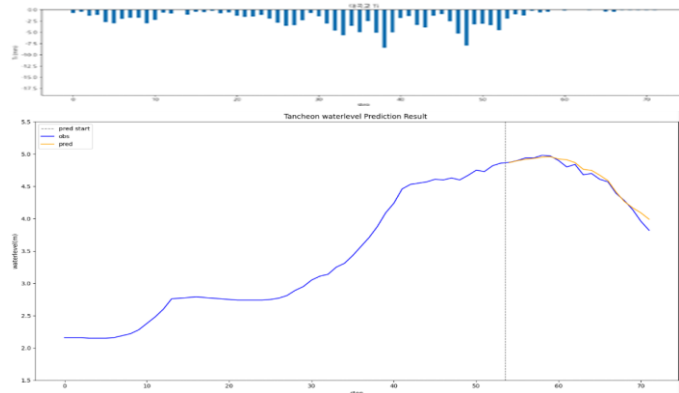
testdata_240723

R2 : -4.2174



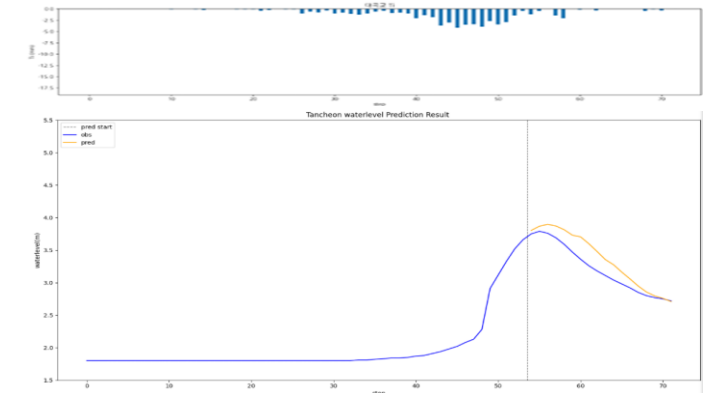
testdata_220712

R2 : 0.9659



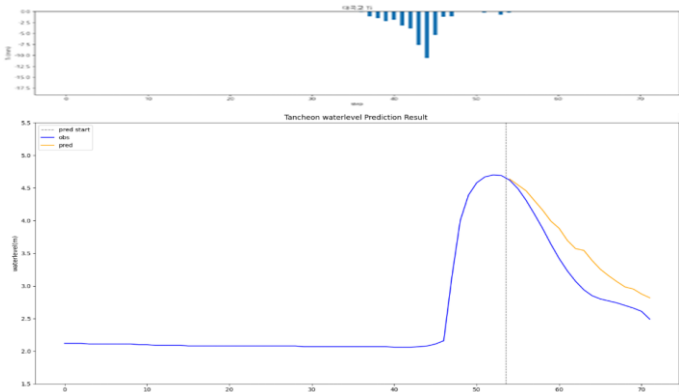
testdata_250716

R2 : 0.7288



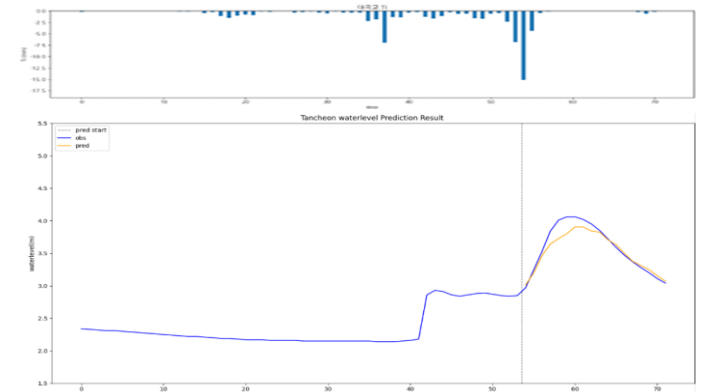
testdata_230711

R2 : 0.7131



testdata_250814

R2 : 0.8953

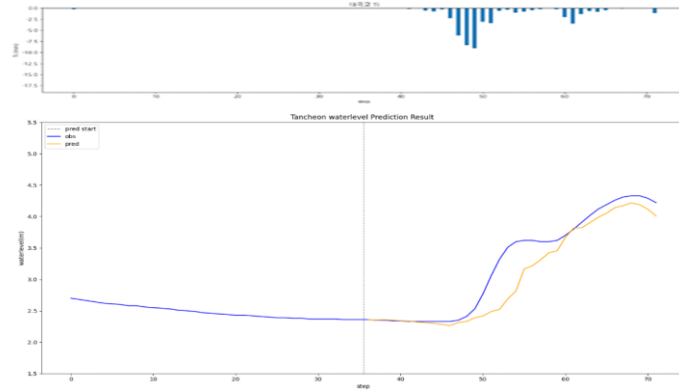


대곡교 - 관심수위 (수위 X 0.98)

* model_(gn/dg)_0.98

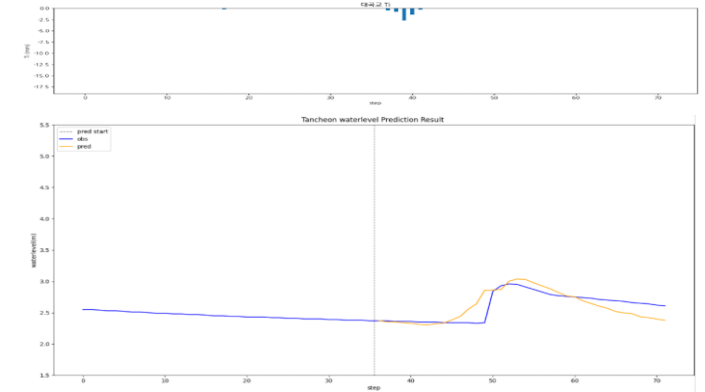
testdata_200802

R2 : 0.8579



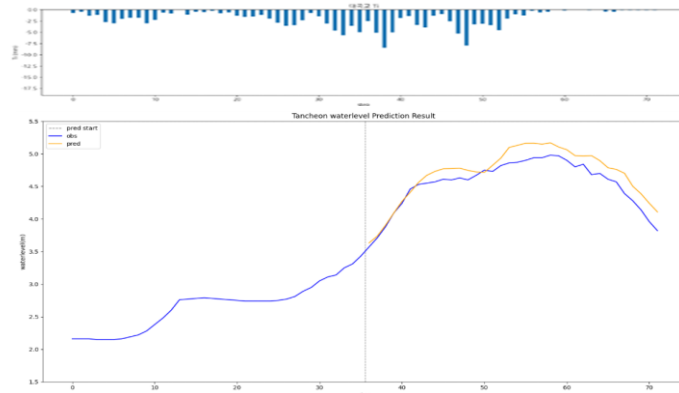
testdata_240723

R2 : 0.5221



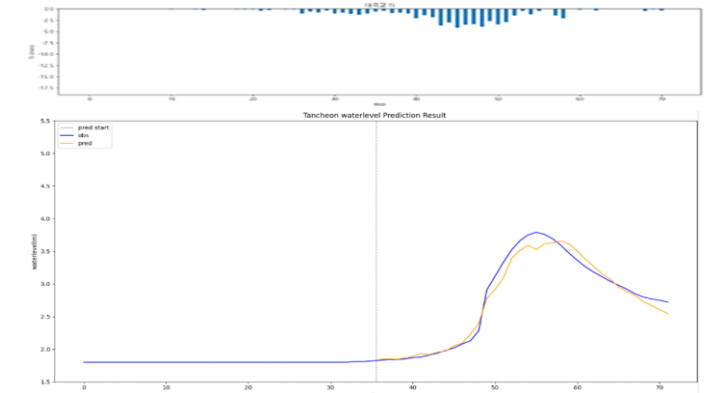
testdata_220712

R2 : 0.7718



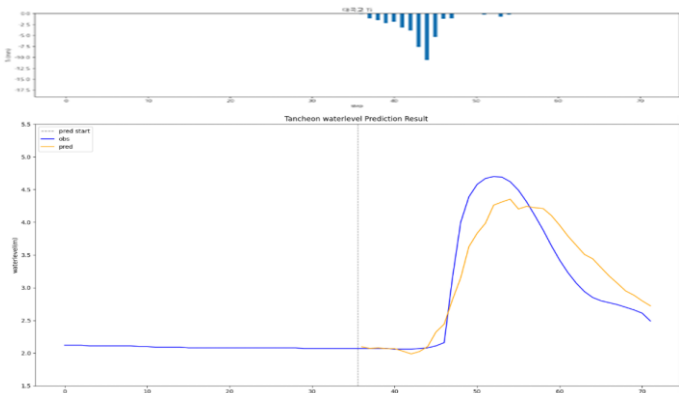
testdata_250716

R2 : 0.9736



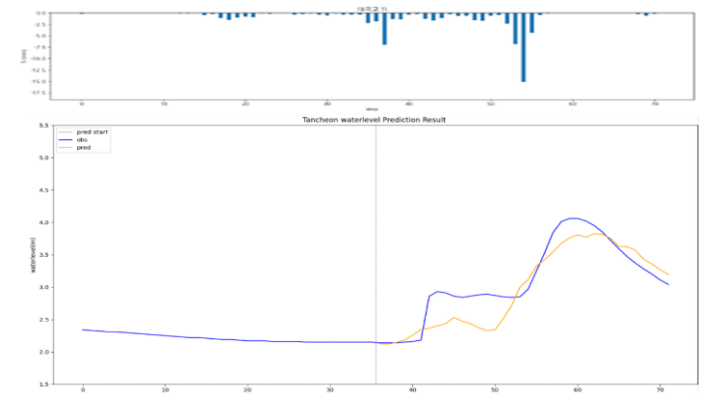
testdata_230711

R2 : 0.8246



testdata_250814

R2 : 0.7718

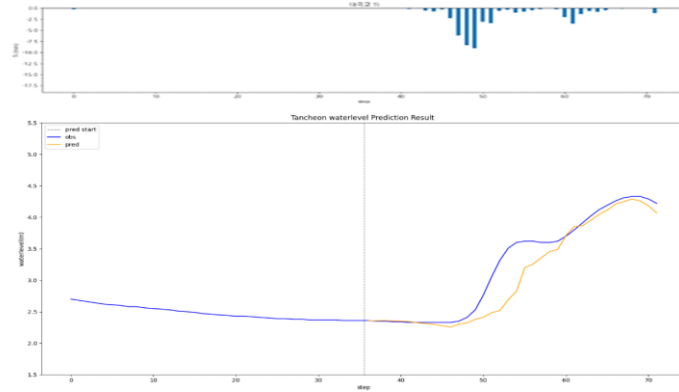


대곡교 - 관심수위 (수위 X 1.02)

* model_(gn/dg)_1.02

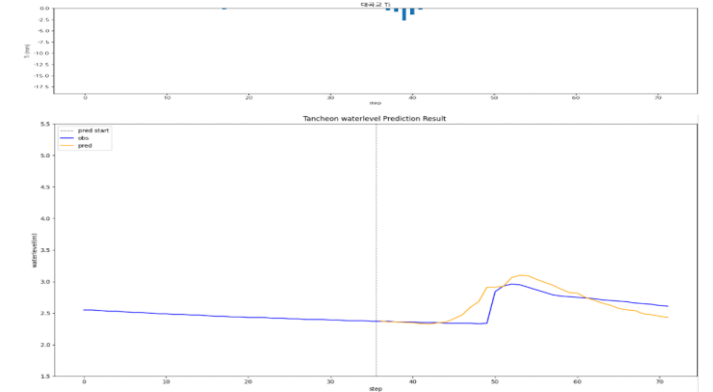
testdata_200802

R2 : 0.8688



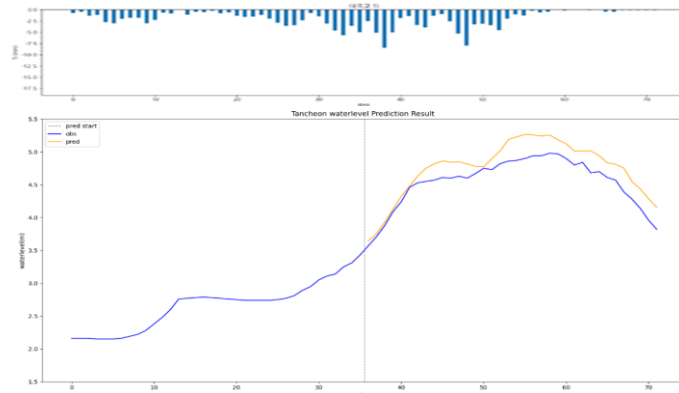
testdata_240723

R2 : 0.4905



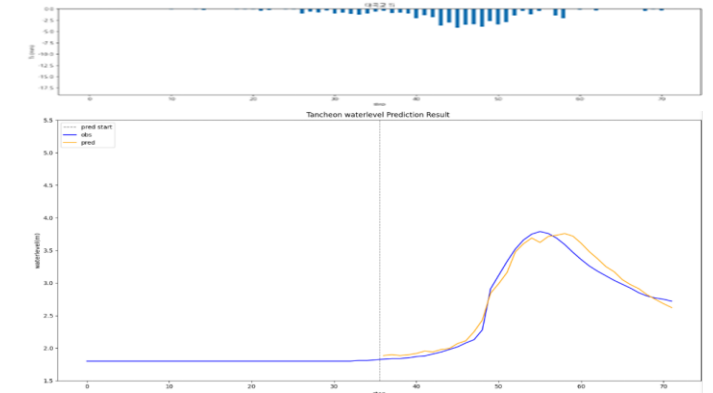
testdata_220712

R2 : 0.5974



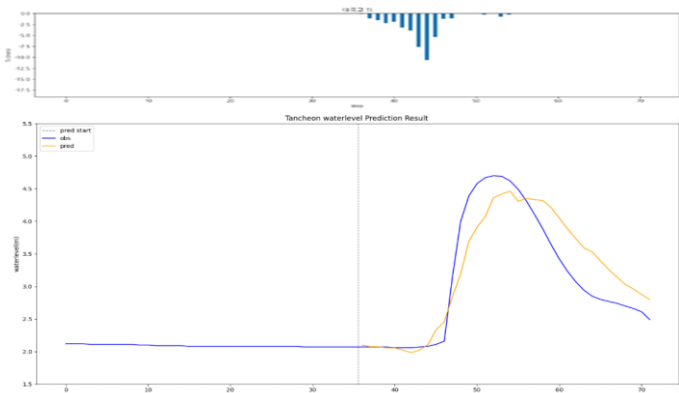
testdata_250716

R2 : 0.9726



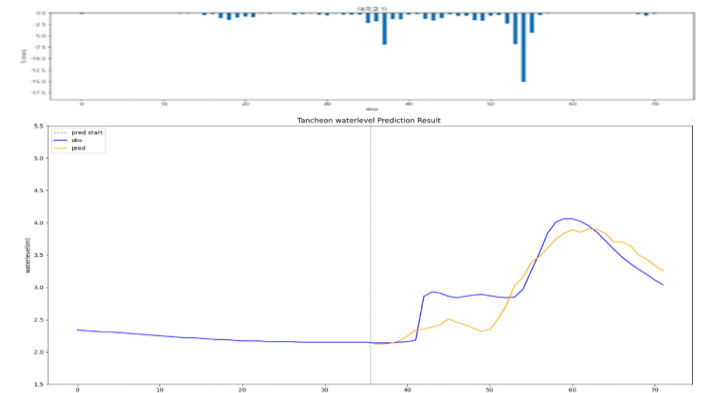
testdata_230711

R2 : 0.8086



testdata_250814

R2 : 0.7639

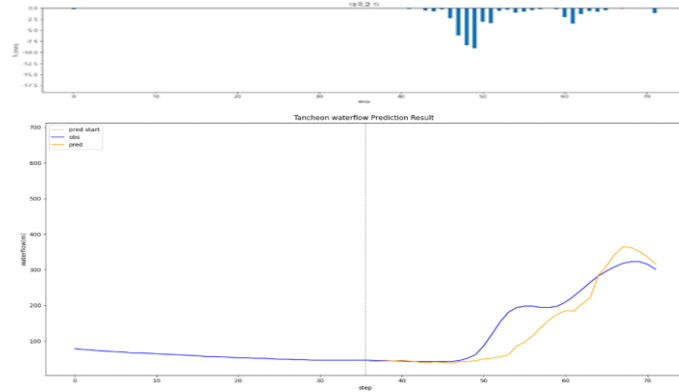


대곡교 - 관심수위 (유량 예측)

* model_(gn/dg)_flow

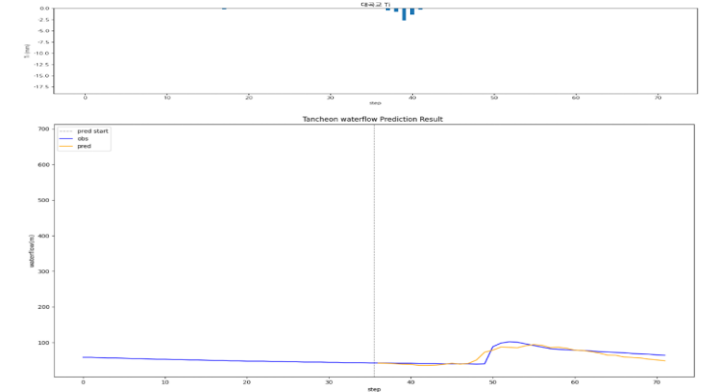
testdata_200802

R2 : 0.8104



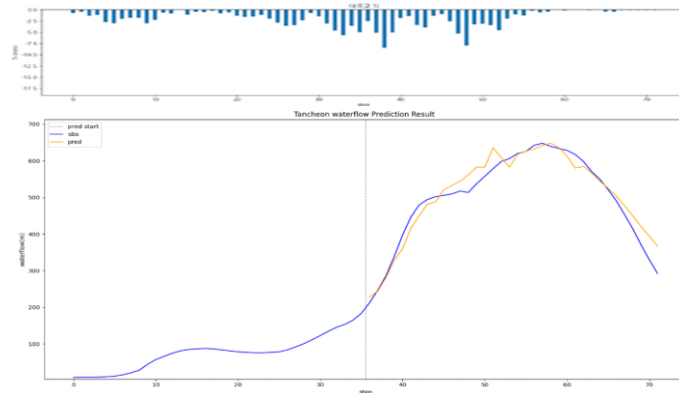
testdata_240723

R2 : 0.7932



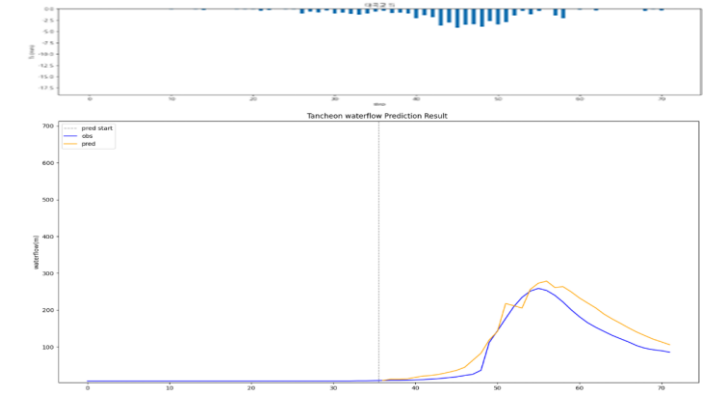
testdata_220712

R2 : 0.9411



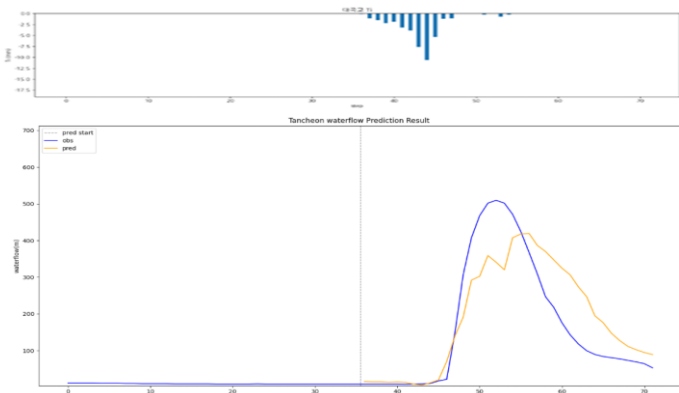
testdata_250716

R2 : 0.8734



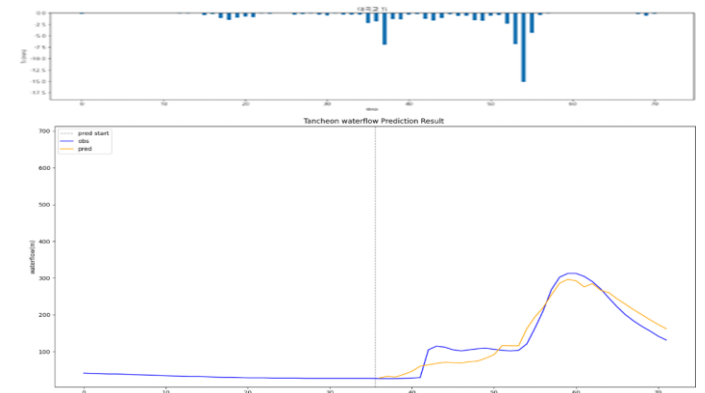
testdata_230711

R2 : 0.7145



testdata_250814

R2 : 0.9158

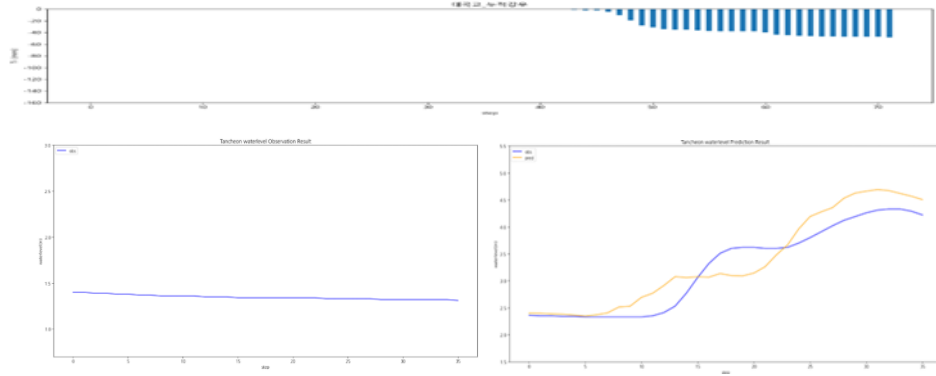


대곡교 - 관심수위 (누적 강우 110mm 이상 4)

* model_(gn/dg)_110_4_2

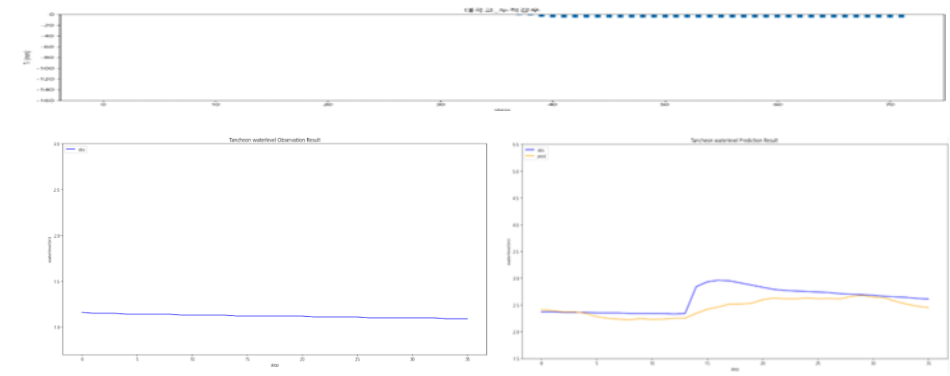
testdata_200802

R2 : 0.8350



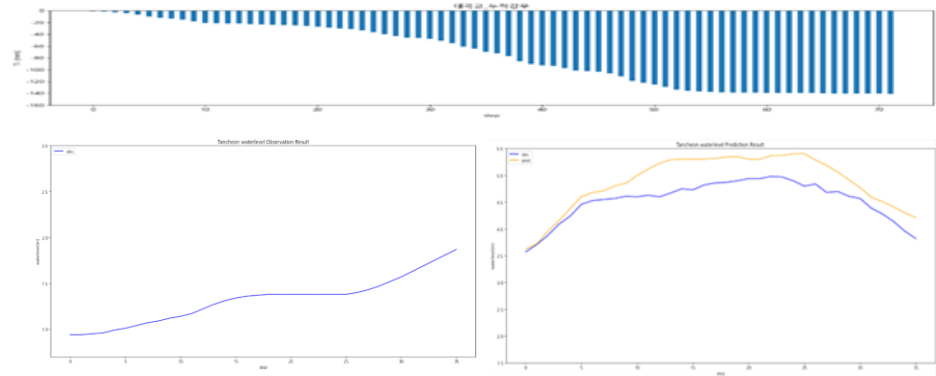
testdata_240723

R2 : 0.0871



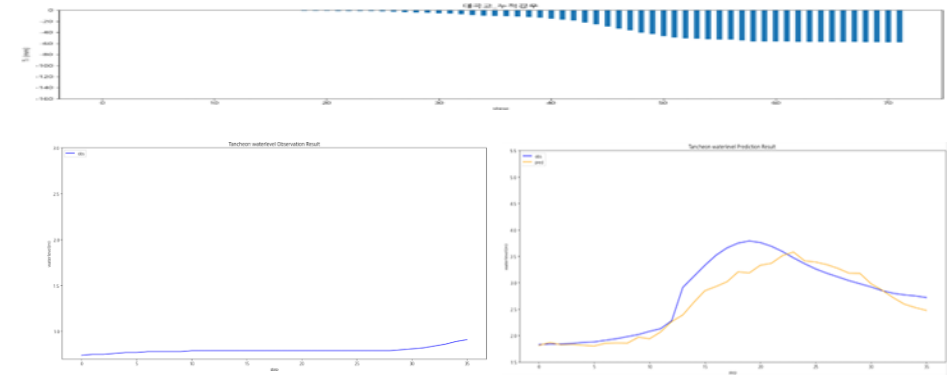
testdata_220712

R2 : -0.0424



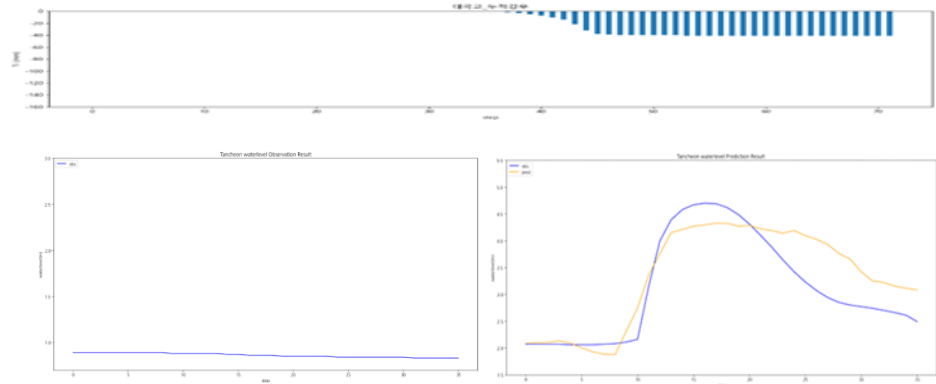
testdata_250716

R2 : 0.8305



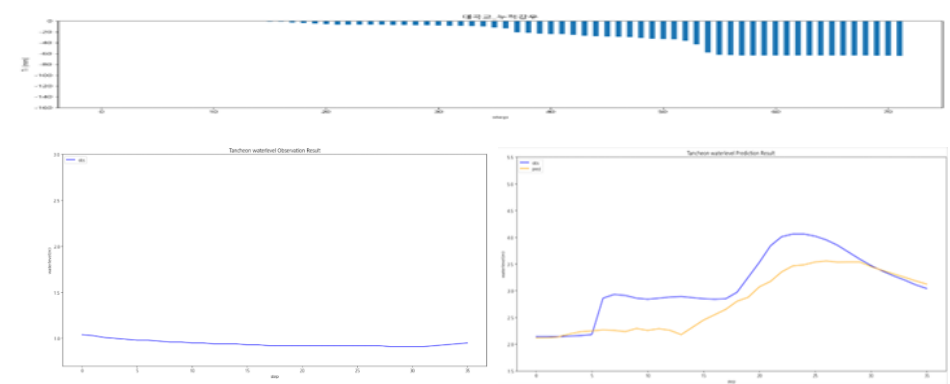
testdata_230711

R2 : 0.7456



testdata_250814

R2 : 0.5011

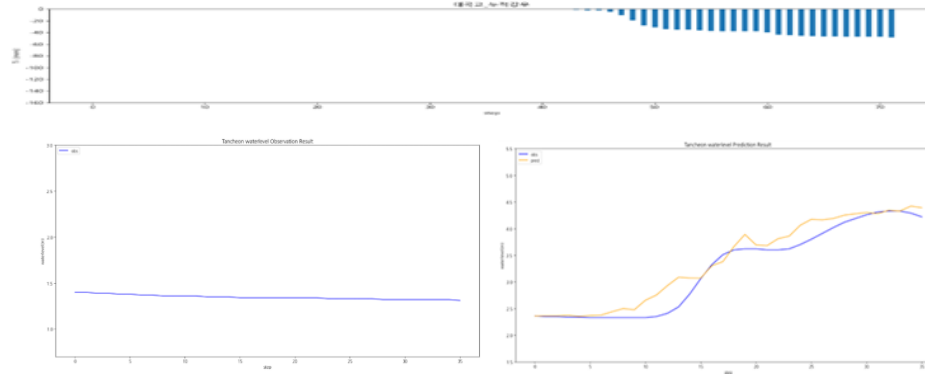


대곡교 - 관심수위 (누적 강우 110mm 이상 5)

* model_(gn/dg)_110_5_2

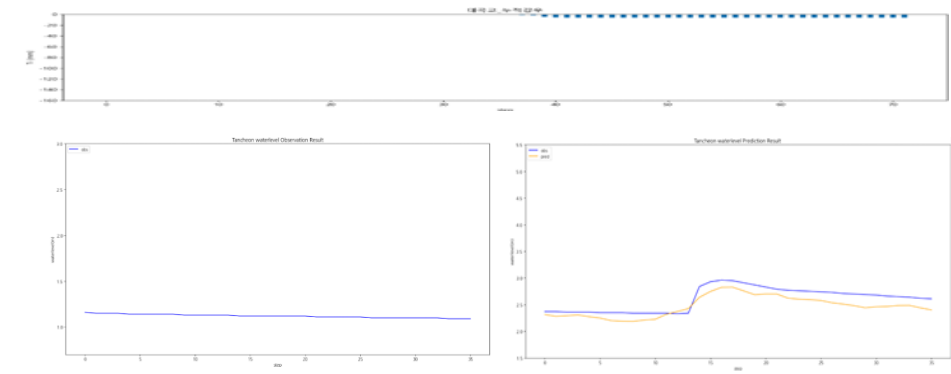
testdata_200802

R2 : 0.9246



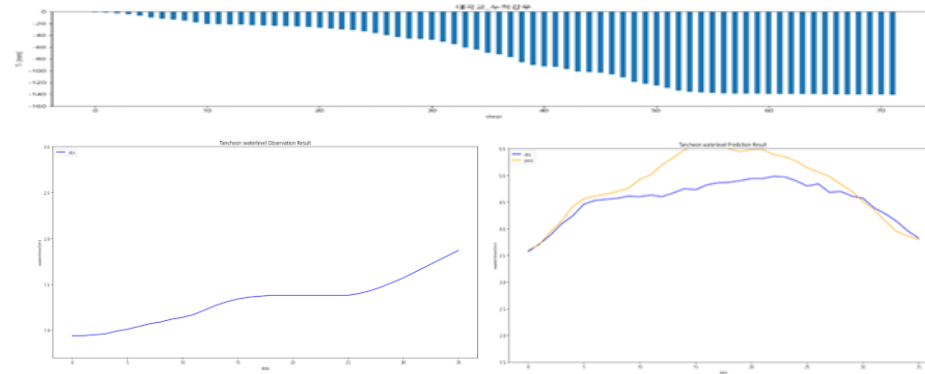
testdata_240723

R2 : 0.5130



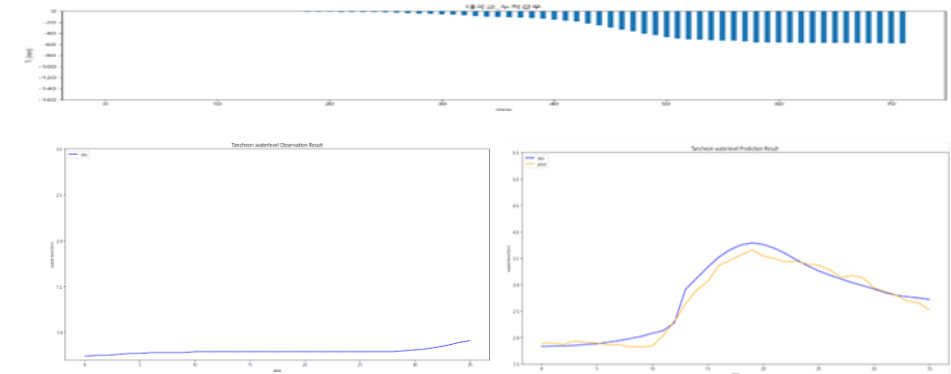
testdata_220712

R2 : -0.1000



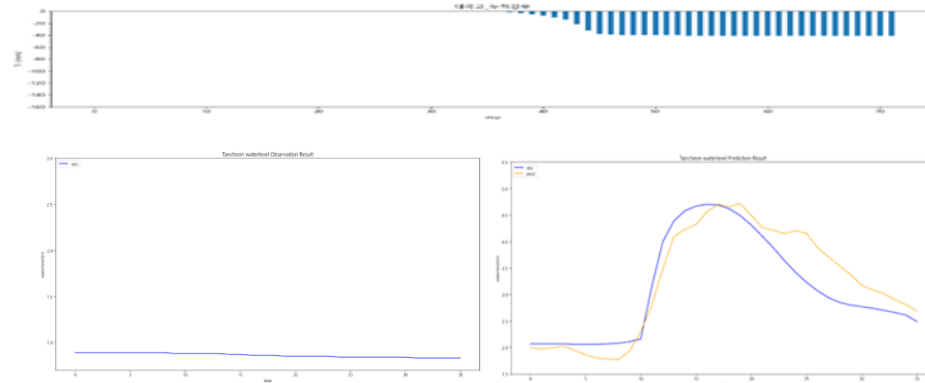
testdata_250716

R2 : 0.9577



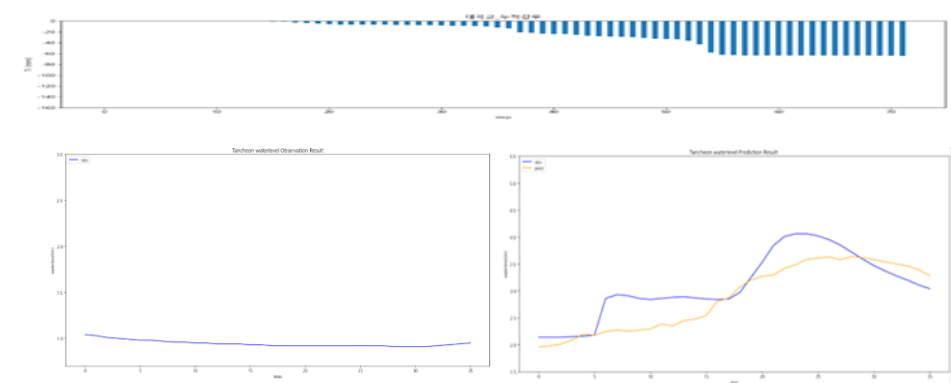
testdata_230711

R2 : 0.8276



testdata_250814

R2 : 0.6115

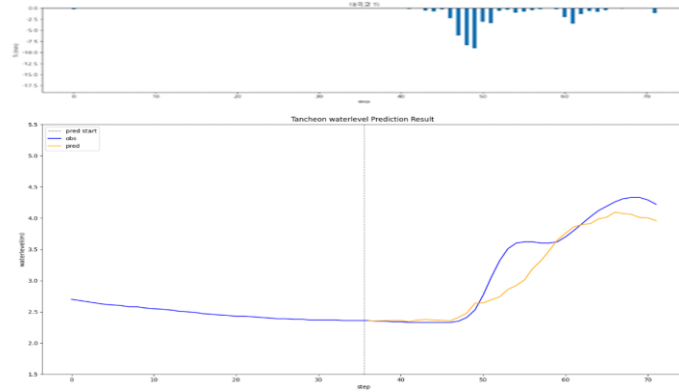


대곡교 - 관심수위 (20년 이후)

* model_(gn/dg)_20

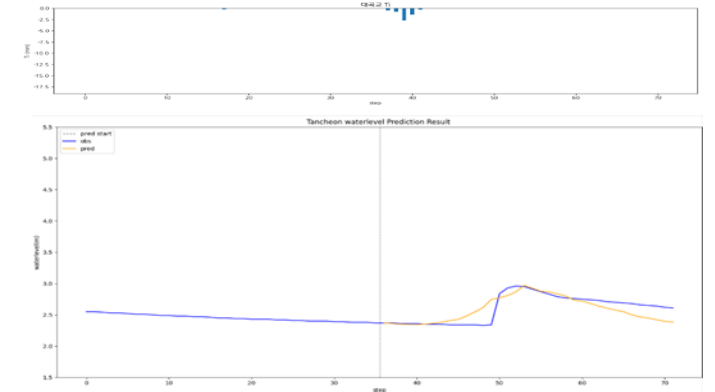
testdata_200802

R2 : 0.8845



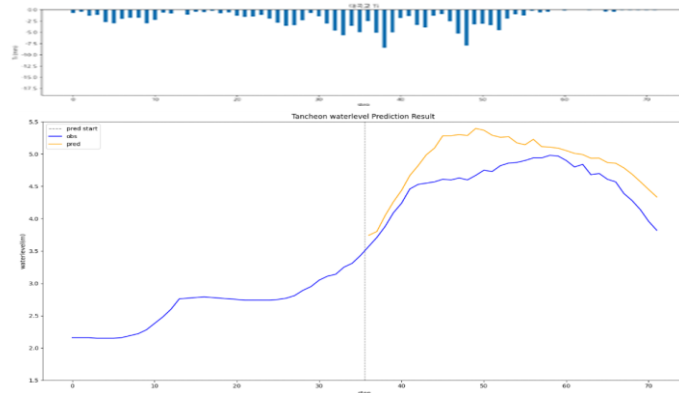
testdata_240723

R2 : 0.6077



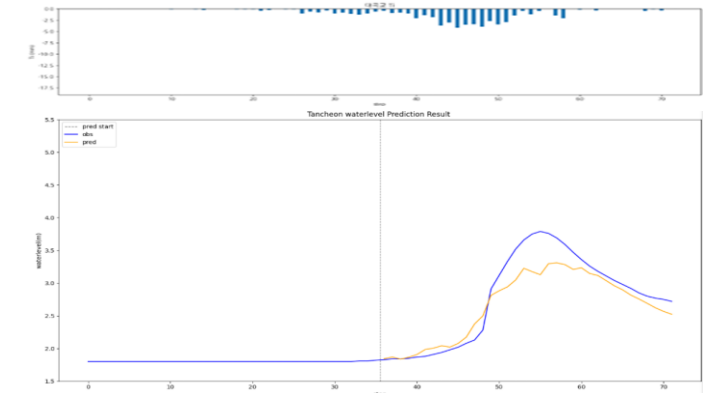
testdata_220712

R2 : -0.1561



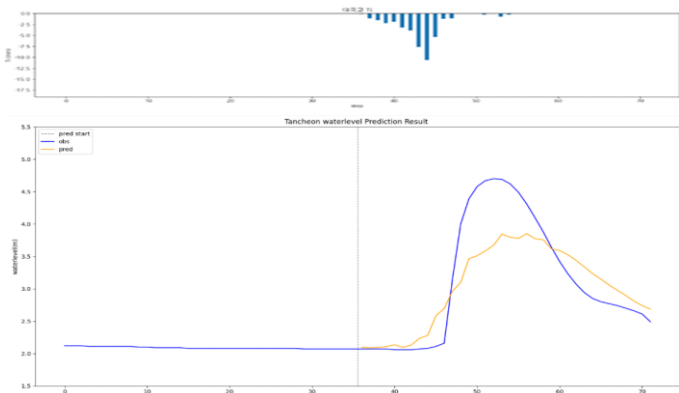
testdata_250716

R2 : 0.8647



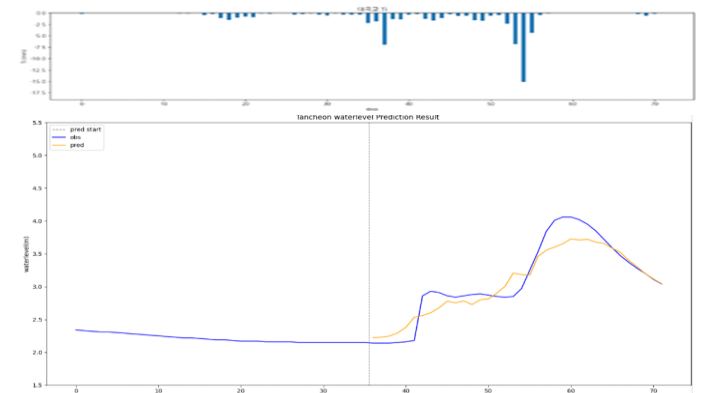
testdata_230711

R2 : 0.7304



testdata_250814

R2 : 0.8840

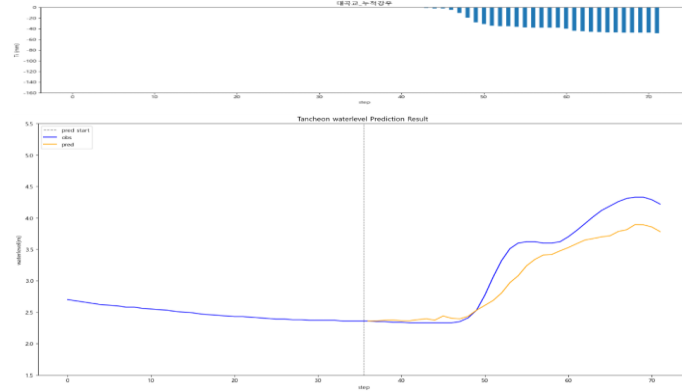


대곡교 - 관심수위 (20년 이후 + 누적 강우)

* model_(gn/dg)_rf20

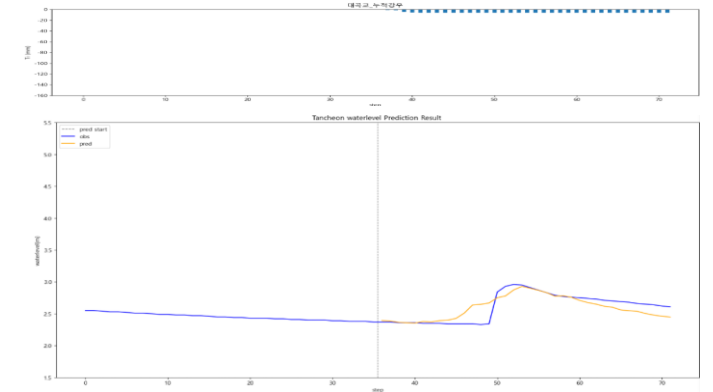
testdata_200802

R2 : 0.8523



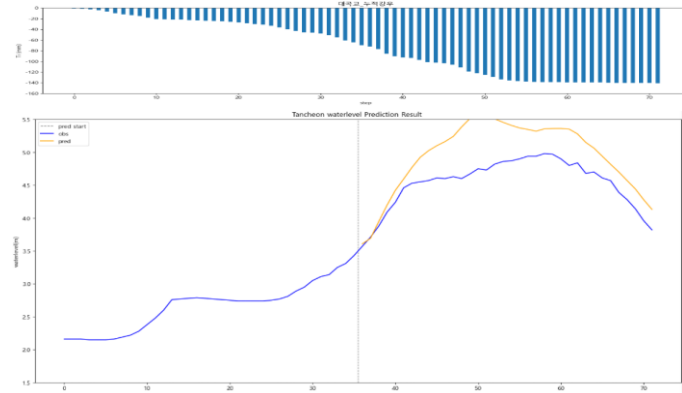
testdata_240723

R2 : 0.6739



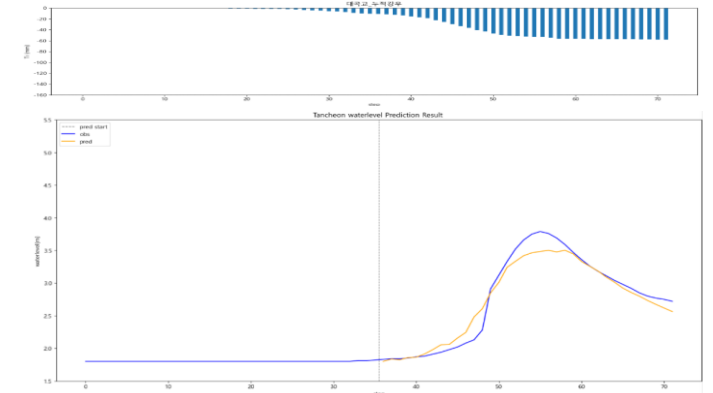
testdata_220712

R2 : -0.5266



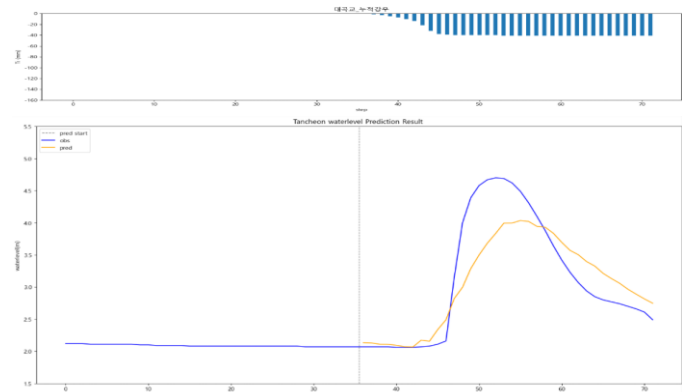
testdata_250716

R2 : 0.9528



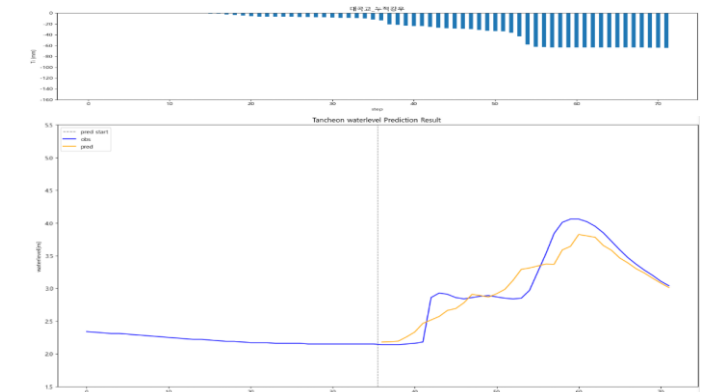
testdata_230711

R2 : 0.7527



testdata_250814

R2 : 0.8631



2. 단일 LSTM 개발 결과

학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

+) 누적 강우 컬럼 추가

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교 _Ti	궁내교 _Ti	대곡교 누적강우	궁내교 누적강우
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0	0.0	0.0

관심 수위 전체 데이터

- 학습 , 검증 : 총 67개 中 61개 데이터 파일 사용
- 테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개) - shuffle
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 20년 이후 데이터

- 학습 , 검증 : 총 40개 中 35개 데이터 파일 사용
- 테스트 : 학습 데이터에서 제외한 5개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2020 ~ 20240920 (25개) - shuffle
- 검증 : 20240921 ~ 250919 85번 (7개)
- 테스트 : 250919 86번 ~ 251006 (3개)

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

- learning_rate = 0.0003
- Loss_function = MSE
- Seq_length = 36 (6시간)
- Dense = 1 (36번 처리)
- 학습 이벤트 순서 shuffle

전체 강우 학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0003
 - Loss_function = MSE
 - Seq_length = 36 (6시간)
 - Dense = 1 (36번 처리)
 - 학습 이벤트 순서 shuffle

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

- 학습 : 총 1420개 中 1412개 데이터 파일 학습에 사용 (결측 5개 + 검증 3개)
- 검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 70% - shuffle
- 검증 : 20%
- 테스트 : 나머지 (10%)

궁내교

- » 전체 강우 기반 학습 시 일반적인 이벤트에서는 높은 R^2
- » 일부 극한 사상에서는 예측 성능이 저하

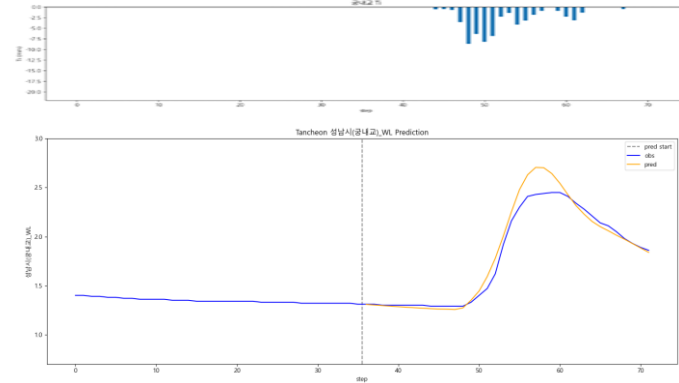
- R^2 평가

전체 강우		관심수위		20년 이후	
R2	궁내교	궁내교		궁내교	
Test data	전체 강우	관심수위	+ 누적강우	20년 이후	+ 누적강우
20.08.02	0.9541	0.9508	0.9176	0.9610	0.9407
22.07.12	0.9273	0.0509	0.4645	0.2350	0.2055
23.07.11	0.9329	0.9096	0.9119	0.7765	0.6441
24.07.23	0.6838	0.7689	0.7989	0.7457	0.6904
25.07.16	0.4005	0.5051	0.4844	0.6535	0.7551
25.08.14	0.8120	0.7659	0.7933	0.8250	0.9146

궁내교 - 전체강우

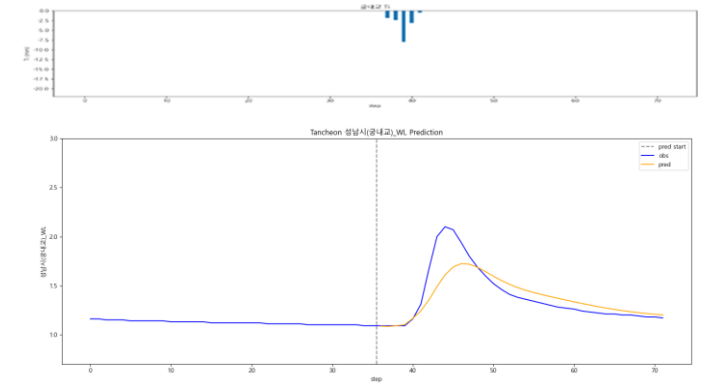
testdata_200802

R2 : 0.9541



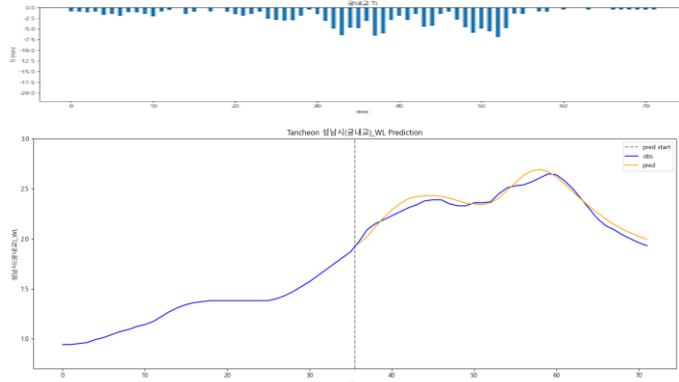
testdata_240723

R2 : 0.6838



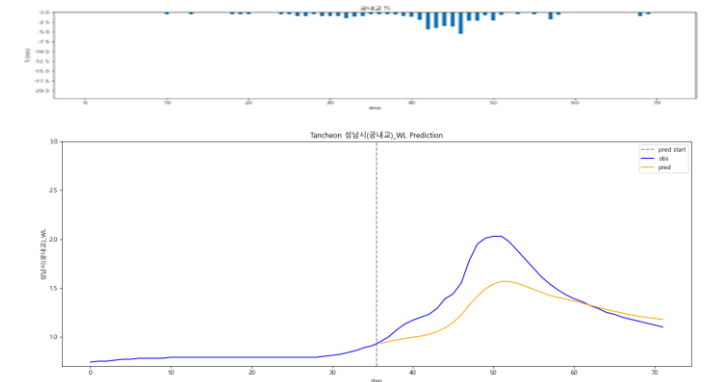
testdata_220712

R2 : 0.9273



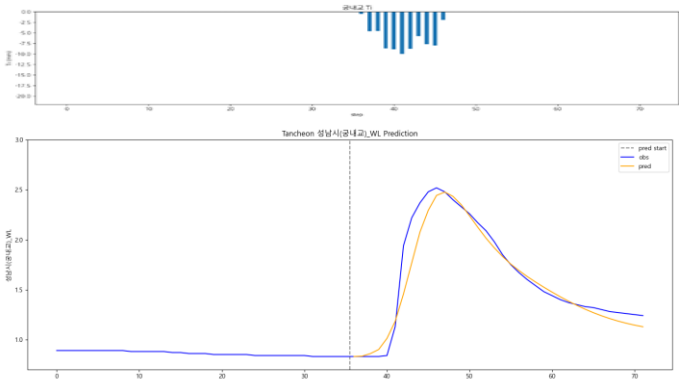
testdata_250716

R2 : 0.4005



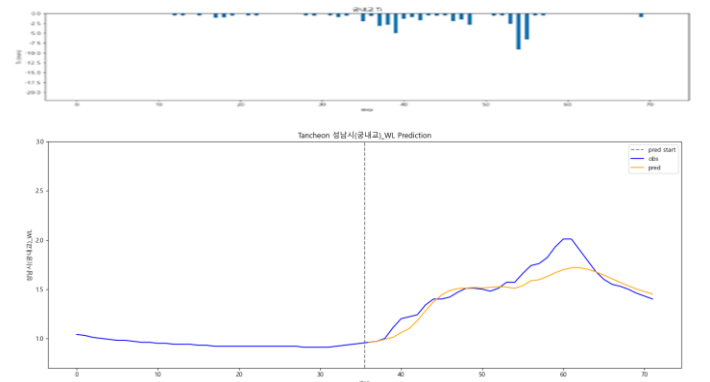
testdata_230711

R2 : 0.9329



testdata_250814

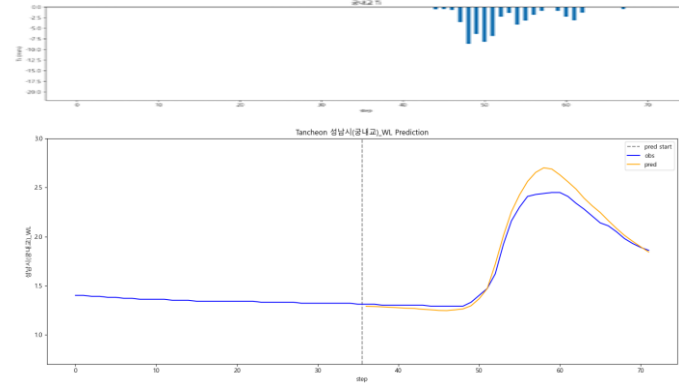
R2 : 0.8120



궁내교 - 관심수위

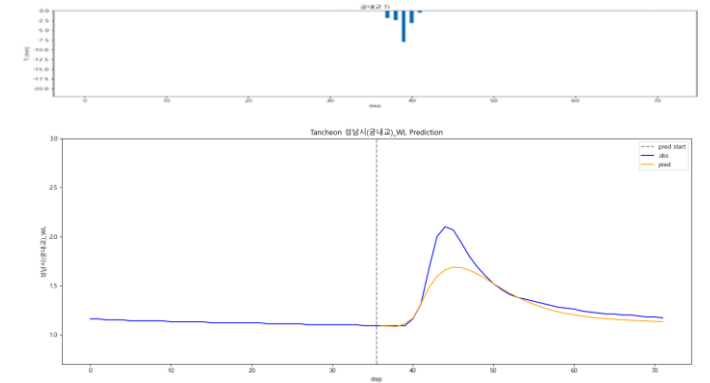
testdata_200802

R2 : 0.9508



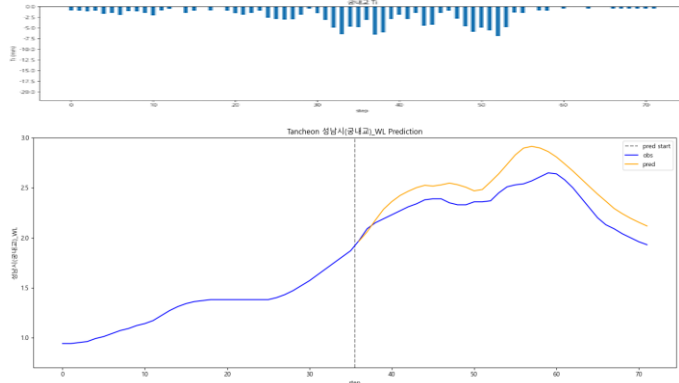
testdata_240723

R2 : 0.7689



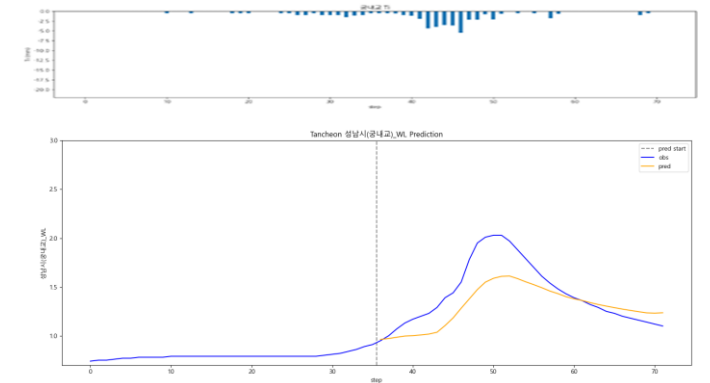
testdata_220712

R2 : 0.0509



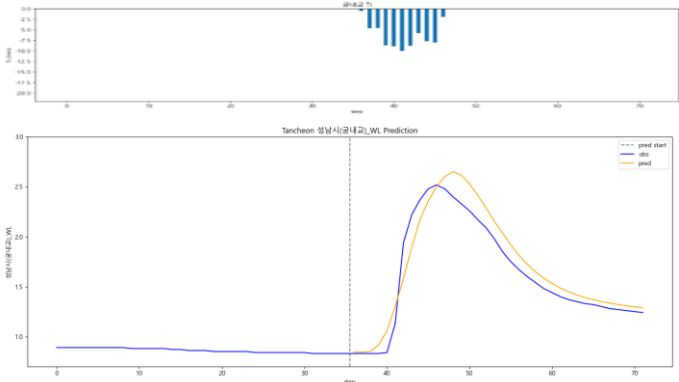
testdata_250716

R2 : 0.5051



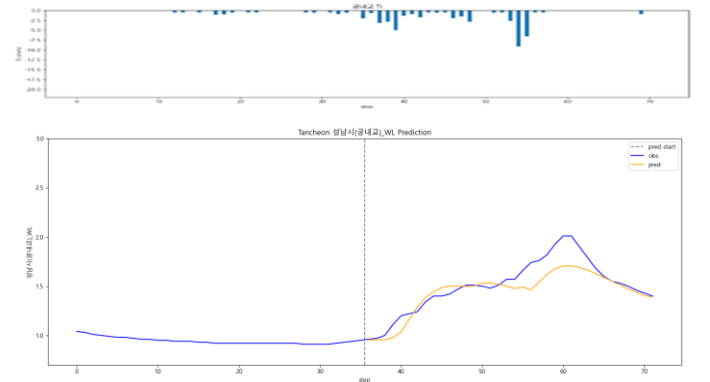
testdata_230711

R2 : 0.9096



testdata_250814

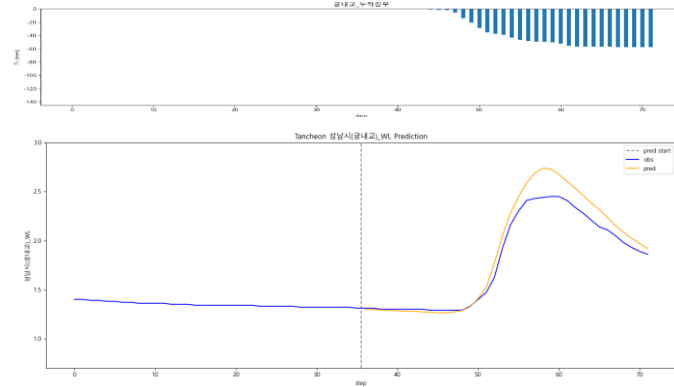
R2 : 0.7659



궁내교 - 관심수위 (+ 누적강우)

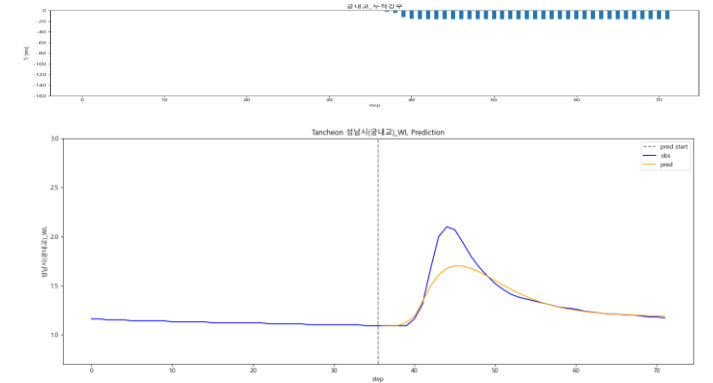
testdata_200802

R2 : 0.9176



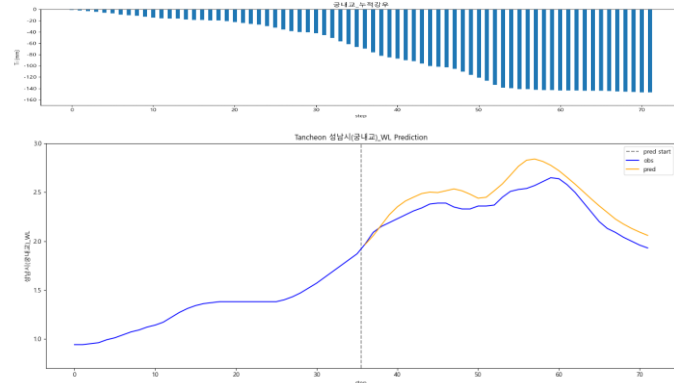
testdata_240723

R2 : 0.7989



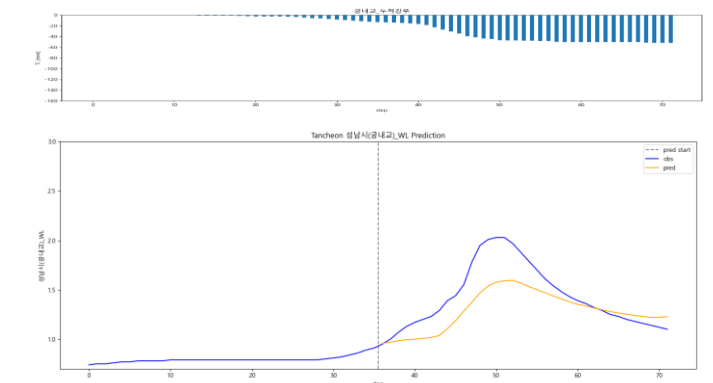
testdata_220712

R2 : 0.4645



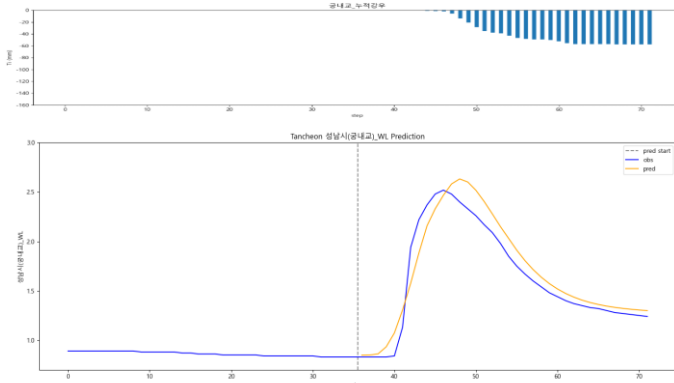
testdata_250716

R2 : 0.4844



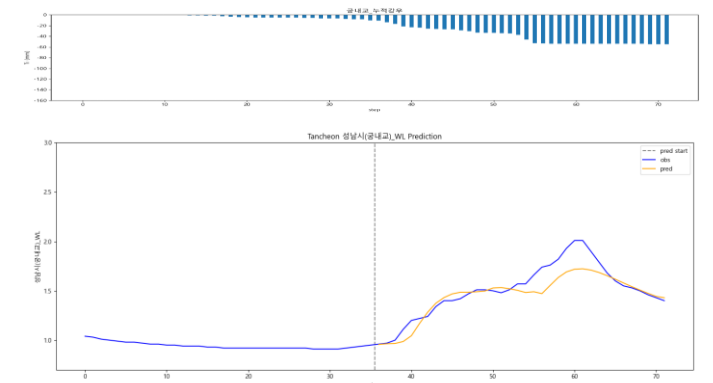
testdata_230711

R2 : 0.9119



testdata_250814

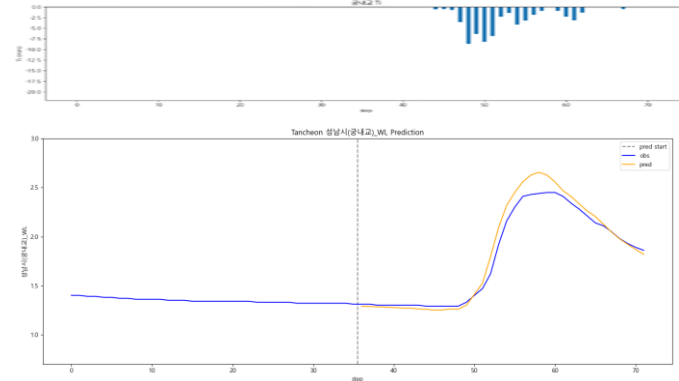
R2 : 0.7933



궁내교 - 관심수위 (20년 이후)

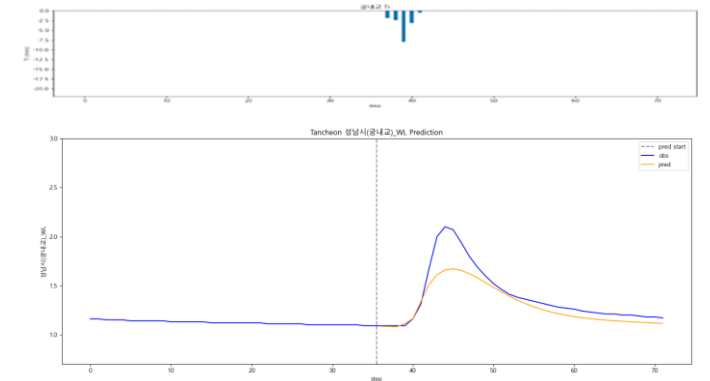
testdata_200802

R2 : 0.9610



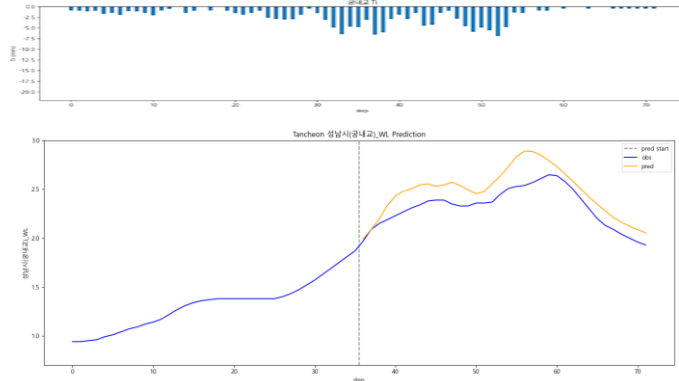
testdata_240723

R2 : 0.7457



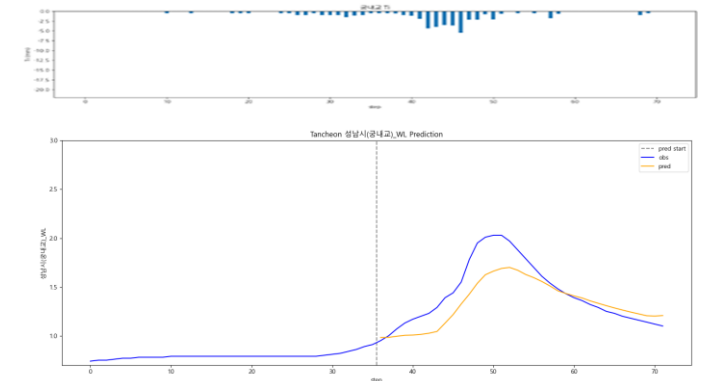
testdata_220712

R2 : 0.2350



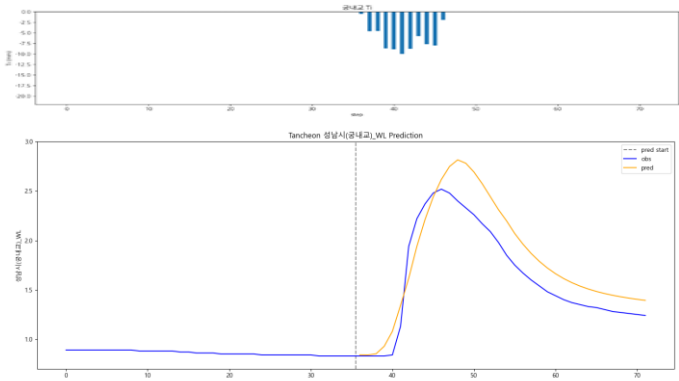
testdata_250716

R2 : 0.6535



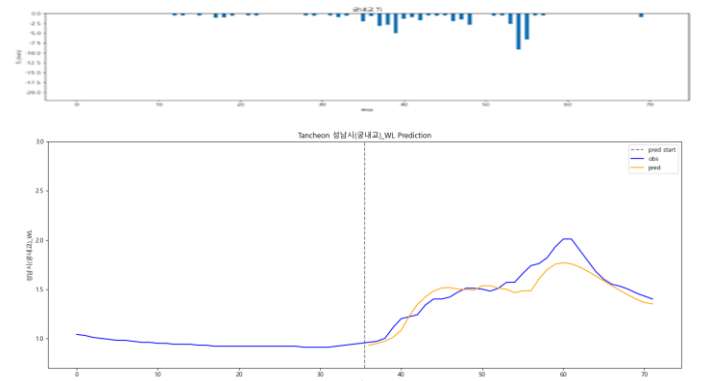
testdata_230711

R2 : 0.7765



testdata_250814

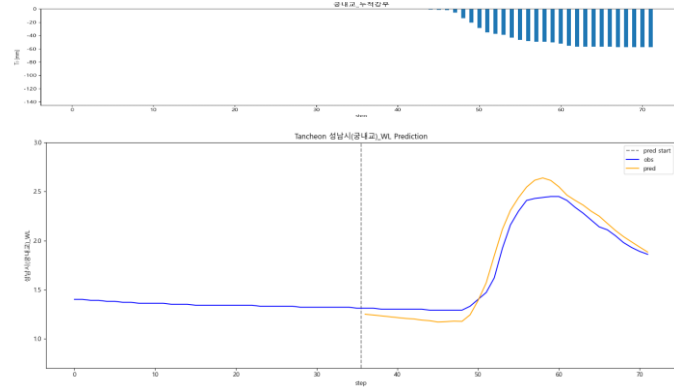
R2 : 0.8250



궁내교 - 관심수위 (20년 이후 + 누적강우)

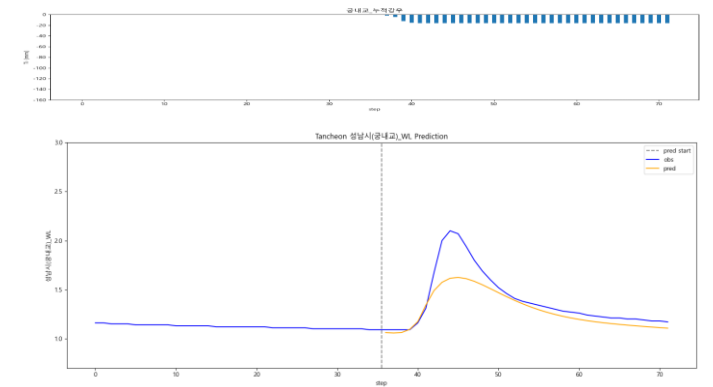
testdata_200802

R2 : 0.9407



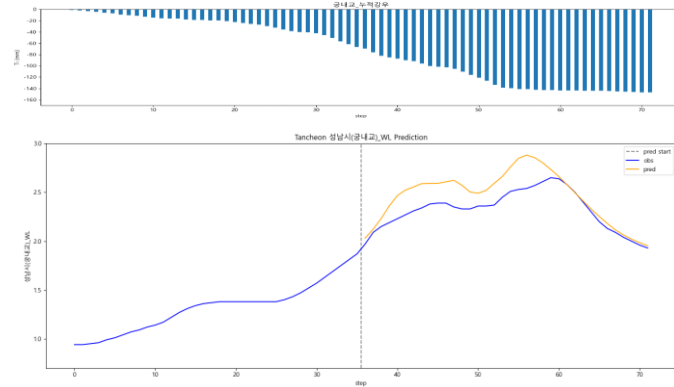
testdata_240723

R2 : 0.6904



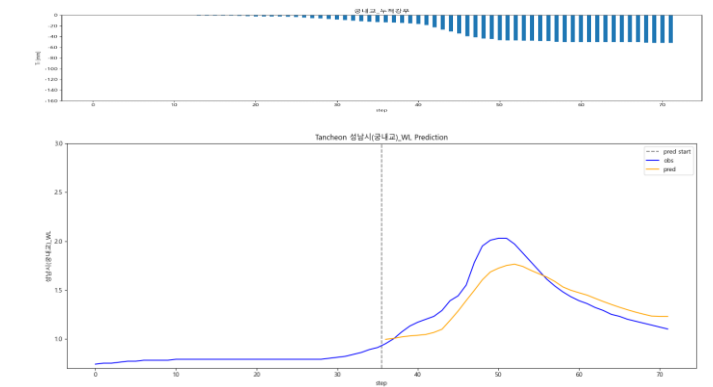
testdata_220712

R2 : 0.2055



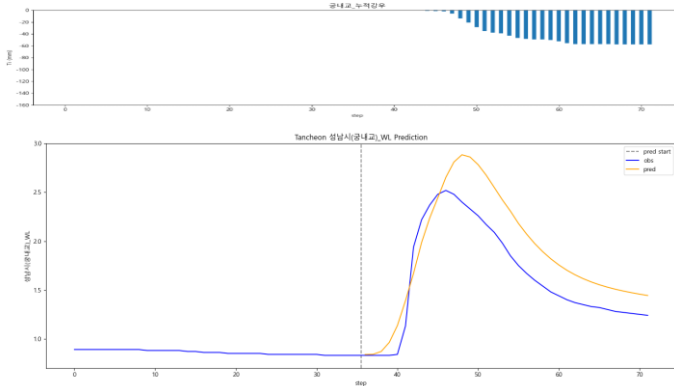
testdata_250716

R2 : 0.7551



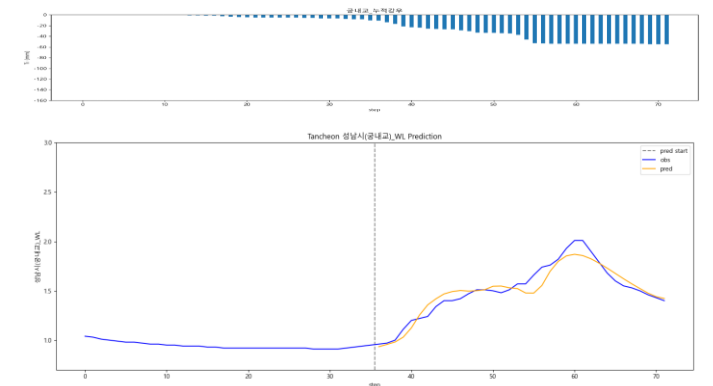
testdata_230711

R2 : 0.6441



testdata_250814

R2 : 0.9146



대곡교

- » 전체 강우 기반 학습 시 일반적인 이벤트에서는 높은 R^2
- » 일부 이벤트에서는 완만한 상승·지연 특성으로 인해 성능 저하

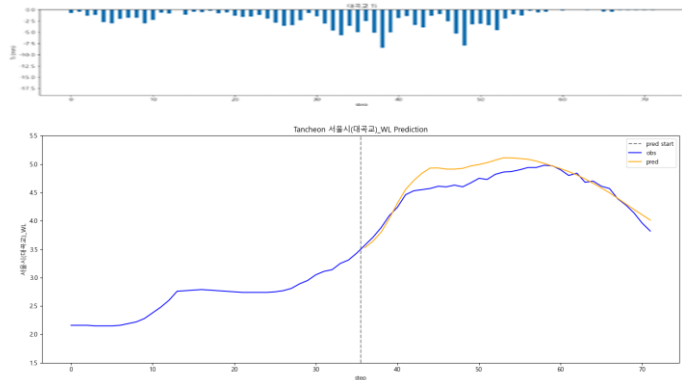
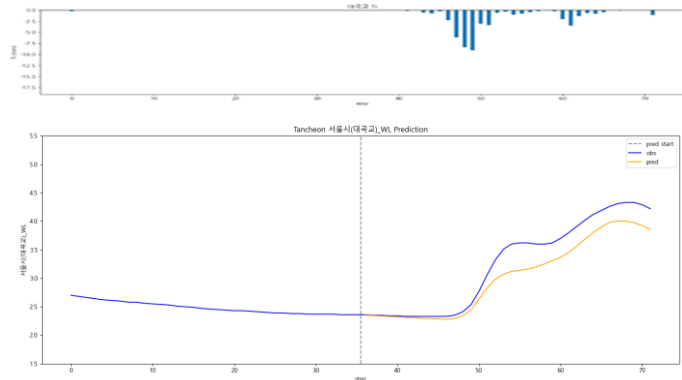
- R^2 평가

전체 강우		관심수위		20년 이후	
R2	대곡교	대곡교		대곡교	
Test data	전체 강우	관심수위	+ 누적강우	20년 이후	+ 누적강우
20.08.02	0.8737	0.9090	0.9422	0.9578	0.9719
22.07.12	0.7555	0.5072	0.6228	0.3842	0.5927
23.07.11	0.7992	0.8436	0.8492	0.7163	0.7324
24.07.23	0.1336	0.5396	0.5213	0.4315	0.3858
25.07.16	0.8864	0.9218	0.9162	0.9106	0.9196
25.08.14	0.8356	0.9005	0.8923	0.9172	0.9006

대곡교 - 전체강우

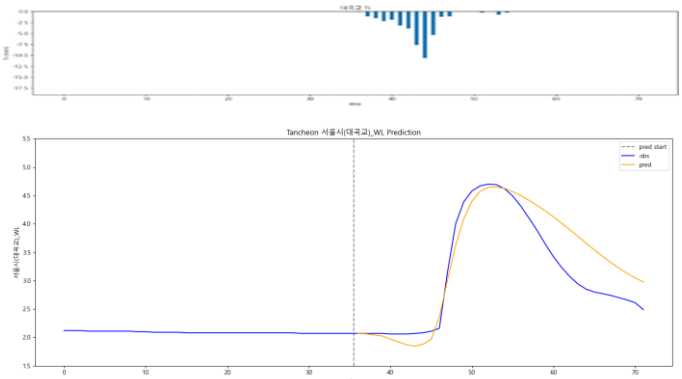
testdata_200802

R2 : 0.8737



testdata_220712

R2 : 0.7555

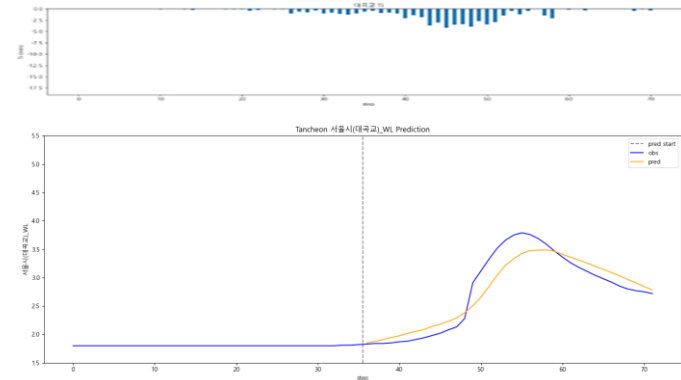
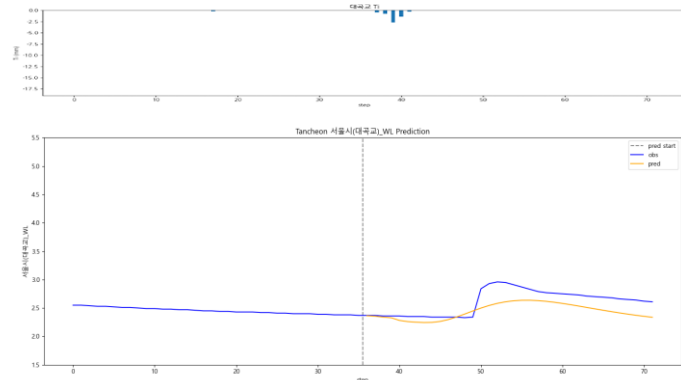


testdata_230711

R2 : 0.7992

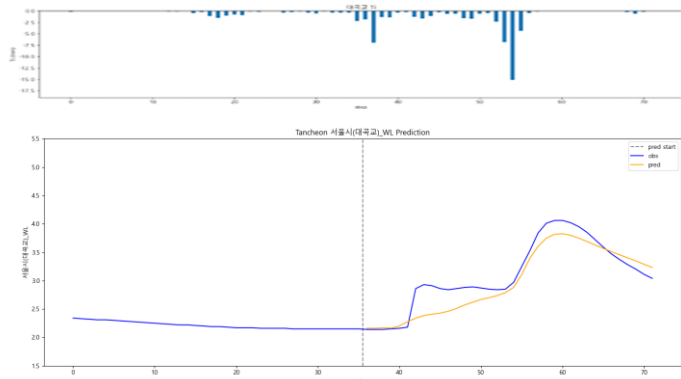
testdata_240723

R2 : 0.1336



testdata_250716

R2 : 0.8864



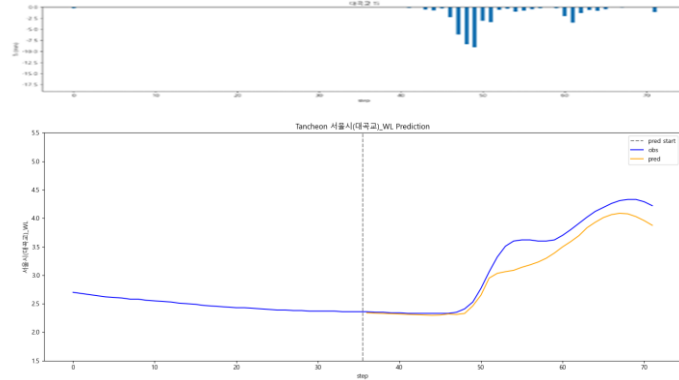
testdata_250814

R2 : 0.8356

대곡교 - 관심수위

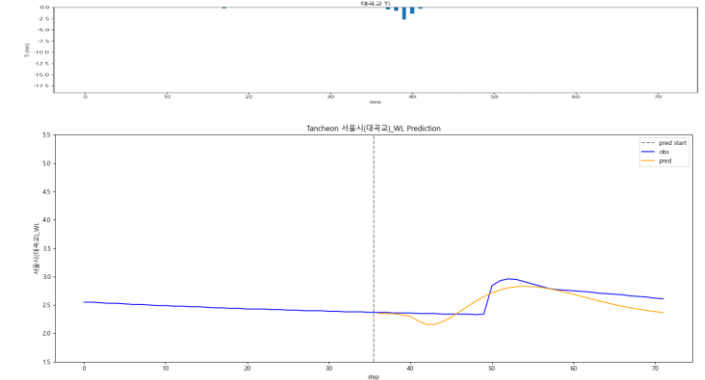
testdata_200802

R2 : 0.9090



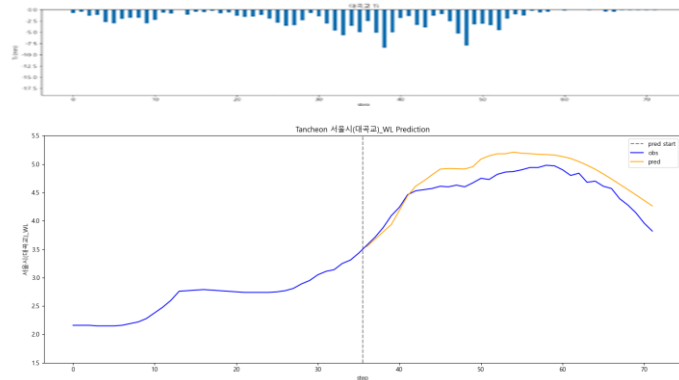
testdata_240723

R2 : 0.5396



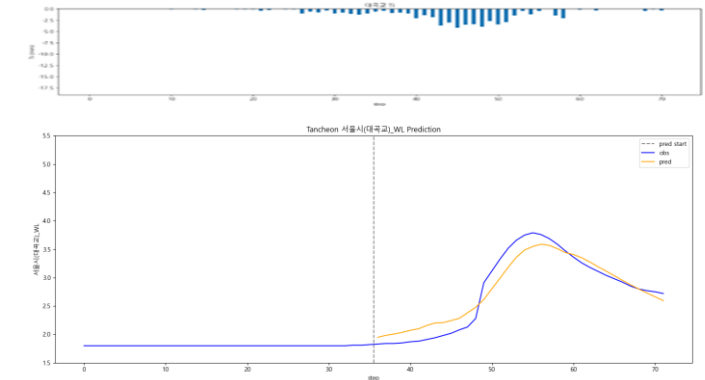
testdata_220712

R2 : 0.5072



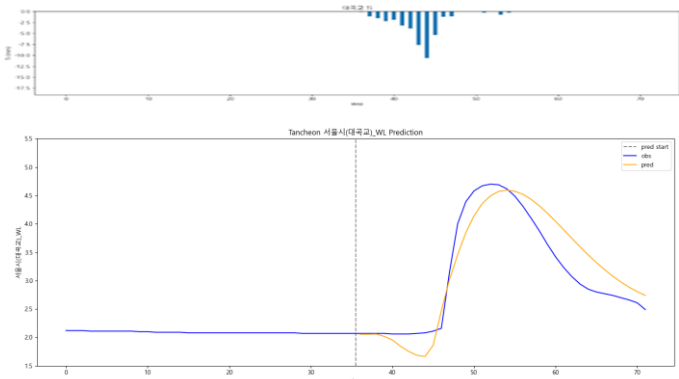
testdata_250716

R2 : 0.9218



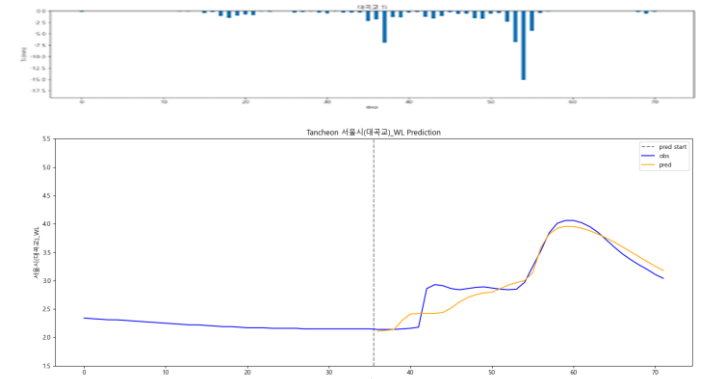
testdata_230711

R2 : 0.8436



testdata_250814

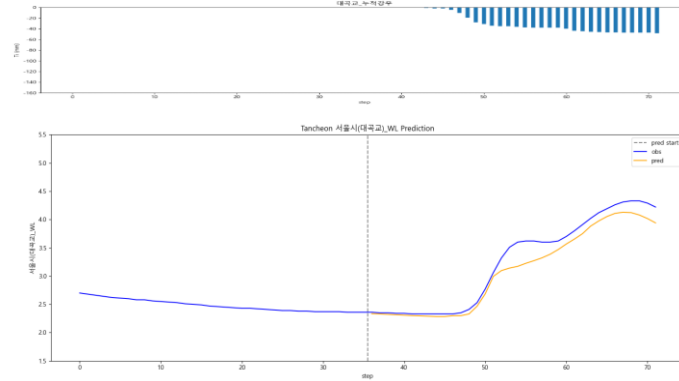
R2 : 0.9005



대곡교 - 관심수위 (+ 누적강우)

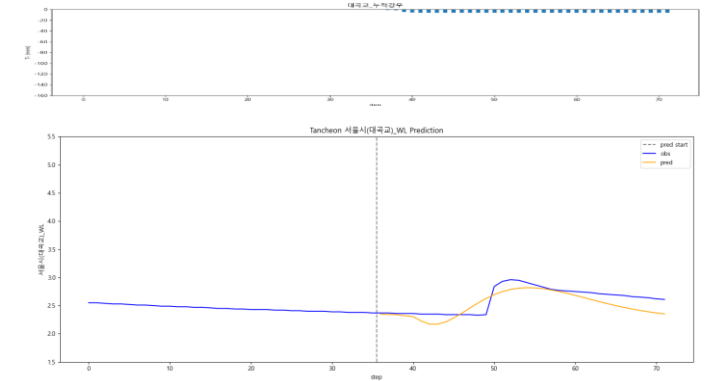
testdata_200802

R2 : 0.9422



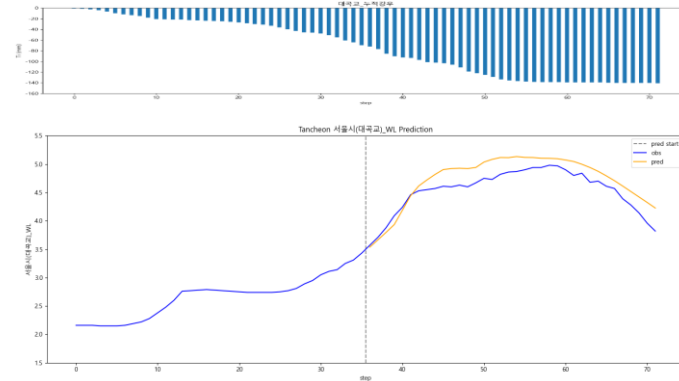
testdata_240723

R2 : 0.5213



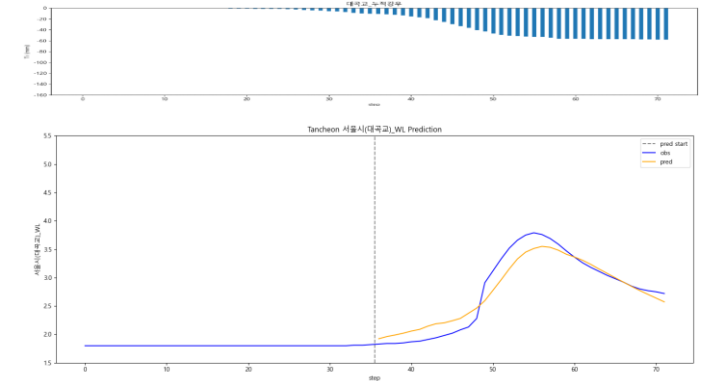
testdata_220712

R2 : 0.6228



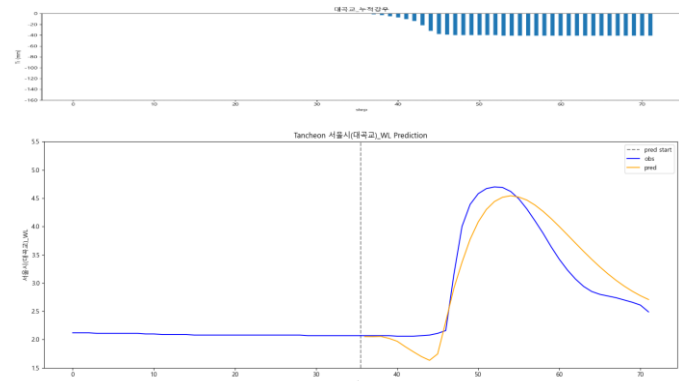
testdata_250716

R2 : 0.9162



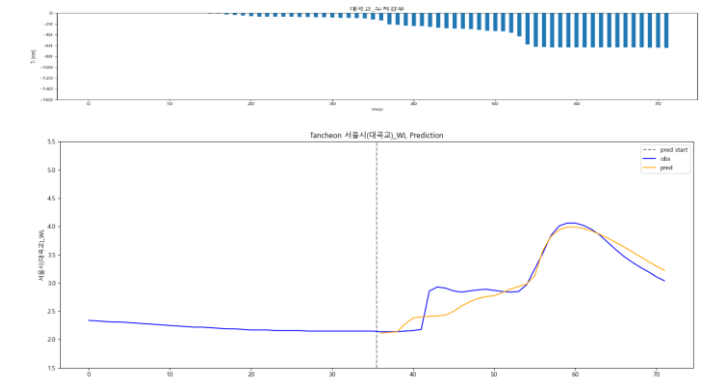
testdata_230711

R2 : 0.8492



testdata_250814

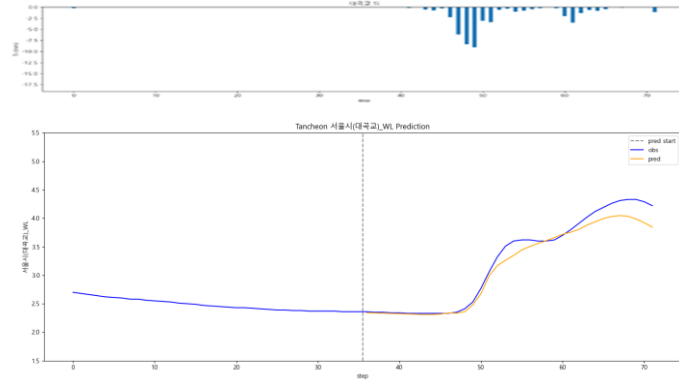
R2 : 0.8923



대곡교 - 관심수위 (20년 이후)

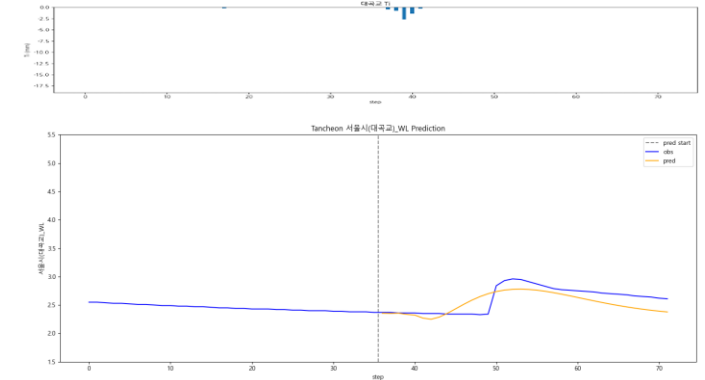
testdata_200802

R2 : 0.9578



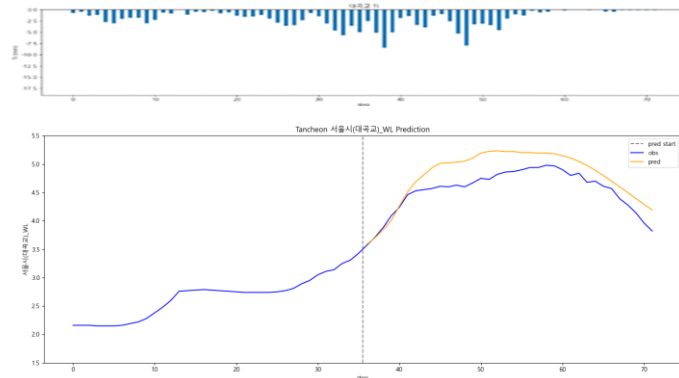
testdata_240723

R2 : 0.4315



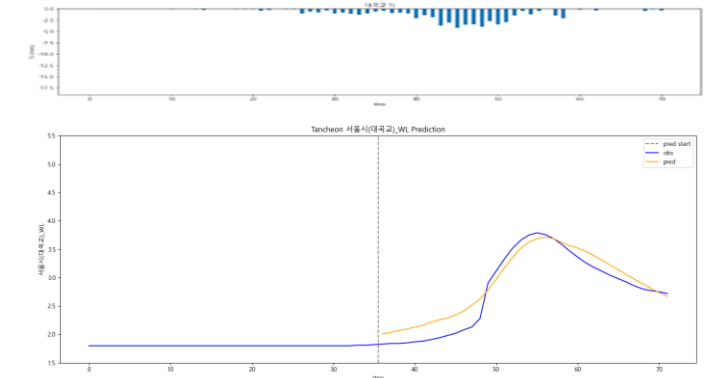
testdata_220712

R2 : 0.3842



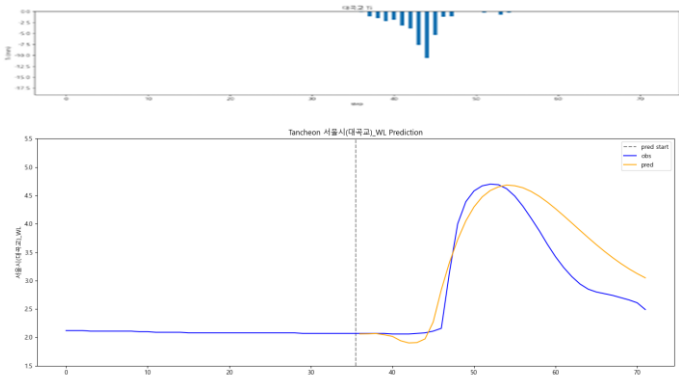
testdata_250716

R2 : 0.9106



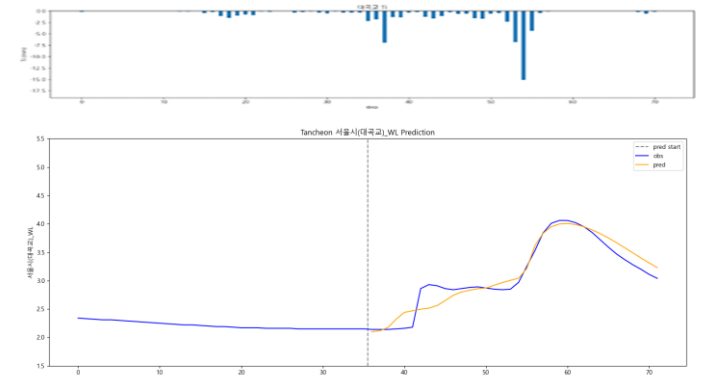
testdata_230711

R2 : 0.7163



testdata_250814

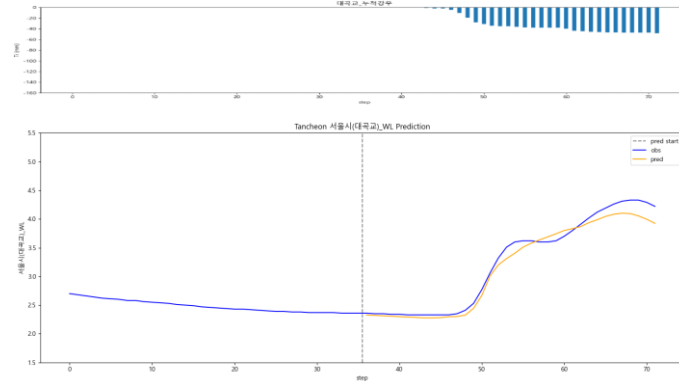
R2 : 0.9172



대곡교 - 관심수위 (20년 이후 + 누적강우)

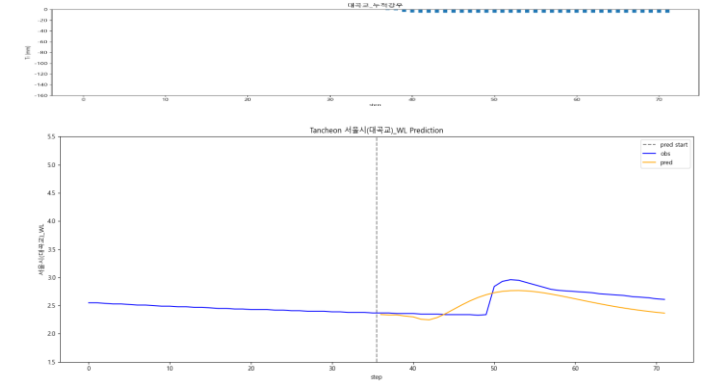
testdata_200802

R2 : 0.9719



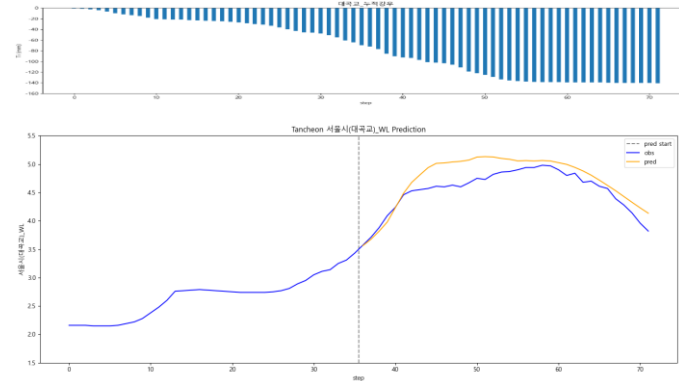
testdata_240723

R2 : 0.3858



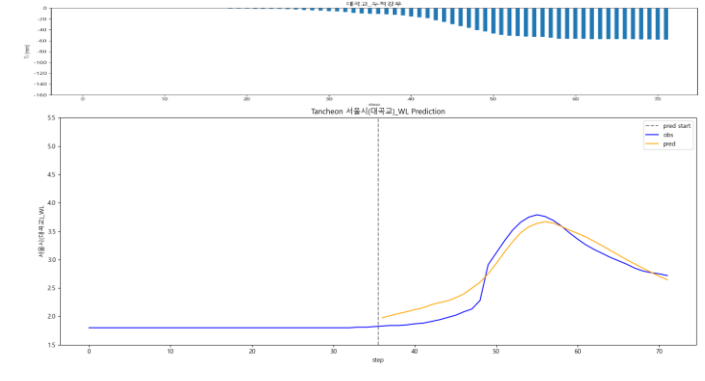
testdata_220712

R2 : 0.5927



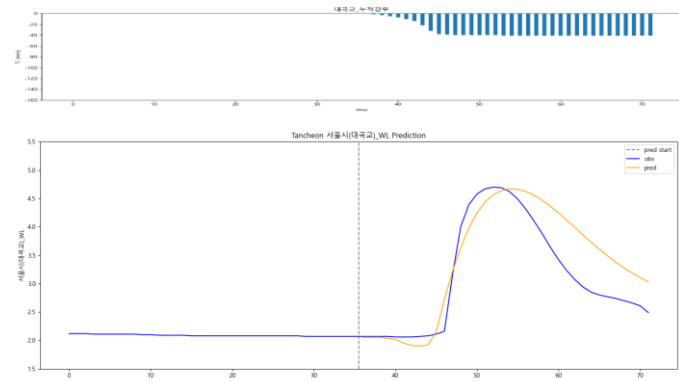
testdata_250716

R2 : 0.9196



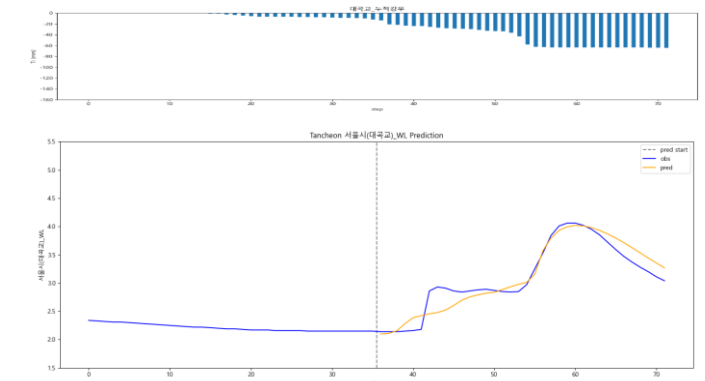
testdata_230711

R2 : 0.7324



testdata_250814

R2 : 0.9006



3. 결론

결론

- Parallel LSTM > 단일 LSTM
 - 관심수위 + 2020년 이후 학습 이 상대적으로 안정적인 성능
- 22.07.12 유형 사상 : 궁내교 , 대곡교 두 지점 동시에 성능이 저하

예측 구간 별 특징

예측 잘 맞는 구간

- 주로 20.08.02
- 단봉형 (피크 1개)
- 상승/하강이 비교적 매끈한 이벤트
- 6시간 윈도우 내에 주요 상승이 다 들어오는 이벤트
- 기저수위 (예측 시작 직전 수위) 가 안정적인 이벤트
- 선행강우 - 수위 상태가 “ 학습에서 흔히 본 ” 범위
- 이때 Lag (지연시간) $\approx 0 \sim \pm 20$ 분 ,
Amplitude (피크) 도 대체로 0.95 ~ 1.05 근처로 유지

예측이 잘 틀리는 구간

- 다봉형 (피크 2개 이상)
- 플래토 (계단식)
- 급격한 상승
- 기저수위 (예측 시작 직전 수위) 가 이벤트 중간에 비정상적으로 이동
(배수/ 펌프 / 지류 · 역류 영향)
- 상태가 다른데 같은 강우 - 수위 반응을 강요 하는 상황 : 동일 강우라도 수위 반응이 달라짐
(선행 포화 / 저류 , 관거 / 우수토실 , 하천 시설물 운영 , 역수위 영향 등)

4. 추가 분석

(관심수위 + 2020년 이후 학습)

오차 , 정확도

* 예측수위 - 관측수위 (정확도)

* 오차 : 소수 3 자리 반올림 , 정확도 : 소수 2 자리에서 반올림

- 궁내교

- 대곡교

1시간 뒤 시점

궁내교		1시간 (궁내교)	
Test data	1시간 오차	관측수위	예측수위
20.08.02	+0.07 (94.9 %)	1.3	1.37
22.07.12	+0.22 (90.4 %)	2.31	2.53
23.07.11	-0.50 (74.4 %)	1.94	1.44
24.07.23	-0.21 (87.3 %)	1.67	1.46
25.07.16	-0.09 (93.0 %)	1.23	1.14
25.08.14	+0.08 (93.9 %)	1.24	1.32

대곡교		1시간 (대곡교)	
Test data	1시간	관측수위	예측수위
20.08.02	+0.04 (98.3 %)	2.33	2.37
22.07.12	+0.29 (93.6 %)	4.53	4.82
23.07.11	+0.07 (96.6 %)	2.06	2.13
24.07.23	+0.01 (99.4 %)	2.35	2.36
25.07.16	+0.10 (95.0 %)	1.91	2.01
25.08.14	-0.30 (89.5 %)	2.86	2.56

3시간 뒤 시점

궁내교		3시간 (궁내교)	
Test data	3시간 오차	관측수위	예측수위
20.08.02	+0.01 (99.5 %)	2.16	2.17
22.07.12	+0.43 (83.0 %)	2.51	2.94
23.07.11	-0.08 (95.9 %)	1.85	1.77
24.07.23	-0.01 (99.1 %)	1.36	1.35
25.07.16	-0.10 (94.2 %)	1.79	1.69
25.08.14	+0.08 (94.6 %)	1.57	1.65

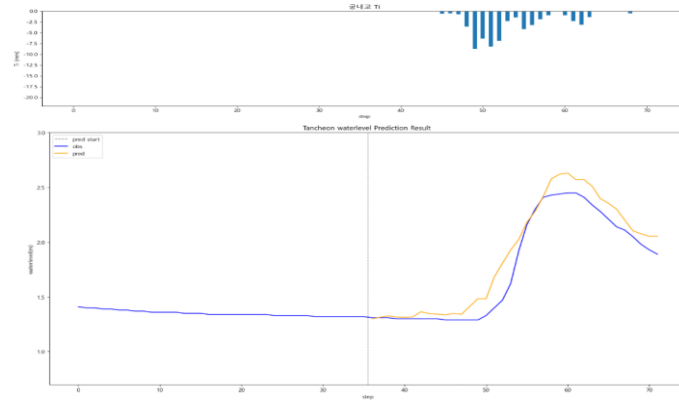
대곡교		3시간 (대곡교)	
Test data	3시간	관측수위	예측수위
20.08.02	-0.68 (81.1 %)	3.6	2.92
22.07.12	+0.30 (93.8 %)	4.87	5.17
23.07.11	-0.82 (82.2 %)	4.62	3.8
24.07.23	+0.01 (99.6 %)	2.91	2.92
25.07.16	-0.58 (84.7 %)	3.75	3.17
25.08.14	+0.21 (92.8 %)	2.97	3.18

궁내교 - 10분단위

* model_(gn/dg)_20

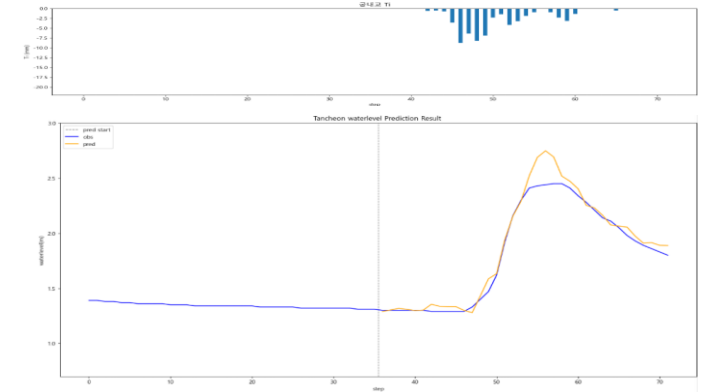
10분 전

R2 : 0.9138



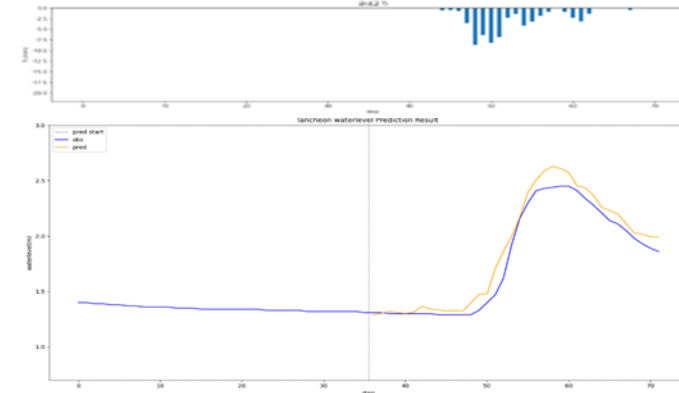
20분 후

R2 : 0.9579



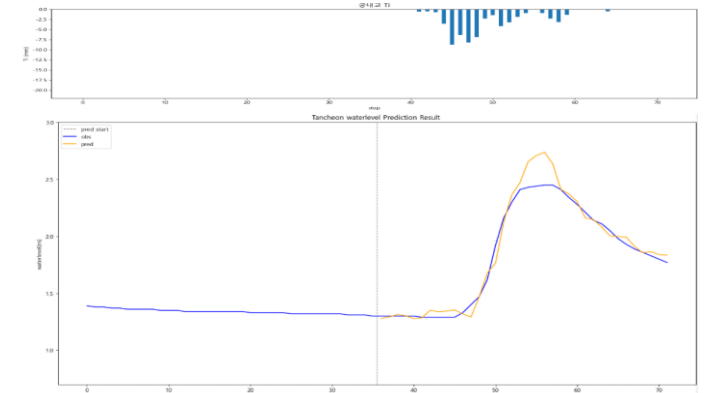
testdata_ 200802 (02)

R2 : 0.9513



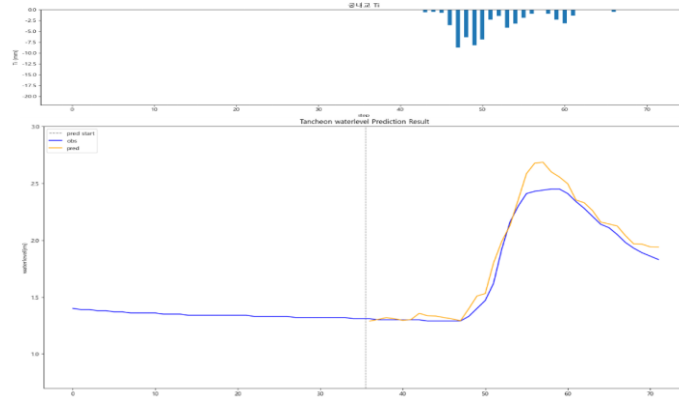
30분 후

R2 : 0.9529



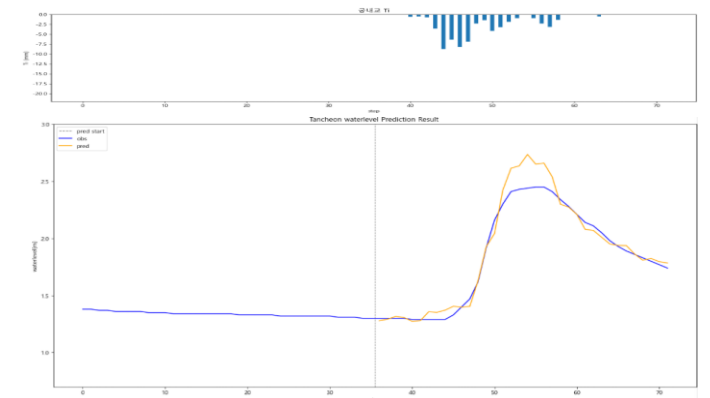
10분 후

R2 : 0.9585



40분 후

R2 : 0.9478

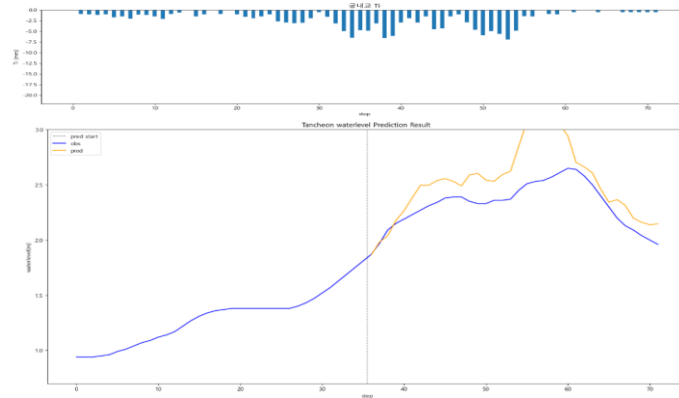


궁내교 - 10분단위

* model_(gn/dg)_20

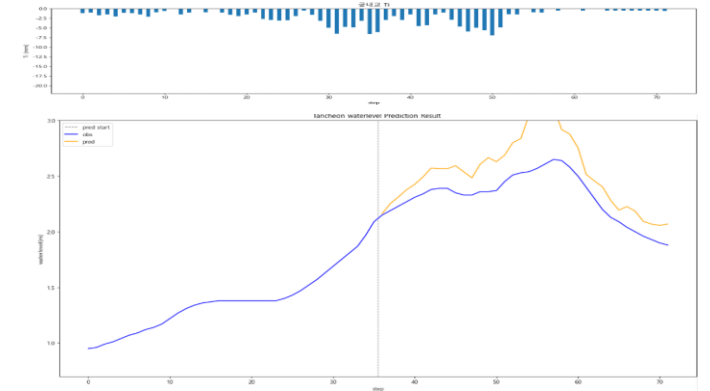
10분 전

R2 : -0.7405



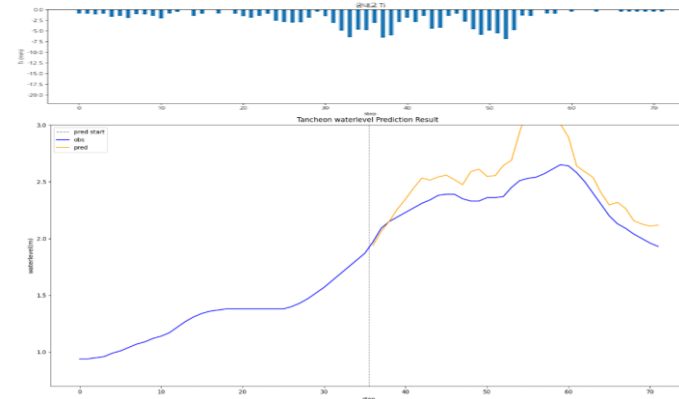
20분 후

R2 : -0.4821



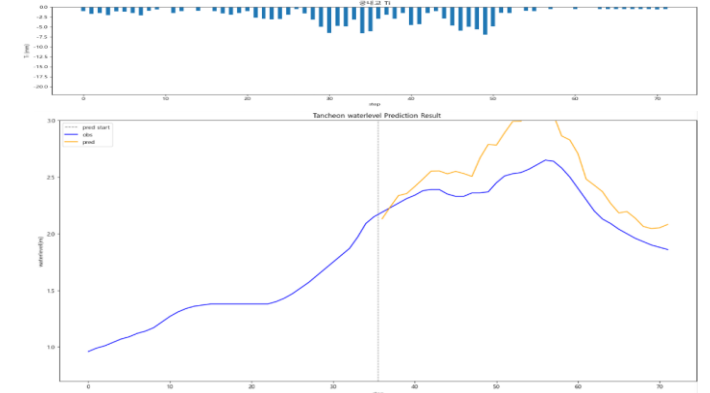
testdata_ 220712 (16)

R2 : -0.8548



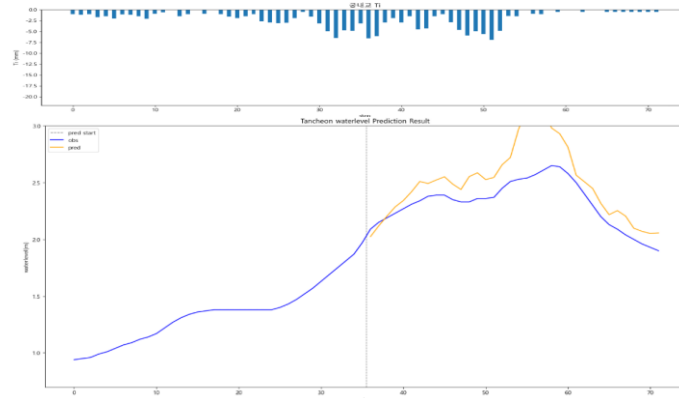
30분 후

R2 : -0.5463



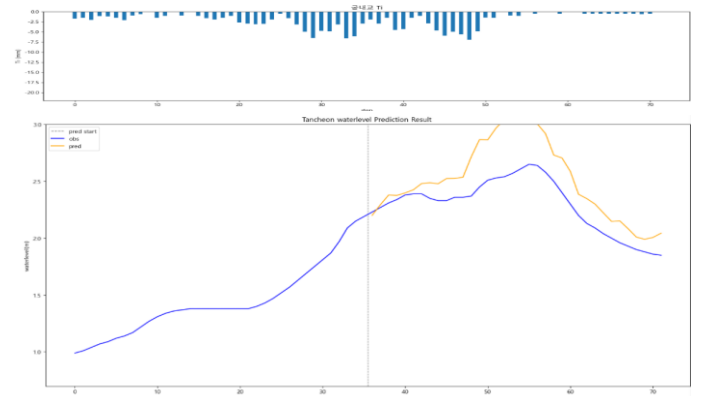
10분 후

R2 : -0.4408



40분 후

R2 : -0.2182

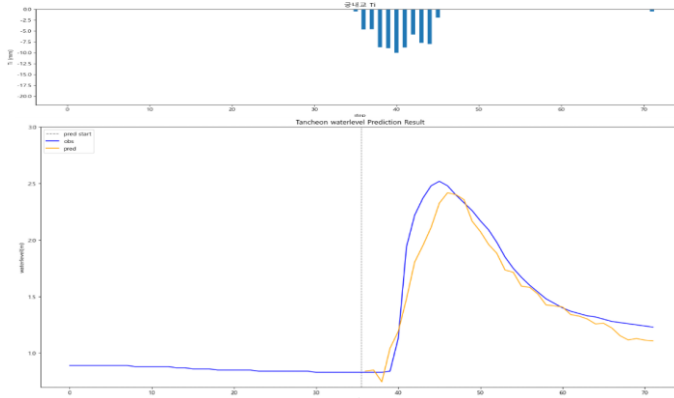
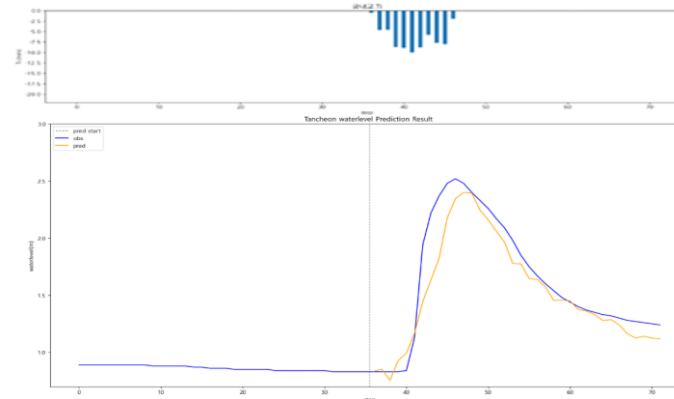
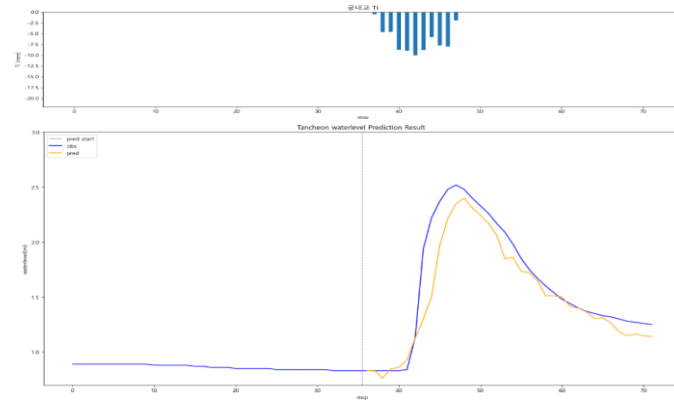


궁내교 - 10분단위

* model_(gn/dg)_20

10분 전

R2 : 0.8657

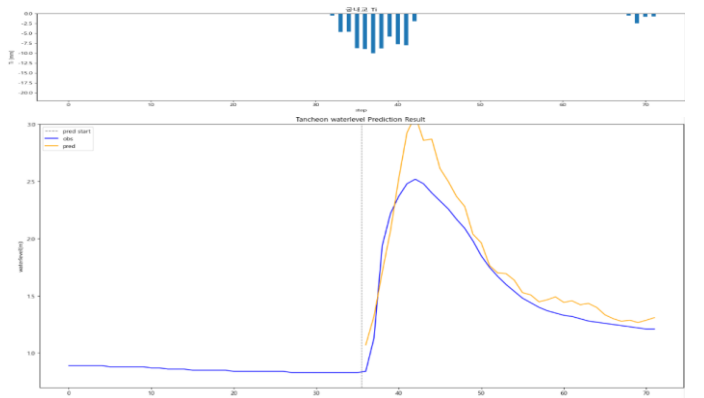
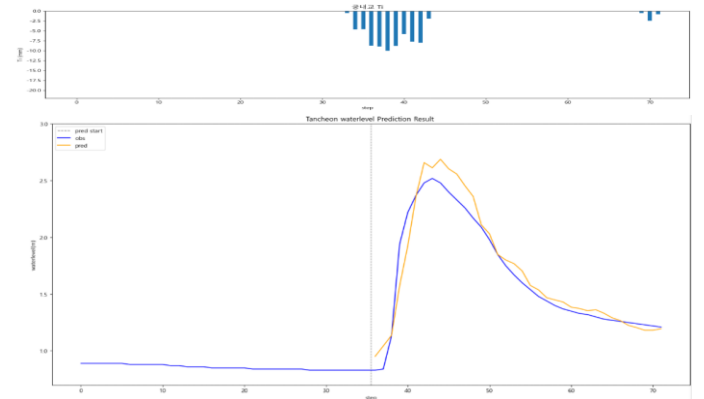
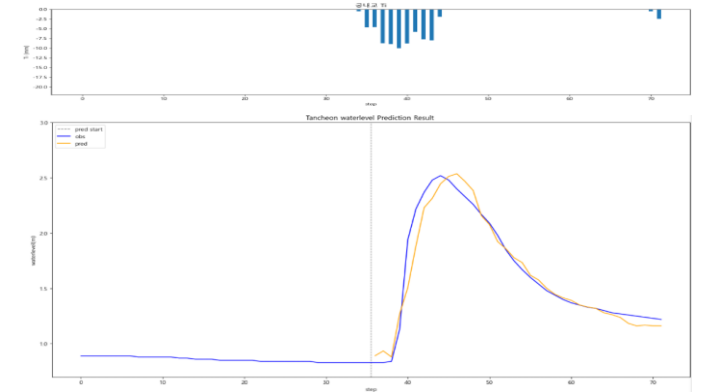


10분 후

R2 : 0.9012

20분 후

R2 : 0.9471



30분 후

R2 : 0.9322

40분 후

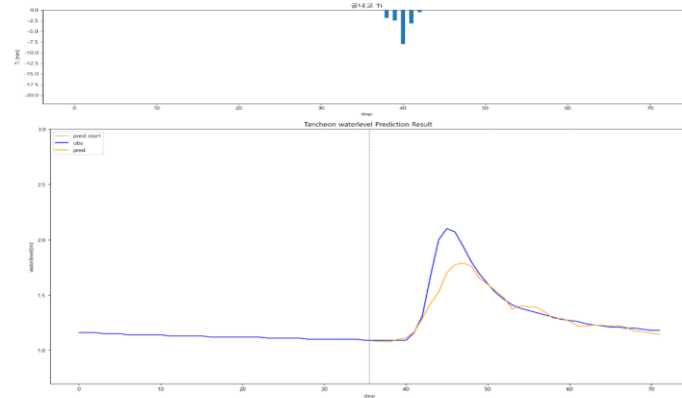
R2 : 0.8200

궁내교 - 10분단위

* model_(gn/dg)_20

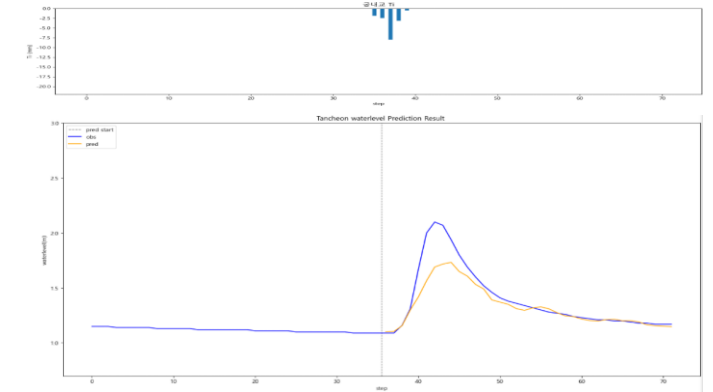
10분 전

R2 : 0.8064



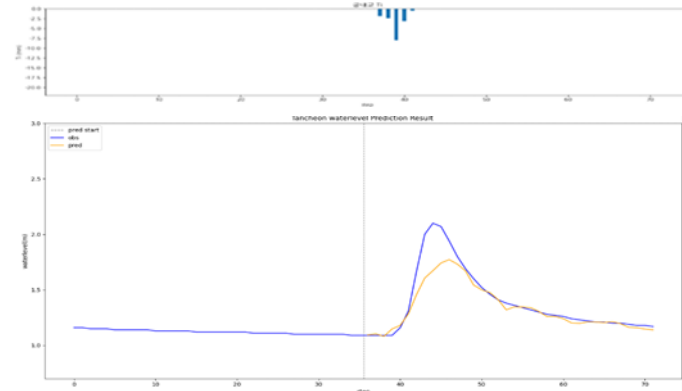
20분 후

R2 : 0.7748



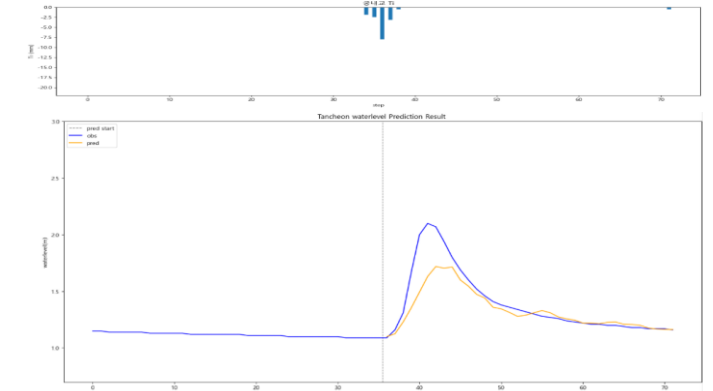
testdata_ 240723 (18)

R2 : 0.8136



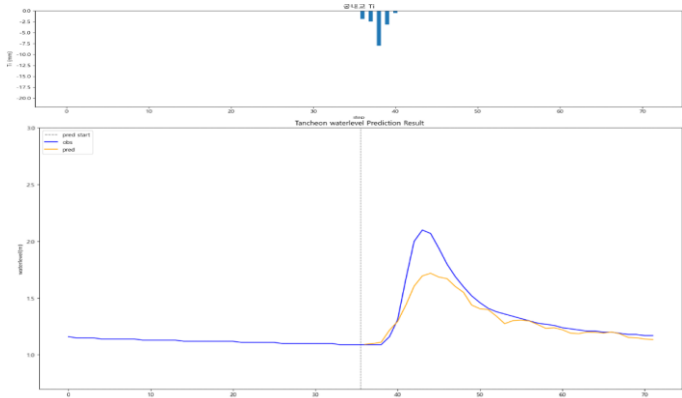
30분 후

R2 : 0.7132



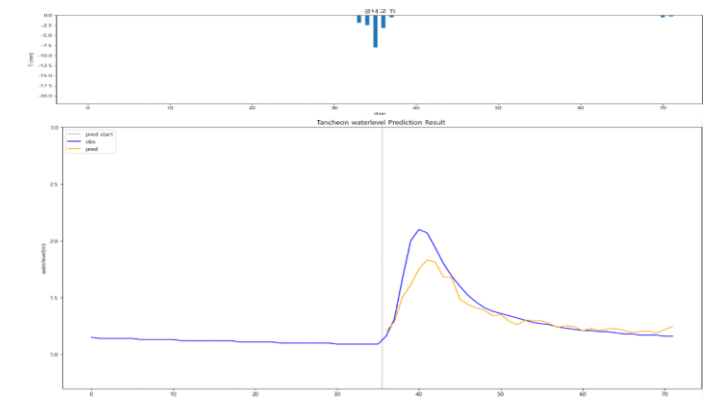
10분 후

R2 : 0.7844



40분 후

R2 : 0.8407

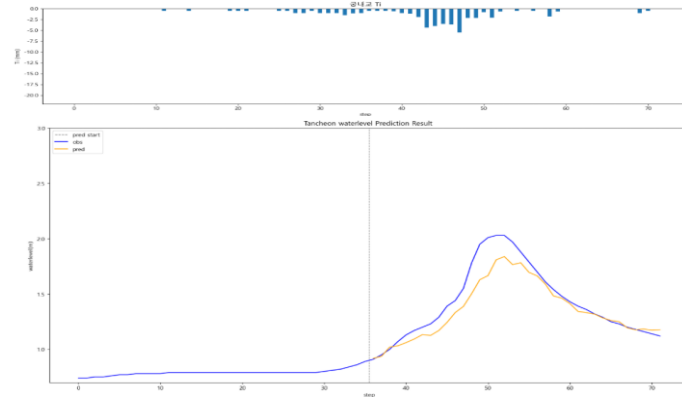


궁내교 - 10분단위

* model_(gn/dg)_20

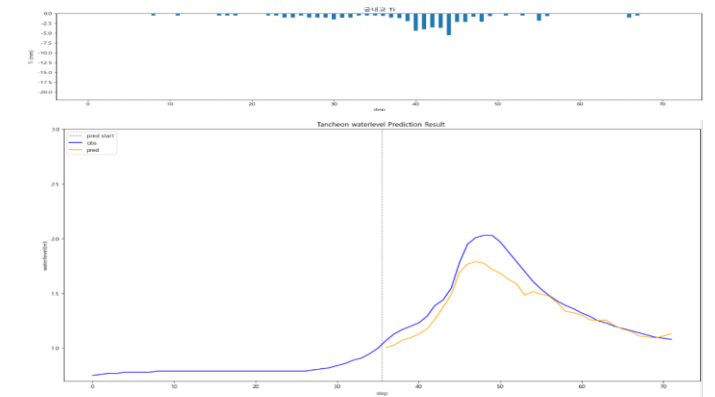
10분 전

R2 : 0.8512



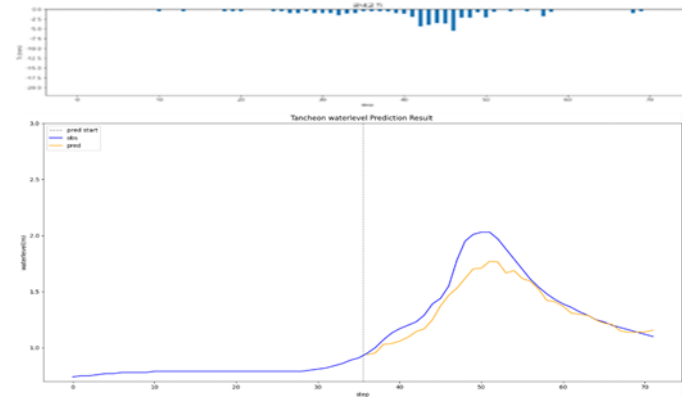
20분 후

R2 : 0.8289



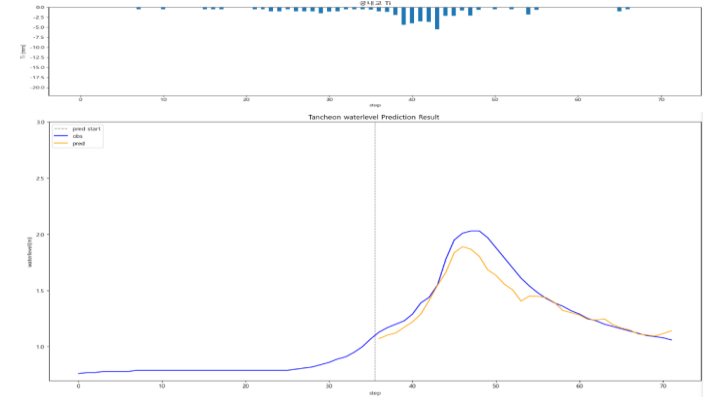
testdata_ 250716 (17)

R2 : 0.8213



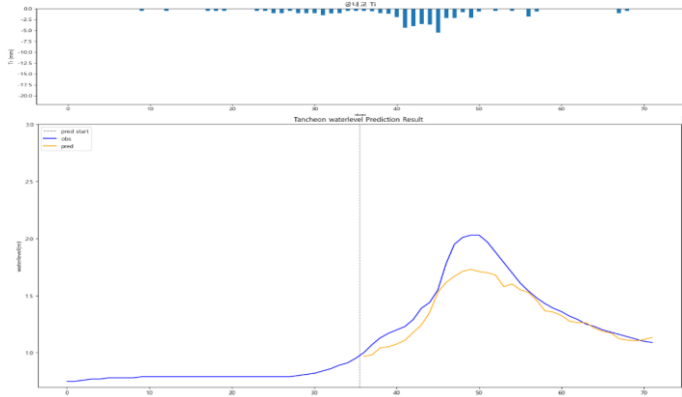
30분 후

R2 : 0.8696



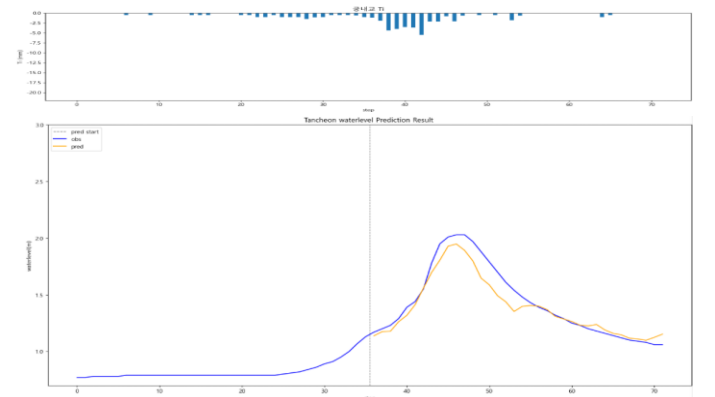
10분 후

R2 : 0.8073



40분 후

R2 : 0.9051

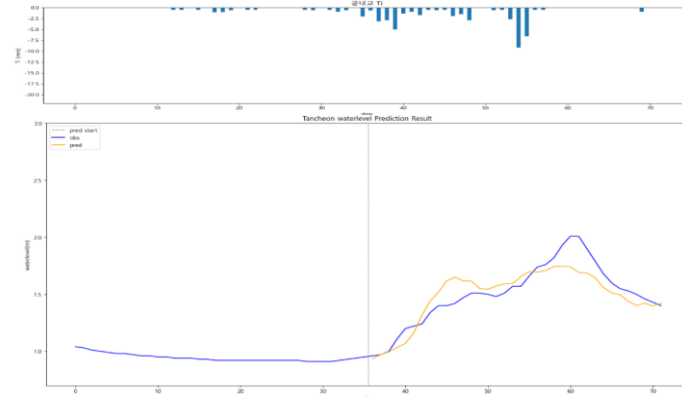
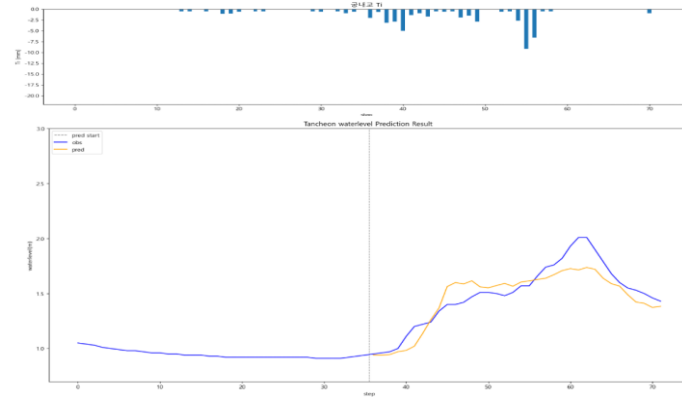


궁내교 - 10분단위

* model_(gn/dg)_20

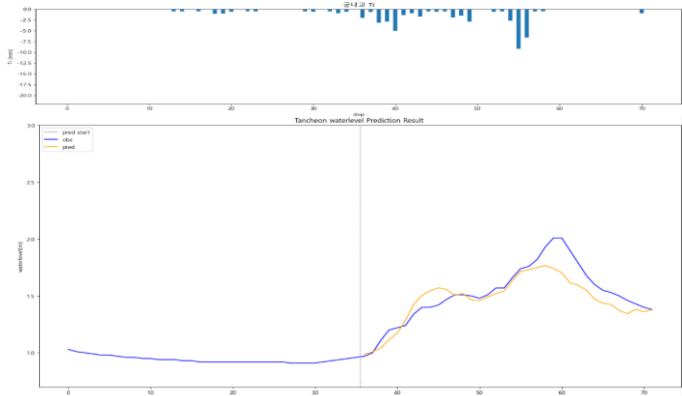
10분 전

R2 : 0.8109



testdata_250814 (05)

R2 : 0.7835

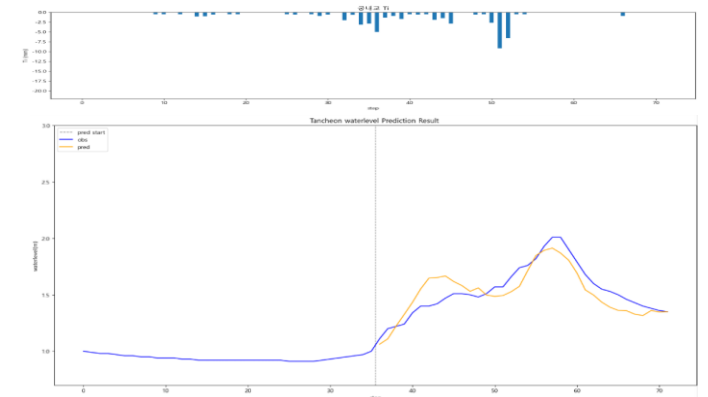
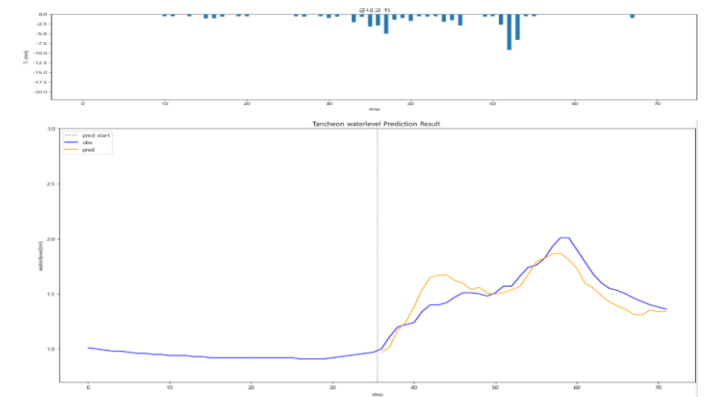


10분 후

R2 : 0.7776

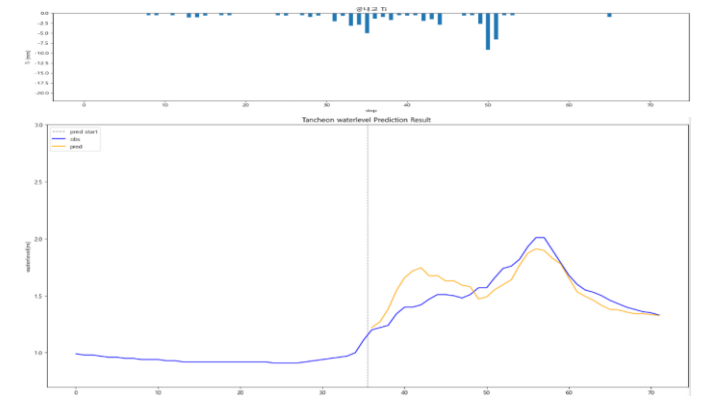
20분 후

R2 : 0.7068



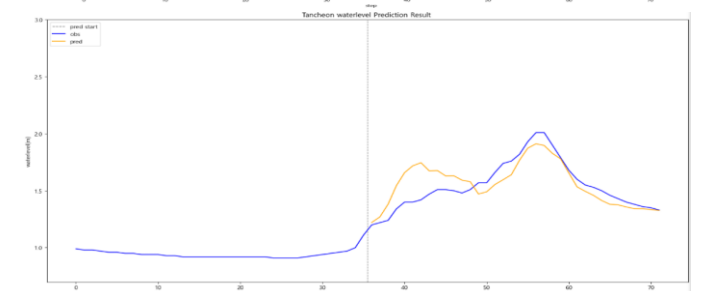
30분 후

R2 : 0.7407



40분 후

R2 : 0.6295

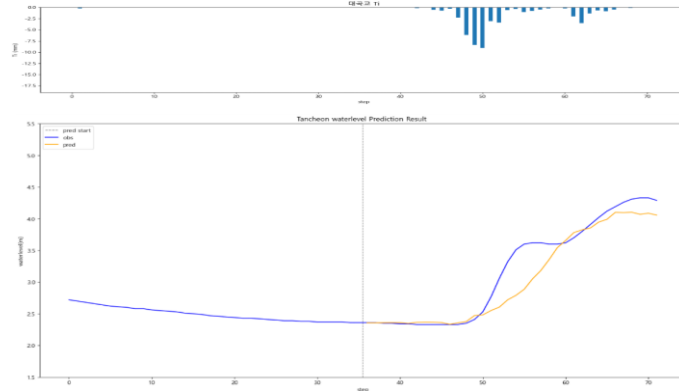


대곡교 - 10분 단위

* model_(gn/dg)_20

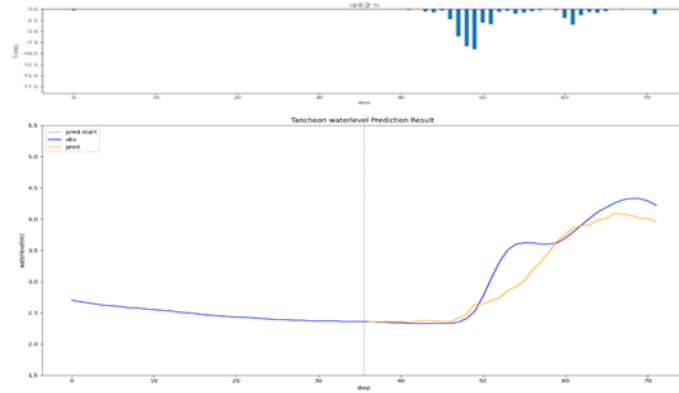
10분 전

R2 : 0.8834



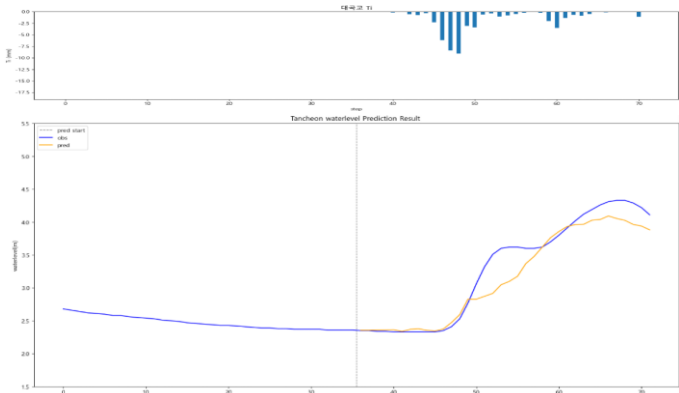
testdata_ 200802 (02)

R2 : 0.8845



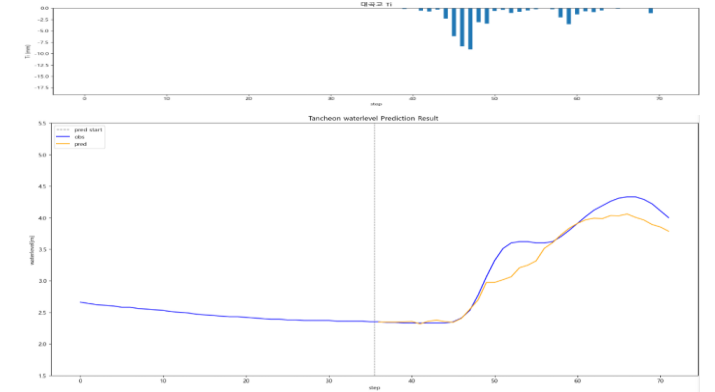
10분 후

R2 : 0.9069



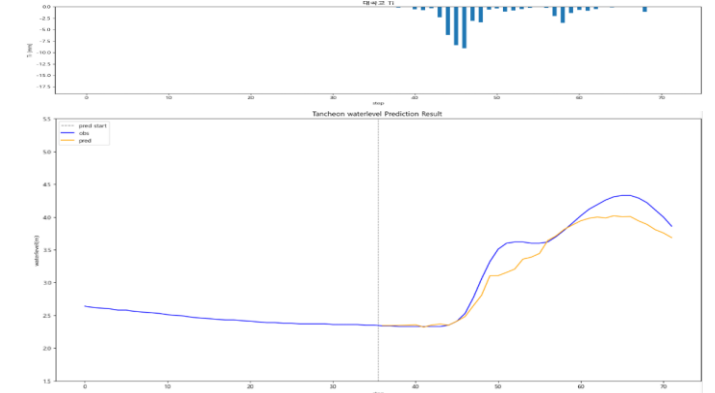
20분 후

R2 : 0.9173



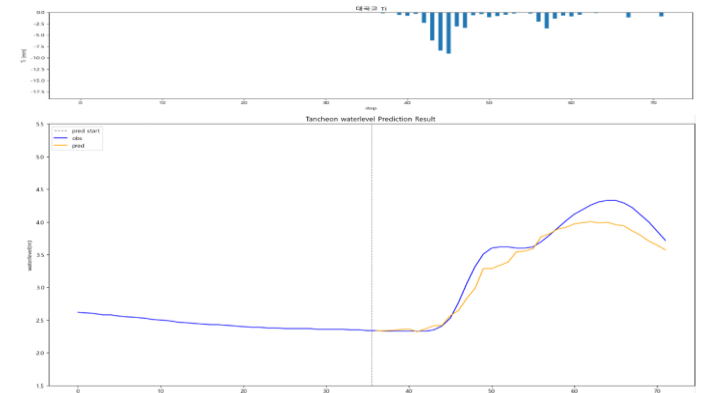
30분 후

R2 : 0.9201



40분 후

R2 : 0.9249

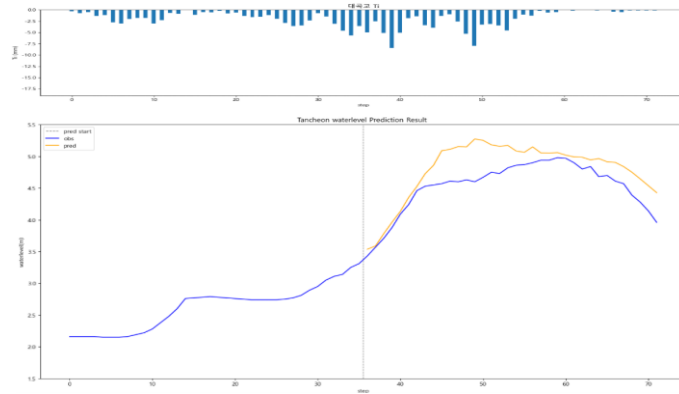


대곡교 - 10분 단위

* model_(gn/dg)_20

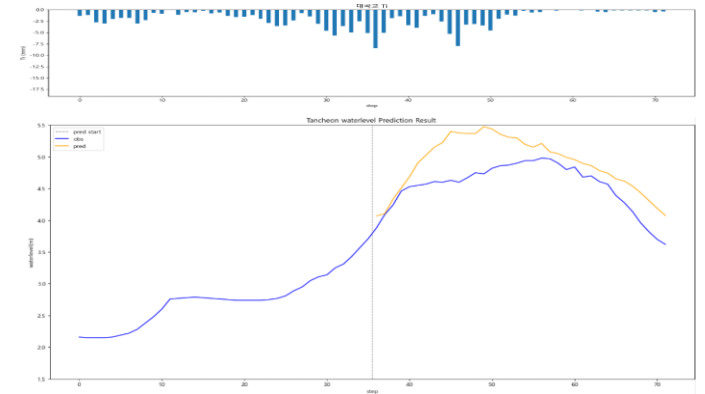
10분 전

R2 : 0.3503



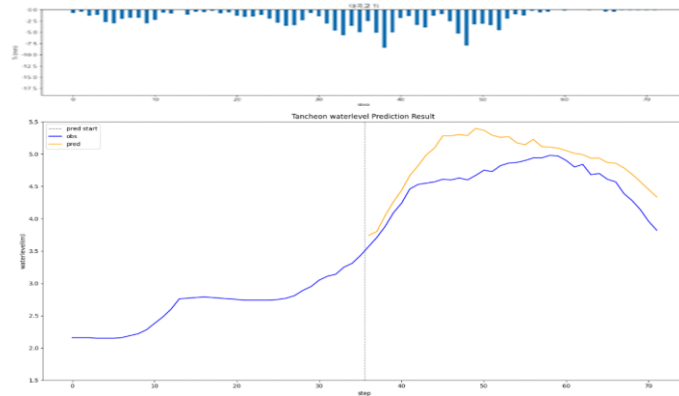
20분 후

R2 : -0.2738



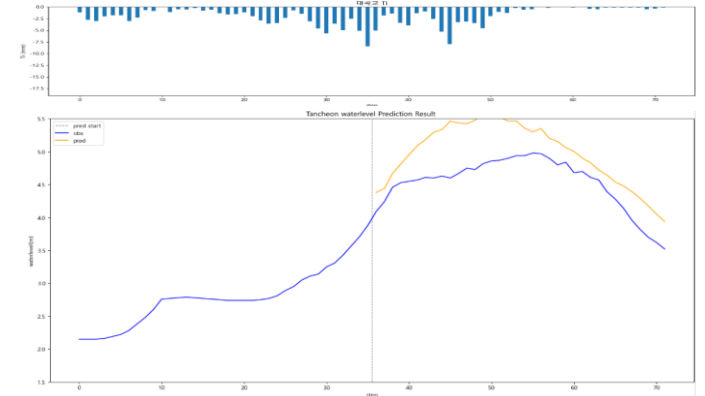
testdata_220712 (16)

R2 : -0.1561



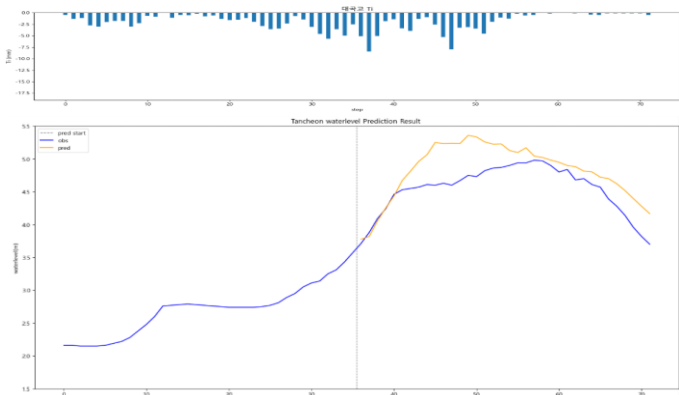
30분 후

R2 : -0.5470



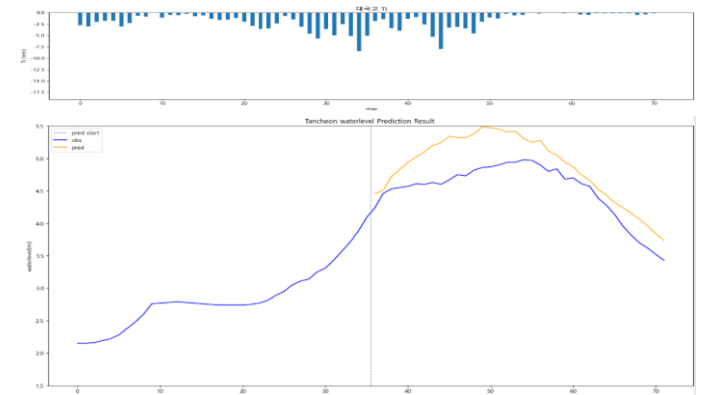
10분 후

R2 : 0.0810



40분 후

R2 : 0.1284

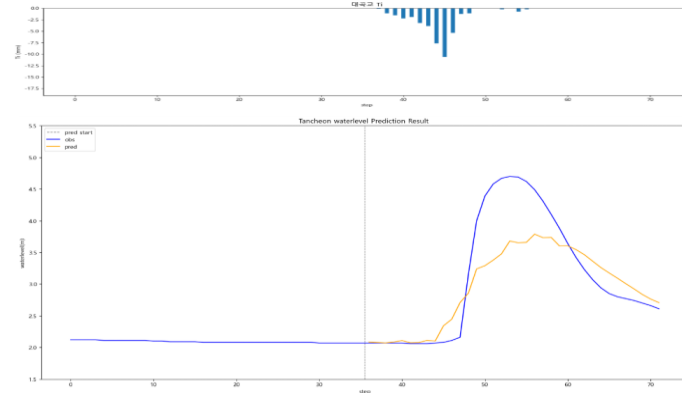


대곡교 - 10분 단위

* model_(gn/dg)_20

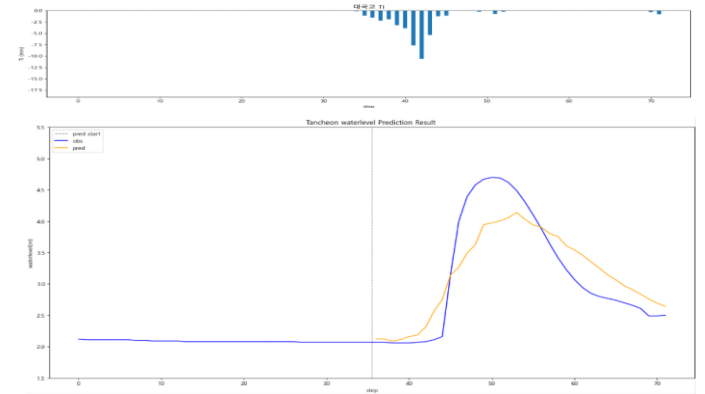
10분 전

R2 : 0.7046



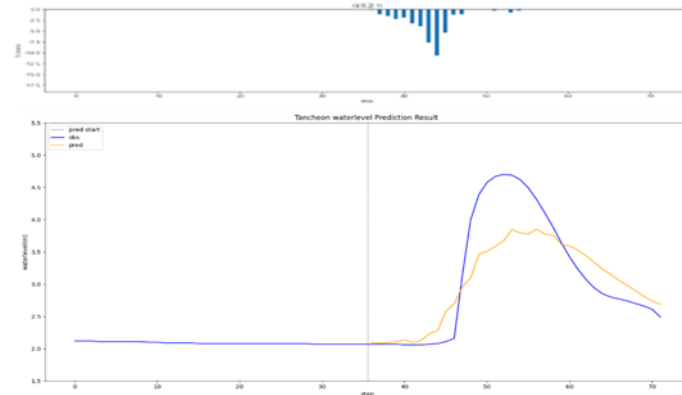
20분 후

R2 : 0.7840



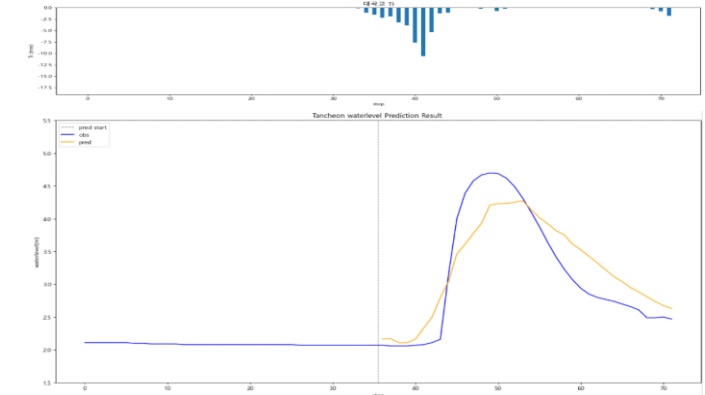
testdata_230711 (07)

R2 : 0.7304



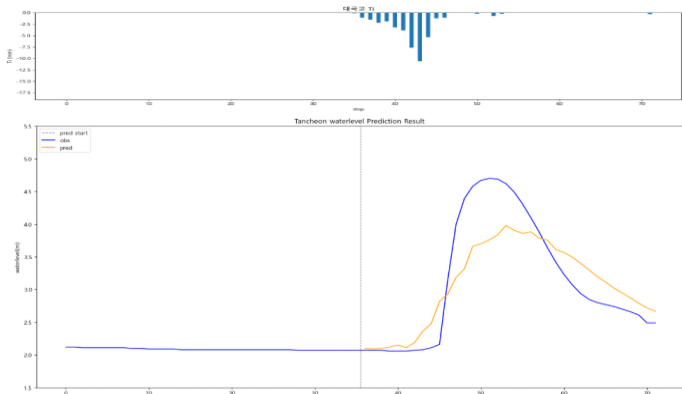
30분 후

R2 : 0.8028



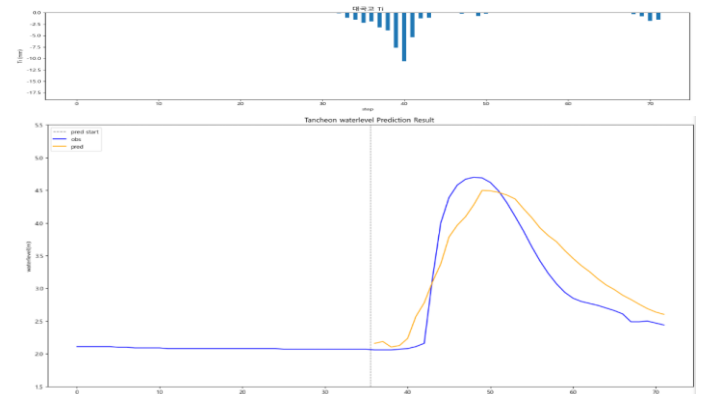
10분 후

R2 : 0.7413



40분 후

R2 : 0.8056

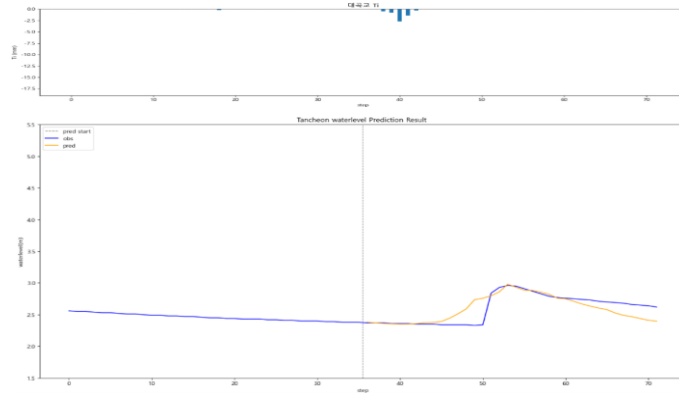


대곡교 - 10분 단위

* model_(gn/dg)_20

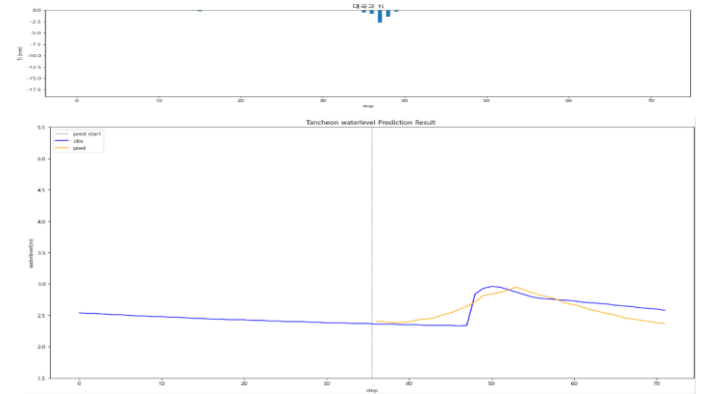
10분 전

R2 : 0.5765



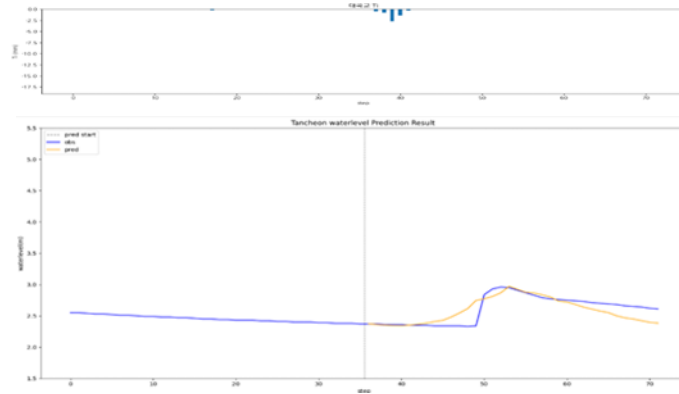
20분 후

R2 : 0.5821



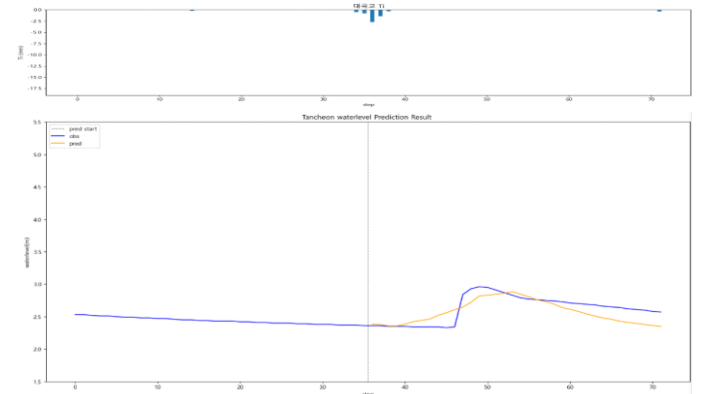
testdata_240723 (18)

R2 : 0.6077



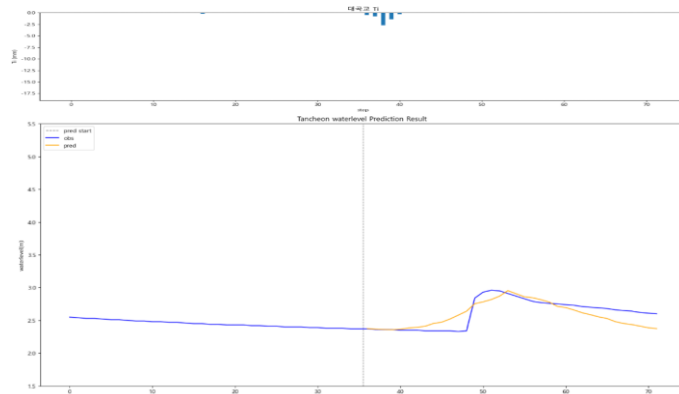
30분 후

R2 : 0.5114



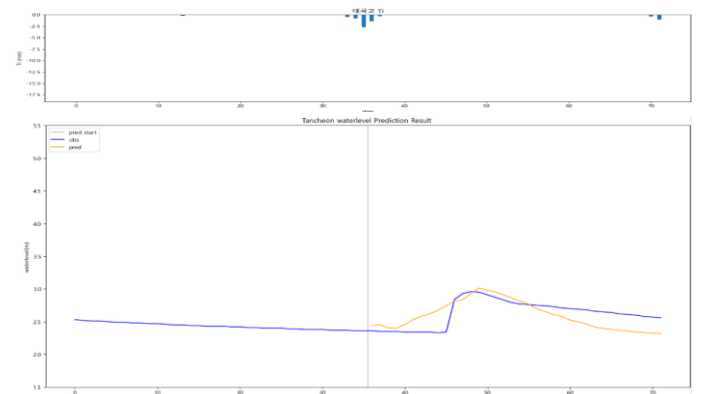
10분 후

R2 : 0.6268



40분 후

R2 : 0.1481

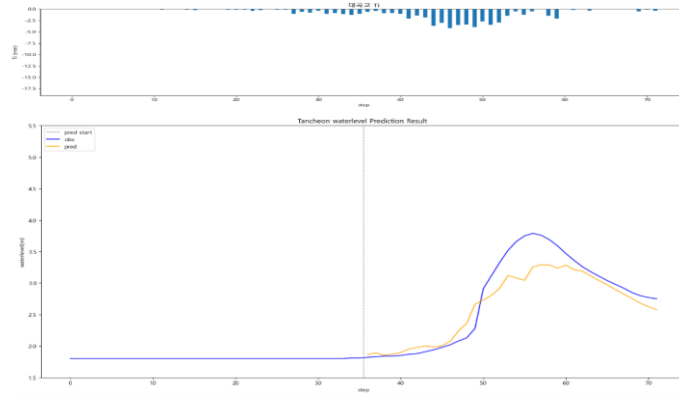


대곡교 - 10분 단위

* model_(gn/dg)_20

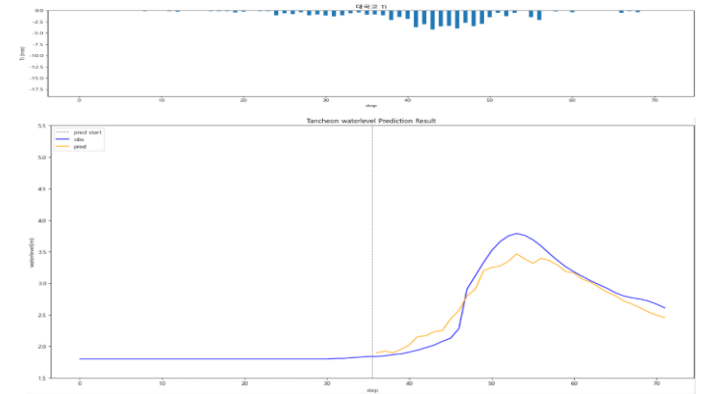
10분 전

R2 : 0.8560



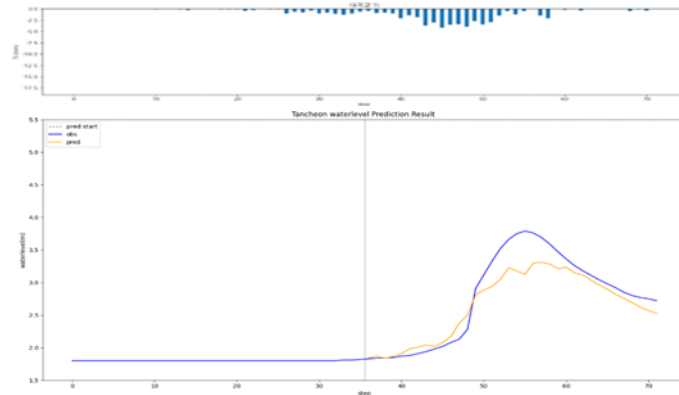
20분 후

R2 : 0.9062



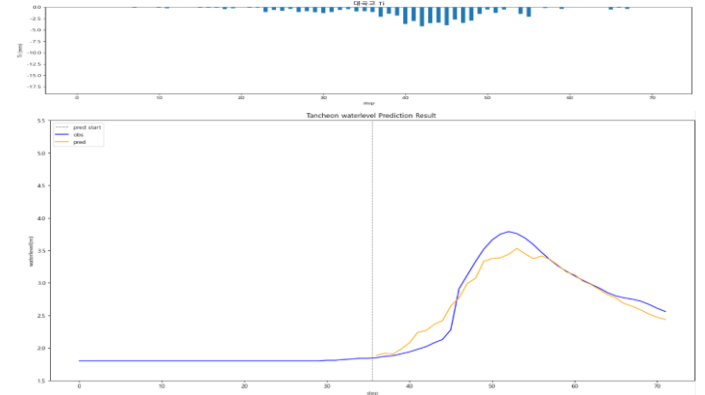
testdata_ 250716 (17)

R2 : 0.8647



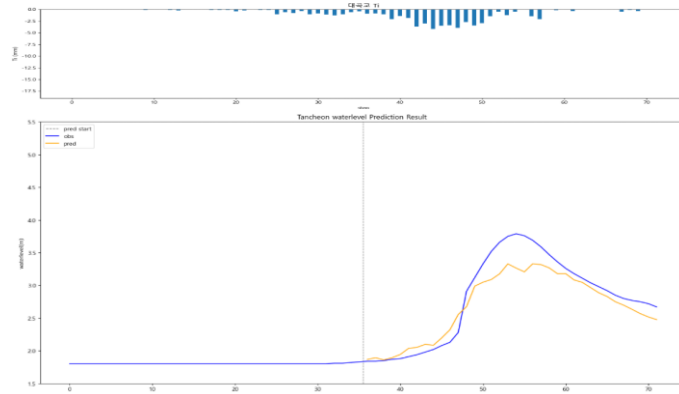
30분 후

R2 : 0.9148



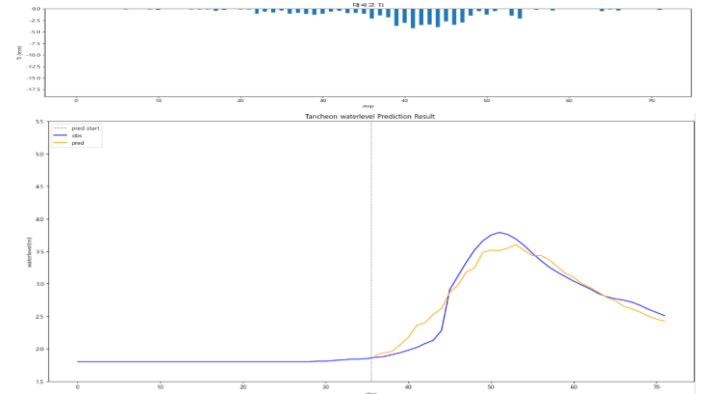
10분 후

R2 : 0.8740



40분 후

R2 : 0.9243

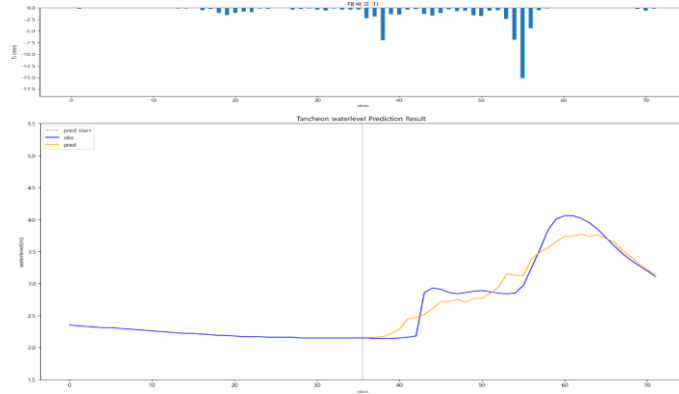


대곡교 - 10분 단위

* model_(gn/dg)_20

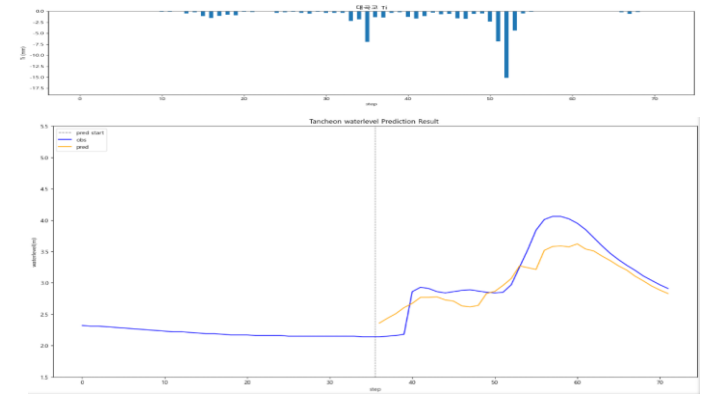
10분 전

R2 : 0.9033



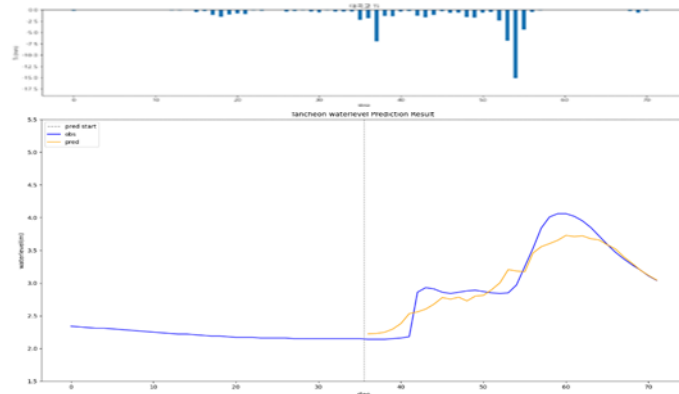
20분 후

R2 : 0.7671



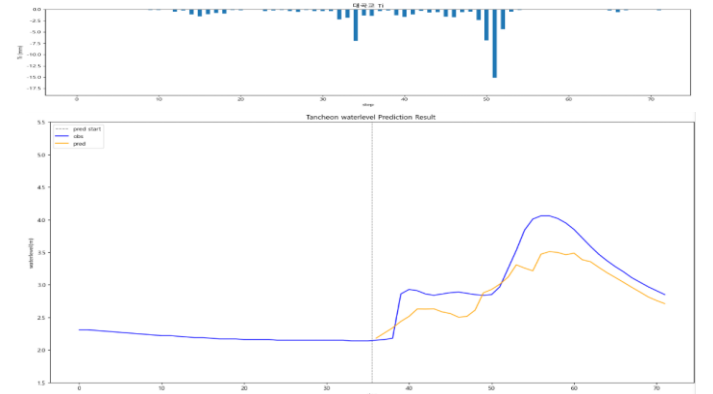
testdata_250814 (05)

R2 : 0.8840



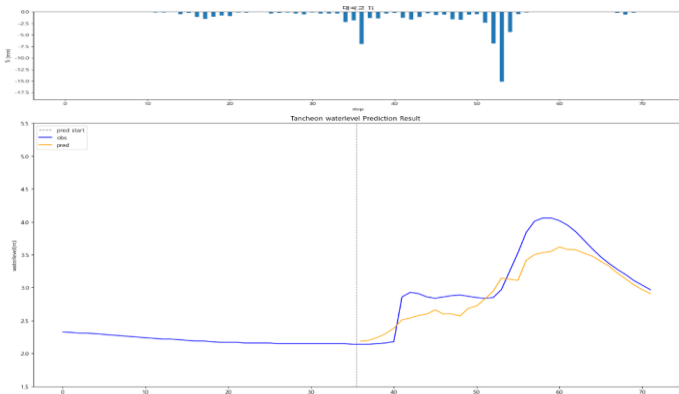
30분 후

R2 : 0.6061



10분 후

R2 : 0.7832



40분 후

R2 : 0.6646

